

Ideal-Fluid Flow

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Chapter 1

Introduction

1.1 Ideal-Fluid

Definition 1.1.1.

$$\text{Ideal-fluid} \left\{ \begin{array}{l} \text{Inviscid } (\mu = 0) \\ \text{Incompressible flow } (\rho = \text{constant}) \end{array} \right. \text{fluid}$$

The flow field generated by such flow $\implies$ *Ideal flow*

Note.

More commonly, ideal flow deals with flows which are *inviscid* ($\mu = 0$), *incompressible* ($\rho = \text{constant}$), and also *irrotational* ($\nabla \times \mathbf{u} = \mathbf{0}$). Such flows are called *Potential Flows*.

- Justifications:

For real fluids, more and less, $\mu \neq 0$, $\rho \neq \text{constant}$.

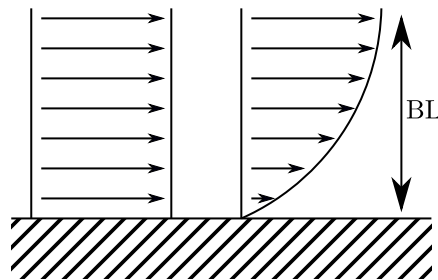
<i> Cases when flows can be treated as *inviscid*. Air and

1.1. IDEAL-FLUID

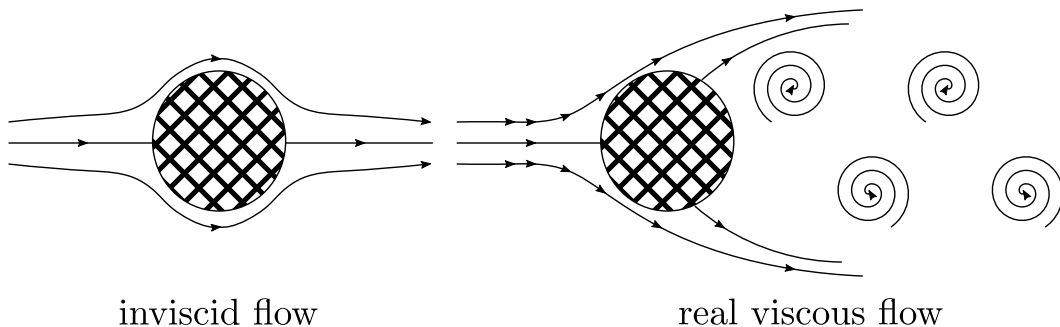
water have negligible viscosity.

$$(\nu = \mu/\rho \sim \mathcal{O}(10^{-5} \sim 10^{-6})\text{m}^2/\text{s})$$

- ① near a solid body (e.g. boundary layer)



- ② wake



- ③ fluid-fluid interface, shock...

In general, viscosity effect can be neglected if

$$Re = UL/\nu \ll 1$$

<ii> Cases when flows can be treated as *incompressible*.

- (a) Almost in all cases, the *liquid* can be taken as *incompressible*.
- (b) For *gases*, we can neglect their compressibility if the variation in density $\Delta\rho/\rho$ in the flow is $< 5\%$.

CHAPTER 1. INTRODUCTION

- ① in the case where variation in density is largely due to *pressure change* from Bernoulli's equation.

$$\begin{aligned}\frac{d(v^2/2)}{ds} &= -\frac{1}{\rho} \frac{dp}{ds} \\ \Rightarrow \frac{\Delta p}{\rho} &\sim \frac{(\Delta v)^2}{2} \sim \frac{U_\infty^2}{2}\end{aligned}\quad (1.1)$$

The sound speed “ a ” is defined:

$$a = \sqrt{\left. \frac{\partial p}{\partial \rho} \right|_s} = \sqrt{\gamma RT}, \quad \gamma = \frac{c_p}{c_v}$$

$$\begin{aligned}\therefore a^2 &= \gamma RT = \gamma \frac{p}{\rho}, \quad \boxed{M_\infty = \frac{U_\infty}{a}} \text{ (Mach number)} \\ \Rightarrow \frac{U_\infty^2}{p/\rho} &= \gamma M_\infty^2\end{aligned}\quad (1.2)$$

substitute (1.2) into (1.1)

$$\Rightarrow \frac{\Delta p}{p} \sim \frac{\gamma}{2} M_\infty^2 \quad (1.3)$$

use $p = \rho RT$, take $T \approx \text{constant}$,

$$\Rightarrow \frac{\Delta p}{p} \sim \frac{\Delta \rho}{\rho} \Rightarrow \boxed{\frac{\Delta \rho}{\rho} \sim \frac{\gamma}{2} M_\infty^2} \quad (1.4)$$

For most gases, $\gamma \approx 1$, hence $\Delta \rho / \rho < 5\%$ if $M_\infty < 0.3$.

1.1. IDEAL-FLUID

- ② in the case where variation in density is largely due to *temperature change*
Assume *adiabatic process*

$$dh = c_p dT, \quad h + \frac{v^2}{2} = \text{constant}$$
$$\implies \Delta T = \frac{(\Delta v)^2}{2c_p} \implies T_\infty = \frac{U_\infty^2}{2c_p} \quad (1.5)$$

also

$$a^2 = \gamma R T_\infty, \quad c_p = \frac{\gamma R}{\gamma - 1}$$

hence

$$(1.5) \implies \frac{\Delta T}{T_\infty} = \frac{\gamma - 1}{2} M_\infty^2 \quad (1.6)$$

For reversible, adiabatic process,

$$p\rho^{-\gamma} = c \quad \text{or} \quad pv^\gamma = c \quad \text{or} \quad T\rho^{1-\gamma} = c$$

$$\therefore \frac{\Delta T}{T_\infty} \sim (\gamma - 1) \frac{\Delta \rho}{\rho_\infty}$$

hence

$$(1.6) \implies \boxed{\frac{\Delta \rho}{\rho_\infty} \sim \frac{M_\infty^2}{2}} \quad (1.7)$$

$$\frac{\Delta \rho}{\rho_\infty} < 5\% \text{ if } \boxed{M_\infty < 0.3}$$

CHAPTER 1. INTRODUCTION

$\left. \begin{array}{l} \textcircled{1} \\ \textcircled{2} \end{array} \right\} \Rightarrow$ In general, the flow can be treated as incompressible if $M_\infty < 0.3\%$.

Combine <i> and <ii>, the flow can be treated as an

Ideal flow if $\begin{cases} Re \gg 1 \\ M < 0 \end{cases}$

except region $\begin{cases} \text{Boundary layer}(BL) \\ \text{Wake} \\ \text{shock} \end{cases}$

- Applications:

- $\textcircled{1}$ Hydraulic (e.g. Bernoulli's equation)
- $\textcircled{2}$ Surface waves
- $\textcircled{3}$ Aerodynamics (e.g. Thin-airfoil Theory)

1.1. IDEAL-FLUID

Chapter 2

Equations for the Motion of an Ideal-Fluid

2.1 Method of Analysis

Two descriptions methods for fluid motion

(1) Lagrangian description

–tracing the motion of each fluid element (control-mass approach), i.e., $\mathbf{x} = \mathbf{x}(\boldsymbol{\xi}, t)$ where $\boldsymbol{\xi}$ specifies the initial position of the fluid element, then

$$\mathbf{v} \equiv \left(\frac{\partial \mathbf{x}}{\partial t} \right)_{\boldsymbol{\xi}} \quad \text{and} \quad \mathbf{a} \equiv \left(\frac{\partial^2 \mathbf{x}}{\partial t^2} \right)_{\boldsymbol{\xi}} \quad \text{etc.}$$

(2) Eulerian description

2.2. REYNOLDS TRANSPORT THEOREM (RTT)

–recording fluid motion at each fixed location (field description, control-volume approach), i.e., $\mathbf{v} = \mathbf{v}(\mathbf{x}, t)$ where $\mathbf{x}$ are fixed spatial coordinates.

$$\therefore \mathbf{v} \neq \frac{\partial \mathbf{x}}{\partial t}.$$

Analysis can be classified as:

- ① Integral analysis– Conservation laws being applied to a fixed region.
- ② Differential analysis– Conservation laws being applied, in a limiting case, to a point. This requires flow properties to be differentiable.

2.2 Reynolds Transport Theorem (RTT)

Laws of Mechanics are applied to a control-mass system (Lagrangian description). RTT converts the system analysis to an equivalent expression in control-volume approach (Eulerian description).

- Substantial derivative (time rate of change of properties following fluid motion)

CHAPTER 2. EQUATIONS FOR THE MOTION OF AN IDEAL-FLUID

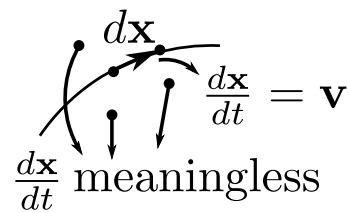
Let $B(\mathbf{x}, t)$ be a scalar (or vector) function, then

$$dB = d\mathbf{x} \cdot \nabla B + \frac{\partial B}{\partial t} dt$$

$$\therefore \frac{dB}{dt} = \mathbf{v} \cdot \nabla B + \frac{\partial B}{\partial t}$$

or

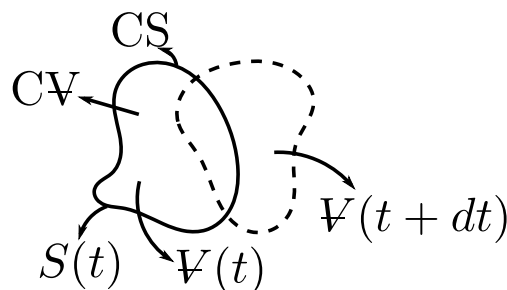
$$\underbrace{\frac{DB}{Dt}}_{\text{Lagrangian derivative}} = \underbrace{\mathbf{v} \cdot \nabla B + \frac{\partial B}{\partial t}}_{\text{equivalent Eulerian expression}}$$



(2.1)

- RTT

Let $V(t)$ be a closed region moving with the fluid (i.e. material volume) with boundary surface denoted by $S(t)$ and let $B(\mathbf{x}, t)^*$ denote any *intensive* property of the fluid field, then



*could be scalar or vector function

2.3. DIFFERENTIAL EQUATIONS FOR AN IDEAL-FLUID MOTION

$$\begin{aligned} \frac{D}{Dt} \iiint_{V(t)} \rho B dV & \stackrel{\dagger}{=} \iiint_{V(t)} \frac{D(\rho B)}{Dt} dV + \rho B \frac{D(dV)}{Dt} \\ & = \iiint_{CV} \frac{D(\rho B)}{Dt} dV + \rho B (\nabla \cdot \mathbf{v}) dV \quad (2.2a) \end{aligned}$$

$$= \iiint_{CV} \left[\frac{\partial(\rho B)}{\partial t} + \nabla \cdot (\rho B \mathbf{v}) \right] dV \quad (2.2b)$$

$$= \iiint_{CV} \frac{\partial(\rho B)}{\partial t} dV + \iint_{CS} \mathbf{n} \cdot (\rho B \mathbf{v}) dS \quad (2.2c)$$

2.3 Differential Equations for an Ideal-Fluid Motion

- Conservation of mass

$$\frac{D}{Dt} \iiint_{V(t)} \rho dV = 0$$

Apply RTT (set $B \equiv 1$)

$$\left\{ \begin{aligned} \iiint_{CV} \frac{\partial \rho}{\partial t} dV + \underbrace{\iint_{CS} \mathbf{n} \cdot (\rho \mathbf{v}) dS}_{\text{net mass flow rate across CS (mass flux)}} &= 0 \end{aligned} \right. \quad (2.3a)$$

$$\left\{ \begin{aligned} \iiint_{CV} \left[\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) \right] dV &= 0 \end{aligned} \right. \quad (2.3b)$$

[†]Recall $\nabla \cdot \mathbf{v} = \frac{1}{dV} \frac{D(dV)}{Dt}$, then $\frac{D(dV)}{Dt} = (\nabla \cdot \mathbf{v}) dV$

CHAPTER 2. EQUATIONS FOR THE MOTION OF AN IDEAL-FLUID

Since CV is *arbitrary*

$$\implies \boxed{\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0} \text{ everywhere} \quad (2.4a)$$

$$\begin{aligned} \text{or } \underbrace{\frac{\partial \rho}{\partial t} + \mathbf{v} \cdot \nabla \rho + \rho \nabla \cdot \mathbf{v}}_{\equiv \frac{D\rho}{Dt}} &= 0 \\ \implies \frac{D\rho}{Dt} + \rho \nabla \cdot \mathbf{v} &= 0 \end{aligned} \quad (2.4b)$$

For an ideal fluid, $\rho = \text{constant}$

$$\implies \frac{D\rho}{Dt} = 0 \quad \text{or} \quad \boxed{\nabla \cdot \mathbf{v} = 0} \text{ everywhere} \quad (2.5)$$

- Conservation of Momentum

$$\frac{D}{Dt} \iiint_{V(t)} \rho \mathbf{v} dV = \sum \mathbf{F} = \iiint_{V(t)} \mathbf{f}_b dV + \iint_{S(t)} \mathbf{f}_s dS \quad (*)$$

where

$\mathbf{f}_b$: body force/volume, $\mathbf{f}_b = \rho \mathbf{g}$

$\mathbf{f}_s$: surface force/area, $\mathbf{f}_s = \mathbf{n} \cdot \boldsymbol{\tau}$

$$(*) \implies \frac{D}{Dt} \iiint_{V(t)} \rho \mathbf{v} dV = \iiint_{V(t)} \rho \mathbf{g} dV + \iint_{S(t)} \mathbf{n} \cdot \boldsymbol{\tau} dS$$

2.3. DIFFERENTIAL EQUATIONS FOR AN IDEAL-FLUID MOTION

Apply RTT (set $B = \mathbf{v}$)

$$\left\{ \begin{array}{l} \iiint_{CV} \frac{\partial(\rho \mathbf{v})}{\partial t} dV + \underbrace{\iint_{CS} \mathbf{n} \cdot (\rho \mathbf{v} \mathbf{v}) dS}_{\text{net momentum flux across CS}} \\ = \iiint_{CV} \rho \mathbf{g} dV + \iint_{CS} \mathbf{n} \cdot \boldsymbol{\tau} dS \end{array} \right. \quad (2.6a)$$

$$\left\{ \begin{array}{l} \iiint_{CV} \frac{\partial(\rho \mathbf{v})}{\partial t} dV + \iint_{CV} \nabla \cdot (\rho \mathbf{v} \mathbf{v}) dV \\ = \iiint_{CV} \rho \mathbf{g} dV + \iiint_{CV} \nabla \cdot \boldsymbol{\tau} dV \end{array} \right. \quad (2.6b)$$

Since CV is *arbitrary*

$$\Rightarrow \frac{\partial(\rho \mathbf{v})}{\partial t} + \nabla \cdot (\rho \mathbf{v} \mathbf{v}) = \rho \mathbf{g} + \nabla \cdot \boldsymbol{\tau} \quad (2.7a)$$

Use $\nabla \cdot (\rho \mathbf{v} \mathbf{v}) = [\nabla \cdot (\rho \mathbf{v})] \mathbf{v} + \rho \mathbf{v} \cdot (\nabla \mathbf{v})$

and apply continuity equation

$$\begin{aligned} \Rightarrow \underbrace{\rho \frac{\partial \mathbf{v}}{\partial t} + \rho \mathbf{v} \cdot \nabla \mathbf{v}}_{\substack{D\mathbf{v} \\ = \rho \frac{D\mathbf{v}}{Dt}}} &= \rho \mathbf{g} + \nabla \cdot \boldsymbol{\tau} \end{aligned} \quad (2.7b)$$

For an ideal fluid, $\mu = 0$,

$$\boldsymbol{\tau} = -p\mathbf{I} \text{ only}$$

hence (2.7b) reduce to

$$\boxed{\rho \frac{D\mathbf{v}}{Dt} = \rho \mathbf{g} - \nabla p} \quad (\text{Euler equation}) \quad (2.8)$$

CHAPTER 2. EQUATIONS FOR THE MOTION OF AN IDEAL-FLUID

- Conservation of Energy (1st law of thermodynamics)

$$\begin{aligned}
 \frac{D}{Dt} \iiint_{V(t)} \rho \left(e + \frac{v^2}{2} \right) dV &= \sum \mathbf{F} \cdot \mathbf{v} \\
 &= \iiint_{V(t)} \mathbf{f}_b \cdot \mathbf{v} dV + \iint_{S(t)} \mathbf{f}_s \cdot \mathbf{v} dS - \iint_{S(t)} \mathbf{n} \cdot \dot{\mathbf{q}} dS \\
 &= \iiint_{V(t)} \rho \mathbf{g} \cdot \mathbf{v} dV + \iint_{S(t)} (\mathbf{n} \cdot \boldsymbol{\tau}) \cdot \mathbf{v} dS + \iint_{S(t)} \mathbf{n} \cdot (k \nabla T) dS
 \end{aligned}$$

where

e : internal energy/mass

$\frac{v^2}{2}$: macro kinetic energy/mass

2.3. DIFFERENTIAL EQUATIONS FOR AN IDEAL-FLUID MOTION

Apply RTT (set $B = e + v^2/2$)

$$\left\{ \begin{aligned}
 & \iiint_{CV} \frac{\partial}{\partial t} \left[\rho \left(e + \frac{v^2}{2} \right) \right] dV \\
 & + \underbrace{\iint_{CS} \mathbf{n} \cdot \left[\rho \left(e + \frac{v^2}{2} \right) \right] dS}_{\substack{\text{net energy flux} \\ \text{across CS}}} \\
 & = \iiint_{CV} \rho \mathbf{g} \cdot \mathbf{v} dV + \iint_{CS} (\mathbf{n} \cdot \boldsymbol{\tau}) \cdot \mathbf{v} dS \\
 & + \iint_{CS} \mathbf{n} \cdot (k \nabla T) dS \tag{2.9a} \\
 & \iiint_{CV} \frac{\partial}{\partial t} \left[\rho \left(e + \frac{v^2}{2} \right) \right] dV \\
 & + \iiint_{CV} \nabla \cdot \left[\rho \left(e + \frac{v^2}{2} \right) \right] dV \\
 & = \iiint_{CV} \rho \mathbf{g} \cdot \mathbf{v} dV + \iiint_{CV} \nabla \cdot (\boldsymbol{\tau} \cdot \mathbf{v}) dV \\
 & + \iiint_{CV} \nabla \cdot (k \nabla T) dV \tag{2.9b}
 \end{aligned} \right.$$

CHAPTER 2. EQUATIONS FOR THE MOTION OF AN IDEAL-FLUID

Since CV is *arbitrary*

$$\begin{aligned} \Rightarrow \frac{\partial}{\partial t} \left[\rho \left(e + \frac{v^2}{2} \right) \right] + \nabla \cdot \left[\rho \left(e + \frac{v^2}{2} \right) \right] \\ = \rho \mathbf{g} \cdot \mathbf{v} + \nabla \cdot (\boldsymbol{\tau} \cdot \mathbf{v}) + \nabla \cdot (k \nabla T) \end{aligned} \quad (2.10a)$$

Use continuity/momentum equation

$$\begin{aligned} \Rightarrow \rho \frac{D}{Dt} \left[\rho \left(e + \frac{v^2}{2} \right) \right] \\ = \rho \mathbf{g} \cdot \mathbf{v} + \nabla \cdot (\boldsymbol{\tau} \cdot \mathbf{v}) + \nabla \cdot (k \nabla T) \end{aligned} \quad (2.10b)$$

For an ideal fluid, $\begin{cases} \rho = \text{constant} \Rightarrow \nabla \cdot \mathbf{v} = 0 \\ \mu = 0 \Rightarrow \boldsymbol{\tau} = -p\mathbf{I} \end{cases}$ hence
(2.10b) reduce to

$$\rho \frac{D}{Dt} \left[\rho \left(e + \frac{v^2}{2} \right) \right] = \rho \mathbf{g} \cdot \mathbf{v} - \nabla \cdot (p\mathbf{v}) + \nabla \cdot (k \nabla T) \quad (*)$$

From momentum equation (2.8) perform

$$\mathbf{v} \cdot \left\{ \rho \frac{D\mathbf{v}}{Dt} = \rho \mathbf{g} - \nabla p \right\} \quad (**)$$

2.3. DIFFERENTIAL EQUATIONS FOR AN IDEAL-FLUID MOTION

notice that

$$\begin{aligned}\mathbf{v} \cdot [\mathbf{v} \cdot \nabla \mathbf{v}] &= \mathbf{v} \cdot \left[\nabla \left(\frac{v^2}{2} \right) - \underbrace{\mathbf{v} \times (\nabla \times \mathbf{v})}_{=\boldsymbol{\xi}} \right] \\ &= \mathbf{v} \cdot \nabla \left(\frac{v^2}{2} \right) - \underbrace{\mathbf{v} \cdot (\mathbf{v} \times \boldsymbol{\xi})}_{=0}\end{aligned}$$

hence (**)

$$\implies \rho \frac{D}{Dt} \left(\frac{v^2}{2} \right) = \rho \mathbf{g} \cdot \mathbf{v} - \mathbf{v} \cdot \nabla p \quad (2.11)$$

Subtract (2.11) from (*)

$$\implies \rho \frac{De}{Dt} = -\mathbf{v} \cdot \nabla p + \nabla \cdot (k \nabla T) \quad (2.12)$$

For ideal fluid, $\nabla \cdot \mathbf{v} = 0$ and non-conducting, $k = 0$,

$$\implies \boxed{\frac{De}{Dt} = 0}$$

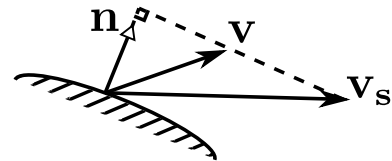
Thus, don't need energy equation when dealing with ideal flows.

CHAPTER 2. EQUATIONS FOR THE MOTION OF AN IDEAL-FLUID

2.4 Boundary Condition for an Ideal-Fluid Motion

- At solid-fluid boundary– No penetration condition

$$\boxed{\mathbf{v} \cdot \mathbf{n} = \mathbf{v}_s \cdot \mathbf{n}} \quad (2.13)$$



If body is fixed, $\mathbf{v}_s = \mathbf{0}$

$$(2.13) \implies \boxed{\mathbf{v} \cdot \mathbf{n} = 0} \quad (2.14)$$

i.e., fluid can only slide on the solid body surface.

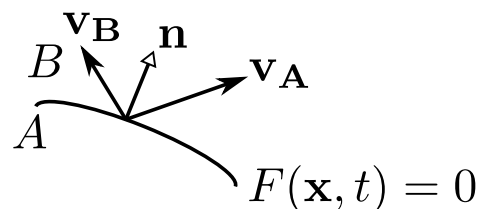
- At fluid-fluid boundary
Assume immiscible situation

(a) Kinematic conditions

Let $F(\mathbf{x}, t) = 0$ represents the instantaneous position of the interface, then

$$\boxed{\mathbf{v}_A \cdot \mathbf{n} = \mathbf{v}_B \cdot \mathbf{n}}$$

$$\text{or } (\mathbf{v}_A - \mathbf{v}_B) \cdot \underbrace{\nabla F}_{\parallel \mathbf{n}} = 0 \quad (2.15)$$

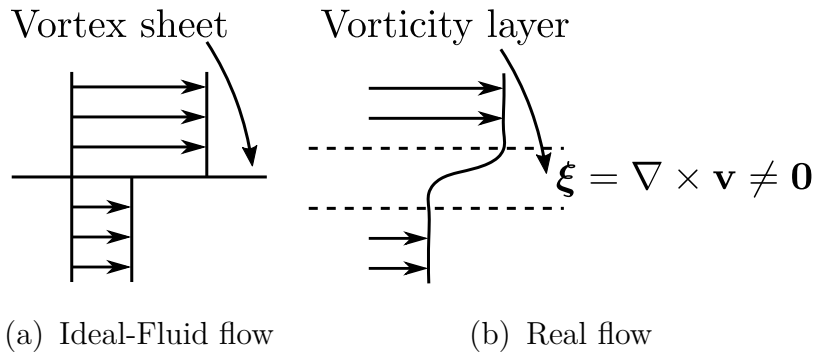


but the tangential components may be different
across the interface $\implies$ surface of tangential

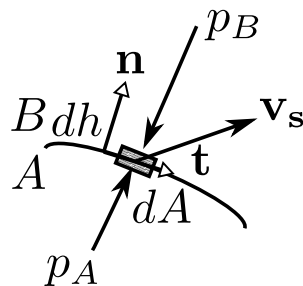
2.4. BOUNDARY CONDITION FOR AN IDEAL-FLUID MOTION

discontinuity.

e.g.



(b) Dynamic condition



Take a small element moving with the interface $dh dA$ over a segment of the interface, the force balance equation in the $\mathbf{n}$ -direction

$$f_{bn} dA dh + (p_A - p_B) dA - \underbrace{\sum a_{\text{rel},n} (\rho dA dh)}_{\text{Apparent force terms}} = 0$$

$$\text{As } dh \rightarrow 0 \implies \boxed{p_A = p_B} \quad (2.16)$$

Chapter 3

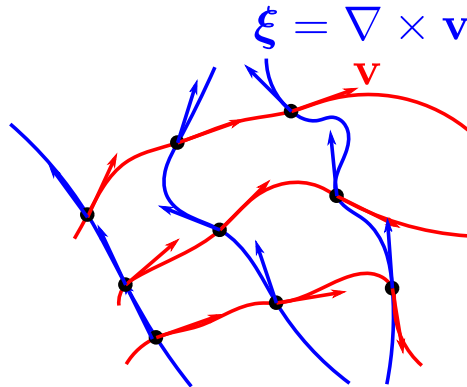
Kinematic and Dynamics of Ideal-Fluid Motion

3.1 Kinematic

3.1.1 Streamlines and Vorticity Lines

- Streamlines: A curve that is everywhere tangent to the (instantaneous) local velocity vectors.
- Vorticity lines: A curve that is everywhere tangent to the (instantaneous) local vorticity vectors.

3.1. KINEMATIC

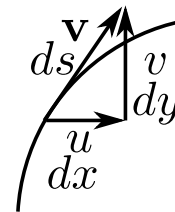


- Equation of streamline

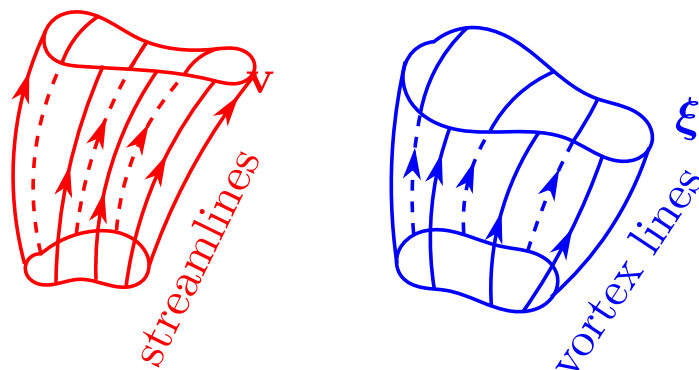
$$\frac{dx}{u} = \frac{dy}{v} = \frac{dz}{w} \quad (3.1)$$

Similarly, for vorticity line

$$\frac{dx}{\xi_x} = \frac{dy}{\xi_y} = \frac{dz}{\xi_z} \quad (3.2)$$



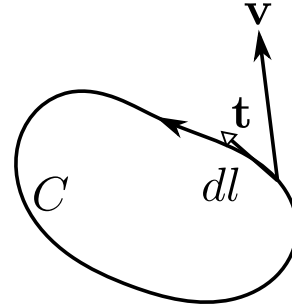
- Stream tube and vorticity tube (3D expression)



- Circulation
Define circulation

CHAPTER 3. KINEMATIC AND DYNAMICS OF IDEAL-FLUID MOTION

$$\Gamma = \oint_C \mathbf{v} \cdot \mathbf{t} dl \quad (3.3)$$



From Stokes' theorem

$$\begin{aligned} \Gamma = \oint_C \mathbf{v} \cdot \mathbf{t} dl &= \iint_S \mathbf{n} \cdot (\nabla \times \mathbf{v}) dS \\ &= \iint_S \mathbf{n} \cdot \boldsymbol{\xi} dS \end{aligned}$$

$\boldsymbol{\xi} = \mathbf{0}$ at all points enclosed in $C \implies \Gamma = 0$ around C .

$\Gamma = 0$ for every loop in the flow field ($\implies$ circulation-free flow field)

$\implies^* \Gamma = 0$ everywhere in the flow field ($\implies$ irrotational flow field)

- Kinematic of vorticity lines

Since $\nabla \cdot (\nabla \times \mathbf{v}) = 0 \implies \boxed{\nabla \cdot \boldsymbol{\xi} = 0}$, there can be no source or sink of vorticity, i.e., vorticity line must either forms a closed loop or must terminate on the boundary. (Part of the Helmholtz Theorem). Furthermore, for a closed region V

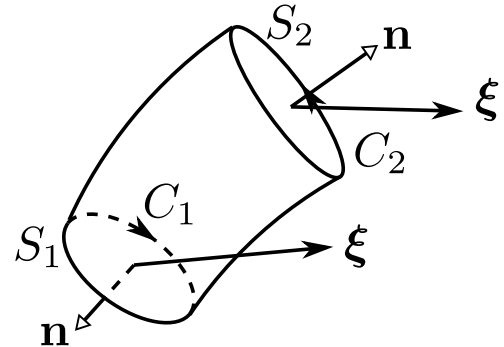
$$\iiint_V \nabla \cdot \boldsymbol{\xi} dV \stackrel{\text{div. thm.}}{=} \oiint_S \mathbf{n} \cdot \boldsymbol{\xi} dS = 0$$

*can be reversed only for simply-connected region

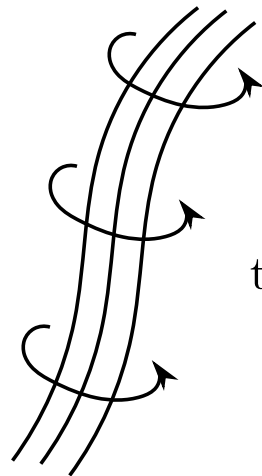
3.1. KINEMATIC

If the closed region is a section of a vorticity tube

$$\begin{aligned} & \iint_{S_1} \mathbf{n} \cdot \boldsymbol{\xi} dS + \iint_{S_2} \mathbf{n} \cdot \boldsymbol{\xi} dS = 0 \\ \Rightarrow & - \oint_{C_1} \mathbf{v} \cdot \mathbf{t} dl + \oint_{C_2} \mathbf{v} \cdot \mathbf{t} dl = 0 \\ \Rightarrow & \boxed{\Gamma_{C_1} = \Gamma_{C_2}} \quad (|\boldsymbol{\xi}_1| S_1 = |\boldsymbol{\xi}_2| S_2) \end{aligned}$$



i.e., the circulation of each cross section of a vorticity tube is a *constant* (Part of the Helmholtz Theorem).



Γ : strength of the vorticity tube

3.1.2 Stream Function and Potential Function

- Potential function

A vector field $\mathbf{v}(\mathbf{x}, t)$ satisfying $\nabla \times \mathbf{v} = \mathbf{0}$ everywhere is called the *irrotational field*.

If $\mathbf{v}$ is velocity field of a flow $\Rightarrow$ *irrotational field*.

CHAPTER 3. KINEMATIC AND DYNAMICS OF IDEAL-FLUID MOTION

Can represent $\mathbf{v}$ by the gradient of a scalar function $\Phi(\mathbf{x}, t)$, i.e.

$$\mathbf{v} = \nabla\Phi \text{ (since } \nabla \times (\nabla\Phi) = \mathbf{0}) \quad (3.4)$$

where Φ is scalar (velocity) potential function.

- Stream function

A vector field $\mathbf{v}(\mathbf{x}, t)$ satisfying $\nabla \cdot \mathbf{v} = 0$ everywhere is called the *solenoidal field*.

If $\mathbf{v}$ is velocity field of a flow $\implies$ *solenoidal field*.

Can represent $\mathbf{v}$ by the curl of a scalar function $\Psi(\mathbf{x}, t)$, i.e.

$$\mathbf{v} = \nabla \times \Psi \text{ (since } \nabla \cdot (\nabla \times \Psi) = 0) \quad (3.5)$$

where Ψ is vector potential function. (Note: Ψ is not uniquely determined by $\mathbf{v}$)

Specially for a 2D incompressible flow

Since $\mathbf{v} = u(x, y, t)\mathbf{e}_x + v(x, y, t)\mathbf{e}_y$ only, can show that (3.5) is satisfied with

$$\Psi = 0\mathbf{e}_x + 0\mathbf{e}_y + \psi(x, y, t)\mathbf{e}_z$$

where ψ is streamfunction for 2D incompressible flows.

Exercise. Show that $u\mathbf{e}_x + v\mathbf{e}_y = \nabla \times \Psi$

and

$$u = \frac{\partial\psi}{\partial y}, \quad v = -\frac{\partial\psi}{\partial x}$$

3.1. KINEMATIC

in polar coordinate, $\psi = \psi(r, \theta, t)$

$$v_r = \frac{1}{r} \frac{\partial \psi}{\partial \theta}, \quad v_\theta = -\frac{\partial \psi}{\partial r}$$

Exercise. Show that the vector potential for axisymmetric incompressible flow can be chosen as

① in cylindrical coordinate (r, θ, z)

$$v_r = v_r(r, z, t), \quad v_z = v_z(r, z, t), \quad v_\theta = 0$$

$$\mathbf{\Psi} = \psi_\theta \mathbf{e}_\theta = \underbrace{\frac{1}{r} \psi(r, z, t)}_{\psi_\theta} \mathbf{e}_\theta \quad (3.6)$$

$$\nabla \times \mathbf{\Psi} = \frac{1}{r} \begin{vmatrix} \mathbf{e}_r & r\mathbf{e}_\theta & \mathbf{e}_z \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial z} \\ 0 & r\psi_\theta & 0 \end{vmatrix} = v_r \mathbf{e}_r + v_z \mathbf{e}_z$$

then $\mathbf{v} = \nabla \times \mathbf{\Psi}$

$$\implies v_r = -\frac{1}{r} \frac{\partial \psi}{\partial z}, \quad v_z = \frac{1}{r} \frac{\partial \psi}{\partial r}$$

where ψ is Stokes' stream function.

② in spherical coordinate (r, θ, ϕ)

$$v_r = v_r(r, \theta, t), \quad v_\theta = v_\theta(r, \phi, t), \quad v_\phi = 0$$

$$\mathbf{\Psi} = \psi_\phi \mathbf{e}_\phi = \underbrace{\frac{1}{r \sin \theta} \psi(r, \theta, t)}_{\psi_\phi} \mathbf{e}_\phi \quad (3.7)$$

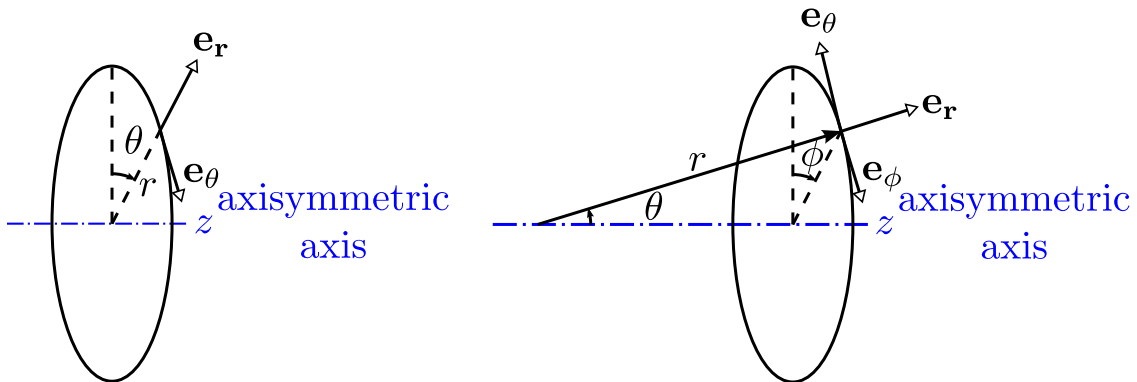
CHAPTER 3. KINEMATIC AND DYNAMICS OF IDEAL-FLUID MOTION

$$\nabla \times \Psi = \frac{1}{r^2 \sin \theta} \begin{vmatrix} \mathbf{e}_r & r\mathbf{e}_\theta & r \sin \theta \mathbf{e}_\phi \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \phi} \\ 0 & 0 & r \sin \theta \psi_\phi \mathbf{e}_\phi \end{vmatrix} = v_r \mathbf{e}_r + v_\theta \mathbf{e}_\theta$$

then $\mathbf{v} = \nabla \times \Psi$

$$\implies v_r = \frac{1}{r^2 \sin \theta} \frac{\partial \psi}{\partial \theta}, \quad v_\theta = -\frac{1}{r} \frac{\partial \psi}{\partial r}$$

where ψ is Stokes' stream function.



(a) cylindrical coordinate (r, θ, z)

(b) spherical coordinate (r, θ, ϕ)

3.2 Dynamics

3.2.1 Bernoulli equation

(for inviscid fluid, $\mu = \lambda = 0$, $\boldsymbol{\tau} = -p\mathbf{I}$ only)

Euler equation

$$\rho \frac{D\mathbf{v}}{Dt} = \rho \mathbf{g} - \nabla p \quad (3.8)$$

3.2. DYNAMICS

Now

$$\frac{D\mathbf{v}}{Dt} = \frac{\partial \mathbf{v}}{\partial t} + \mathbf{v} \cdot \nabla \mathbf{v} = \frac{\partial \mathbf{v}}{\partial t} + \nabla \left(\frac{v^2}{2} \right) - \mathbf{v} \times \boldsymbol{\xi}$$

Exercise. Show that vector identity

$$\mathbf{v} \times (\nabla \times \mathbf{v}) = \nabla \left(\frac{1}{2} \mathbf{v} \cdot \mathbf{v} \right) - \mathbf{v} \cdot \nabla \mathbf{v}$$

and $\nabla \times \mathbf{g} = \mathbf{0}$ (conservative force field)

$\implies \mathbf{g} = -\nabla U$ (U : gravitational force)

$$(3.8) \implies \frac{\partial \mathbf{v}}{\partial t} + \nabla \left(\frac{v^2}{2} \right) - \mathbf{v} \times \boldsymbol{\xi} = -\nabla U - \frac{\nabla p}{\rho} \quad (3.9)$$

- Steady flow along the streamline

Let $d\mathbf{s} \parallel$ a special streamline (i.e. $d\mathbf{s} \perp \mathbf{v}$ along streamline)

$$d\mathbf{s} \cdot (3.9) \implies d\mathbf{s} \cdot \nabla \left(\frac{v^2}{2} \right) - \cancel{d\mathbf{s} \cdot (\mathbf{v} \times \boldsymbol{\xi})}^0 \because (\mathbf{v} \times \boldsymbol{\xi}) \perp d\mathbf{s}$$

$$= -d\mathbf{s} \cdot \nabla U - d\mathbf{s} \cdot \frac{\nabla p}{\rho}$$

$$\implies d \left(\frac{v^2}{2} \right) + \frac{dp}{\rho} + dU = 0$$

integrate (along streamline)

$$\implies \int \frac{dp}{\rho} + \frac{v^2}{2} + U = \text{constant (along streamline)}$$

(3.10a)

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For ideal fluid, $\rho = \text{constant}$

$$(3.10a) \implies \frac{p}{\rho} + \frac{v^2}{2} + U = \text{constant (along streamline)} \quad (3.10b)$$

(Mechanical energy is conserved.)

- Unsteady, *irrotational*, and *barotropic* flow
 irrotational $\implies \boldsymbol{\xi} = \mathbf{0}$ and $\mathbf{v} = \nabla\Phi$
 barotropic $\implies \rho = \rho(p)$ (or $p = p(\rho)$)

Exercise. Show that

$$\implies \frac{\nabla p}{\rho} = \nabla \int \frac{dp}{\rho}$$

(3.8) reduces to

$$\begin{aligned} & \nabla \left\{ \frac{\partial\Phi}{\partial t} + \frac{v^2}{2} + U + \int \frac{dp}{\rho} = 0 \right\} \\ \implies & \frac{\partial\Phi}{\partial t} + \int \frac{dp}{\rho} + \frac{1}{2} \nabla\Phi \cdot \nabla\Phi + U = f(t) \end{aligned} \quad (3.11)$$

For steady and incompressible

$$(3.11) \implies \boxed{\frac{p}{\rho} + \frac{1}{2} \nabla\Phi \cdot \nabla\Phi + U = \text{constant}}$$

(for ideal fluid + irrotation) everywhere

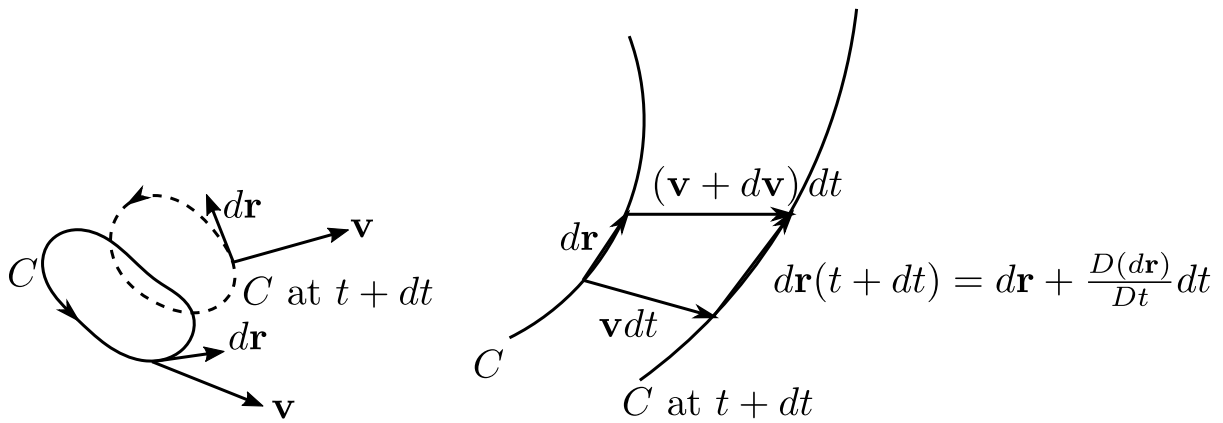
3.2. DYNAMICS

3.2.2 Kelvin's Circulation Theorem

Euler equation

$$\frac{D\mathbf{v}}{Dt} = -\frac{\nabla p}{\rho} + \frac{\mathbf{f}_b}{\rho}$$

where $\mathbf{f}_b$ is body force/mass



$$\begin{aligned} d\mathbf{r} + (\mathbf{v} + d\mathbf{v}) dt &= \mathbf{v} dt + d\mathbf{r} + \frac{D(d\mathbf{r})}{Dt} dt \\ \implies \frac{D(d\mathbf{r})}{Dt} &= d\mathbf{v} \end{aligned}$$

or

$$\frac{D(d\mathbf{r})}{Dt} = d \frac{D\mathbf{r}}{Dt} = d\mathbf{v}$$

CHAPTER 3. KINEMATIC AND DYNAMICS OF IDEAL-FLUID MOTION

Consider around a closed curve curve C

$$\begin{aligned}\Gamma &= \oint_C \mathbf{v} \cdot d\mathbf{r} \\ \frac{D\Gamma}{Dt} &= \frac{D}{Dt} \oint_C \mathbf{v} \cdot d\mathbf{r} \\ &= \oint_C \left[\frac{D\mathbf{v}}{Dt} \cdot d\mathbf{r} + \mathbf{v} \cdot \frac{D(d\mathbf{r})}{Dt} \right] \\ &= \oint_C \left[\frac{D\mathbf{v}}{Dt} \cdot d\mathbf{r} + \mathbf{v} \cdot d\mathbf{v} \right]\end{aligned}$$

Now

①

$$\begin{aligned}\oint_C \frac{D\mathbf{v}}{Dt} \cdot d\mathbf{r} &= \oint_C \left(-\frac{\nabla p}{\rho} + \frac{\mathbf{f}_b}{\rho} \right) \cdot d\mathbf{r} \\ &= \iint_S \mathbf{n} \cdot \mathbf{B} dS\end{aligned}$$

with

$$\mathbf{B} = \nabla \times \left[-\frac{\nabla p}{\rho} + \frac{\mathbf{f}_b}{\rho} \right]$$

②

$$\oint_C \mathbf{v} \cdot d\mathbf{v} = \oint_C d\left(\frac{v^2}{2}\right) = 0$$

as long as no discontinuity in $v^2/2$ existing along the curve C Hence

$$\boxed{\frac{D\Gamma}{Dt} = \iint_S \mathbf{n} \cdot \mathbf{B} dS} \quad (\text{Bjerknes Theorem}) \quad (3.12)$$

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i.e., rate of circulation is caused by the rotationality of either surface force $\nabla p/\rho$ or body force $\mathbf{f}_b/\rho$.

If the flow is barotropic (i.e. $\rho = \rho(p)$ or $p = p(\rho)$ only)

$$\frac{\nabla p}{\rho} = \nabla \int \frac{dp}{\rho}$$

and the body force is conservative, i.e.,

$$\frac{\mathbf{f}_b}{\rho} = -\nabla U$$

then

$$\mathbf{B} = \nabla \times \left[-\nabla \int \frac{dp}{\rho} - \nabla U \right] = \mathbf{0}$$

$$\boxed{\frac{D\Gamma}{Dt} = 0} \quad (\text{Kelvin's Theorem}) \quad (3.13)$$

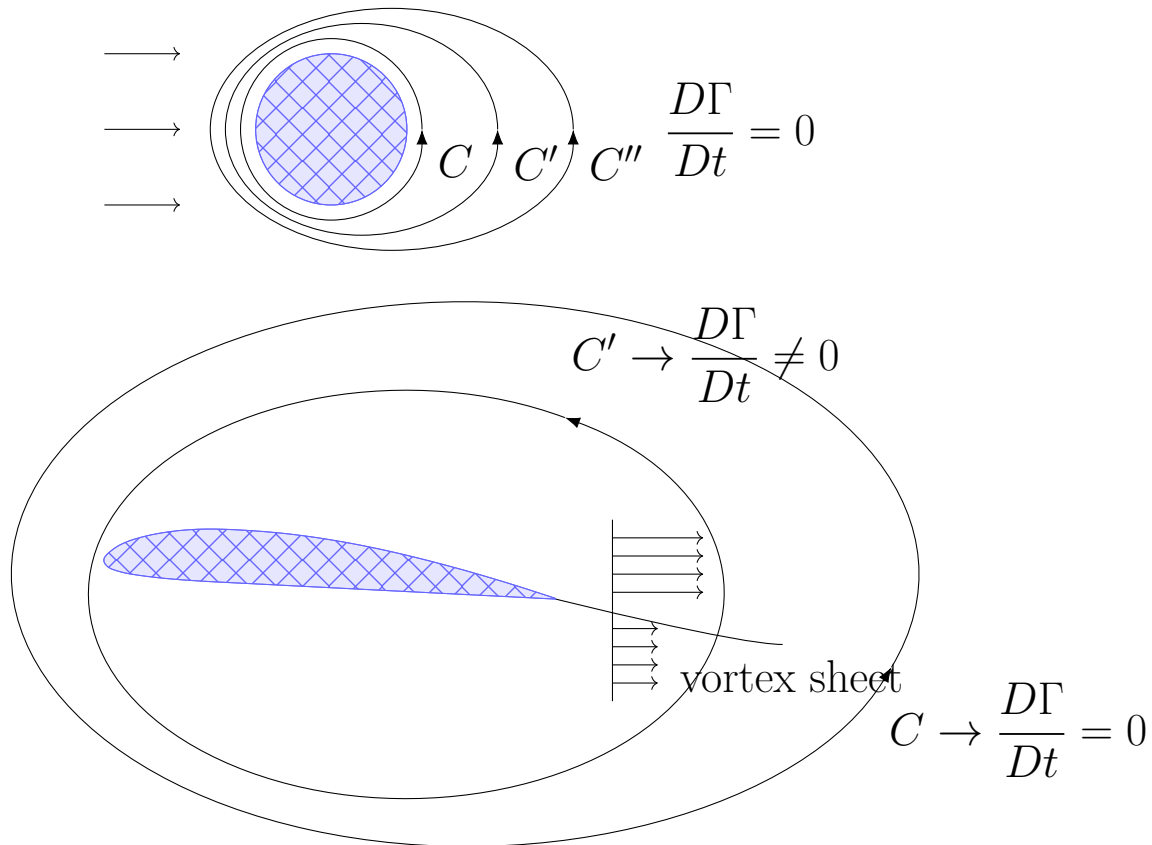
i.e., circulation around any closed material curve is *conserved* in an *inviscid, barotropic* (or $\rho = \text{constant}$), and *conservative body force field*.

Discussion.

- ① The flow does not have to be irrotational ($\boldsymbol{\xi} = \mathbf{0}$)
- ② Kelvin's theorem implies that all motions started from rest (i.e., $\Gamma = 0$ for every loop initially) must be circulation free (i.e. irrotational) at any subsequent instant.

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③ Eq.(3.13) also applies to



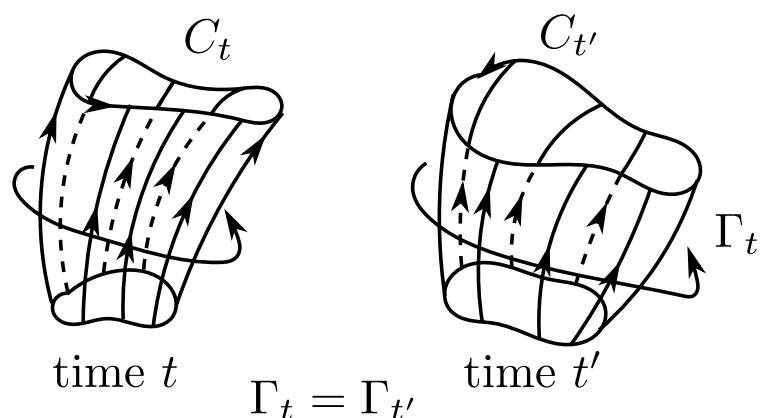
④

- (a) Helmholtz theorem: $\Gamma = \text{constant}$ around the cross-section of a vorticity tube
- (b) Kelvin's theorem: $\Gamma = \text{constant}$ around a material closed curve
- (c) Can show that Kelvin's theorem $\implies$ vorticity line is a material

i.e., once a vorticity line (tube), always a vorticity line (tube).

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Hence (a) + (b) + (c) $\implies$ circulation Γ around a vorticity tube is preserved along its motion (in an inviscid, barotropic, and conservative-body force field)

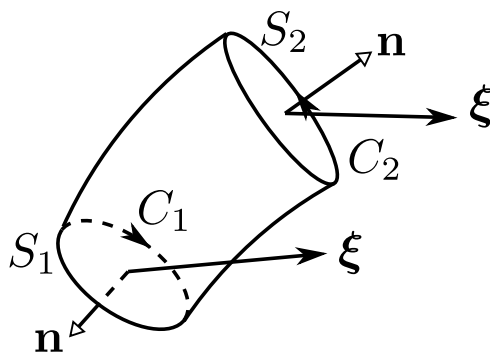


⑤ Summary of the Helmholtz Vortex Theorem

(a) $\nabla \cdot \boldsymbol{\xi} = 0 \implies$ Vorticity line either forms a closed loop or terminate at boundary

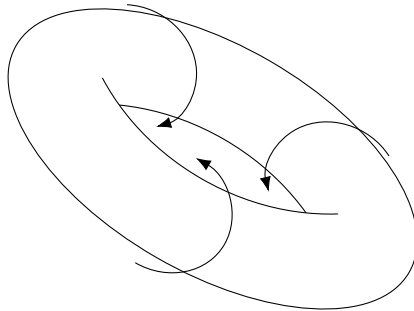
(b)

$$\iint_{S_1} \mathbf{n} \cdot \boldsymbol{\xi} dS = - \iint_{S_2} \mathbf{n} \cdot \boldsymbol{\xi} dS \implies \Gamma_{C_1} = \Gamma_{C_2}$$



CHAPTER 3. KINEMATIC AND DYNAMICS OF IDEAL-FLUID MOTION

- (c) Vorticity line is a material line (in an inviscid, barotropic, and conservative-body force field)
- (d) $\frac{D\Gamma}{Dt} = 0$ along any material closed curve, e.g. vortex ring



- (e) From (b) + (c) + (d) $\implies$ circulation around a vorticity tube is preserved along its motion (in an inviscid, barotropic, and conservative-body force field)

3.2.3 Vorticity Equation

Euler Equation

$$\frac{\partial \mathbf{v}}{\partial t} + \mathbf{v} \cdot \nabla \mathbf{v} = \frac{1}{\rho} \mathbf{g} - \frac{\nabla p}{\rho}$$

$$\implies \frac{\partial \mathbf{v}}{\partial t} + \nabla \left(\frac{v^2}{2} \right) - \mathbf{v} \times \boldsymbol{\xi} = -\nabla U - \frac{\nabla p}{\rho} \quad (*)$$

3.2. DYNAMICS

$$\nabla \times (*) \implies \frac{\partial \boldsymbol{\xi}}{\partial t} + 0 - \nabla \times (\mathbf{v} \times \boldsymbol{\xi}) = 0 - \nabla \times \left(\frac{\nabla p}{\rho} \right) \quad (**)$$

From vector identity

$$\nabla \times (\mathbf{v} \times \boldsymbol{\xi}) = \mathbf{v}(\nabla \cdot \boldsymbol{\xi}) - \underbrace{\boldsymbol{\xi}(\nabla \cdot \mathbf{v})}_{\nabla \cdot \mathbf{v} = -\frac{1}{\rho} \frac{D\rho}{Dt}} + (\boldsymbol{\xi} \cdot \nabla) \mathbf{v} - (\mathbf{v} \cdot \nabla) \boldsymbol{\xi}$$

also

Exercise.

$$\begin{aligned} \nabla \times \left(\frac{\nabla p}{\rho} \right) &= \frac{1}{\rho} \nabla \times \nabla p + \left(\nabla \frac{1}{\rho} \right) \times \nabla p \\ &= -\frac{1}{\rho^2} \nabla \rho \times \nabla p \end{aligned}$$

Exercise.

$$(**) \implies \frac{D}{Dt} \left(\frac{\boldsymbol{\xi}}{\rho} \right) = \frac{\boldsymbol{\xi}}{\rho} \cdot \nabla \mathbf{v} + \frac{1}{\rho^2} \nabla \rho \times \nabla p \quad (3.14)$$

For barotropic flow ($\rho = \rho(p)$ or $p = p(\rho)$)

$$p = p(\rho) \implies \nabla p = \frac{dp}{d\rho} \nabla \rho \implies \nabla p \parallel \nabla \rho \implies \nabla \rho \times \nabla p = 0$$

$$(3.14) \implies \frac{D}{Dt} \left(\frac{\boldsymbol{\xi}}{\rho} \right) = \frac{\boldsymbol{\xi}}{\rho} \cdot \nabla \mathbf{v} \quad (3.15)$$

(in an inviscid, barotropic, and conservative-body force field)

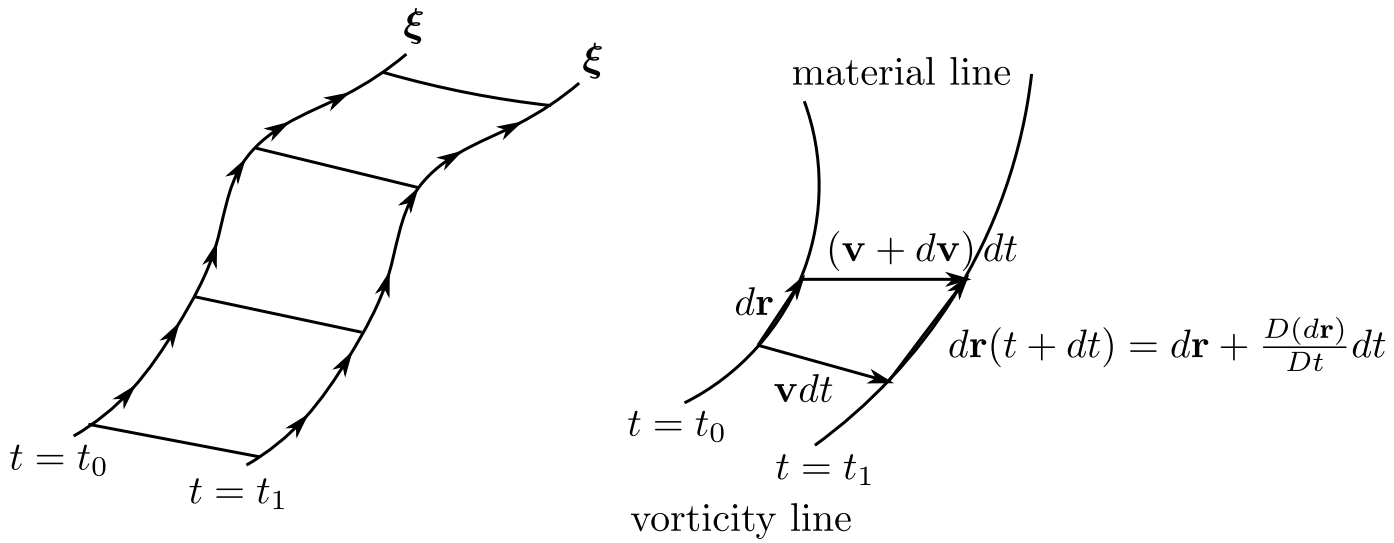
For ideal fluid ($\rho = \text{constant}$)

$$(3.14) \implies \boxed{\frac{D\boldsymbol{\xi}}{Dt} = \boldsymbol{\xi} \cdot \nabla \mathbf{v}} \quad (3.16)$$

CHAPTER 3. KINEMATIC AND DYNAMICS OF IDEAL-FLUID MOTION

- In an inviscid, barotropic (or $\rho = \text{constant}$) and conservative body-force field, a vorticity line is a material line.

$$\implies \frac{D(dr)}{Dt} = d\mathbf{v} = d\mathbf{r} \cdot \nabla \mathbf{v} \quad (3.17)$$



Now let $d\mathbf{r}$ be a segment of a vorticity line at $t = t_0$, i.e.

$$\boldsymbol{\xi} \times d\mathbf{r} = \mathbf{0} \text{ (or } \boldsymbol{\xi} = \lambda d\mathbf{r} \text{) at } t = t_0$$

$$\begin{aligned} \left[\frac{D(\boldsymbol{\xi} \times d\mathbf{r})}{Dt} \right]_{t_0} &= \left(\frac{D\boldsymbol{\xi}}{Dt} \times d\mathbf{r} \right)_{t_0} + \left(\boldsymbol{\xi} \times \frac{D(d\mathbf{r})}{Dt} \right)_{t_0} \\ &= [(\boldsymbol{\xi} \cdot \nabla \mathbf{v}) \times d\mathbf{r}]_{t_0} + [\boldsymbol{\xi} \times d\mathbf{v}]_{t_0} \\ &= [(\lambda d\mathbf{r} \cdot \nabla \mathbf{v}) \times d\mathbf{r}]_{t_0} + [\lambda d\mathbf{r} \times d\mathbf{v}]_{t_0} \\ &= [(\lambda d\mathbf{v}) \times d\mathbf{r}]_{t_0} + [\lambda d\mathbf{r} \times d\mathbf{v}]_{t_0} = \mathbf{0} \\ \implies (\boldsymbol{\xi} \times d\mathbf{r})_{t+dt} &= (\boldsymbol{\xi} \times d\mathbf{r})_{t_0} \end{aligned}$$

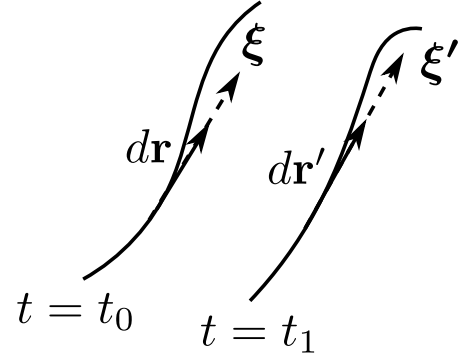
3.2. DYNAMICS

- The term $\boldsymbol{\xi} \cdot \nabla \mathbf{v}$ in vorticity equation

Let $\boldsymbol{\xi} = \lambda d\mathbf{r}$

$\lambda(\mathbf{x}, t) = \frac{|\boldsymbol{\xi}|}{|d\mathbf{r}|}$ vorticity/unit length

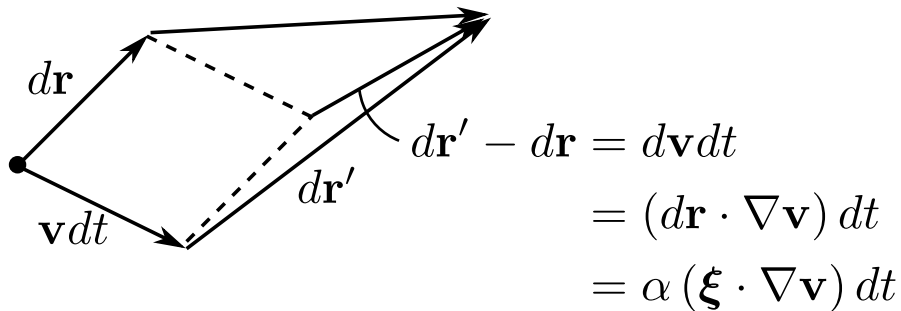
λ : vorticity density



then

$$\begin{aligned} \frac{D\boldsymbol{\xi}}{Dt} &= \boldsymbol{\xi} \cdot \nabla \mathbf{v} \implies \frac{D(\lambda d\mathbf{r})}{Dt} = \lambda d\mathbf{r} \cdot \nabla \mathbf{v} \\ \implies \frac{D\lambda}{Dt} d\mathbf{r} + \lambda \underbrace{\frac{D(d\mathbf{r})}{Dt}}_{d\mathbf{v} = d\mathbf{r} \cdot \nabla \mathbf{v}} &= \lambda d\mathbf{r} \cdot \nabla \mathbf{v} \\ \implies \boxed{\frac{D\lambda}{Dt} = 0} \end{aligned}$$

i.e., λ is conserved when following fluid motion

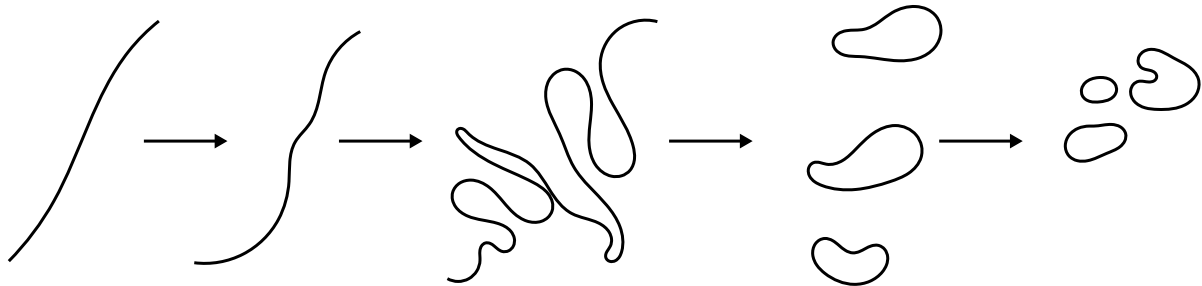


i.e., *stretching* and *turning* of the vorticity line in dt is caused by the *deformation rate field* $\nabla \mathbf{v}$.

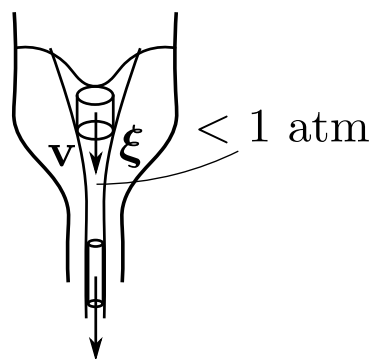
Note.

CHAPTER 3. KINEMATIC AND DYNAMICS OF IDEAL-FLUID MOTION

- ① The term $\boldsymbol{\xi} \cdot \nabla \mathbf{v}$ has important consequence in turbulent flow



- ② vorticity stretching effect



3.2. DYNAMICS

Chapter 4

General Properties of Irrotational Motion in an Ideal Fluid (Potential Flows)

4.1 Basic Formulations of Potential Flows

4.1.1 Governing Equations

Starting from irrotation: $\nabla \times \mathbf{v} = \mathbf{0} \implies \mathbf{v} = \nabla \Phi$

4.1. BASIC FORMULATIONS OF POTENTIAL FLOWS

For an ideal fluid, $\rho = \text{constant}$, continuity equation: $\nabla \cdot \mathbf{v} = 0$

$$\implies \nabla \cdot \nabla \Phi = 0 \implies \boxed{\nabla^2 \Phi = 0} \text{ (Laplace equation)} \quad (4.1)$$

From Kelvin's circulation theorem

$$\frac{D\Gamma}{Dt} = 0 \text{ where } \Gamma = \oint_C \mathbf{v} \cdot \mathbf{t} dl$$

hence if a flow field is circulation free flow (i.e. irrotational flow) initially, it will remain circulation free (i.e. irrotational) thereafter. The dynamics (i.e. pressure) of the flow field can be obtained from Bernoulli's equation, i.e.,

$$\frac{\partial \Phi}{\partial t} + \frac{p}{\rho} + \frac{1}{2} \nabla \Phi \cdot \nabla \Phi + U = f(t) \quad (4.2a)$$

or redefine $\Phi \equiv \Phi - \int f(t) dt$

$$\frac{\partial \Phi}{\partial t} + \frac{p}{\rho} + \frac{1}{2} \nabla \Phi \cdot \nabla \Phi + U = 0 \quad (4.2b)$$

Alternatively, starting from incompressibility:

$$\nabla \cdot \mathbf{v} = 0 \implies \mathbf{v} = \nabla \times \mathbf{\Psi}$$

Use irrotationality: $\nabla \times \mathbf{v} = \mathbf{0}$

$$\implies \nabla \times (\nabla \times \mathbf{\Psi}) = \mathbf{0} \implies -\nabla^2 \mathbf{\Psi} + \nabla (\nabla \cdot \mathbf{\Psi}) = \mathbf{0}$$

Note.

$$\begin{aligned} \tilde{\mathbf{\Psi}} &= \mathbf{\Psi} + \nabla A \implies \nabla \times \tilde{\mathbf{\Psi}} = \nabla \times \mathbf{\Psi} + \cancel{\nabla \times \nabla A}^0 = \mathbf{v} \\ \implies \nabla \cdot \tilde{\mathbf{\Psi}} &= \nabla \cdot \mathbf{\Psi} + \nabla^2 A = 0, \text{ choose } A \text{ such that} \\ \nabla^2 A &= -\nabla \cdot \mathbf{\Psi} \end{aligned}$$

CHAPTER 4. GENERAL PROPERTIES OF IRROTATIONAL MOTION IN AN IDEAL FLUID (POTENTIAL FLOWS)

Hence*

$$\nabla^2 \Psi = 0 \quad (4.3)$$

(a) For 2D flows

$\Psi = \mathbf{e}_z \psi(x, y, t)$ only

$$\implies \nabla \cdot \Psi = \frac{\partial \psi(x, y, t)}{\partial z} = 0$$

therefore, governing equation: $\nabla \Psi = 0 \implies \nabla^2 \psi \mathbf{e}_z = \mathbf{0}$

$$\implies \boxed{\nabla^2 \psi = 0} \quad (4.4)$$

(i.e. $\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} = 0$) (for 2D potential flow only))

Exercise. Show that for axisymmetric flows

Ψ reduces to a vector of one component only and again

$\nabla \cdot \Psi = 0$, i.e.

$$\boxed{\nabla^2 \Psi = 0}$$

- for cylindrical coordinate (r, z)

$$\nabla^2 \left[\mathbf{e}_\theta \frac{1}{r} \psi(r, z, t) \right] = \mathbf{0}$$

expanding:

$$\implies \frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial \psi}{\partial r} \right) + \frac{1}{r} \frac{\partial^2 \psi}{\partial z^2} = 0 \quad (4.5)$$

(not a Laplace equation)

*Note that Ψ is not uniquely determined by $\mathbf{v}$

4.1. BASIC FORMULATIONS OF POTENTIAL FLOWS

- in spherical coordinate (r, θ)

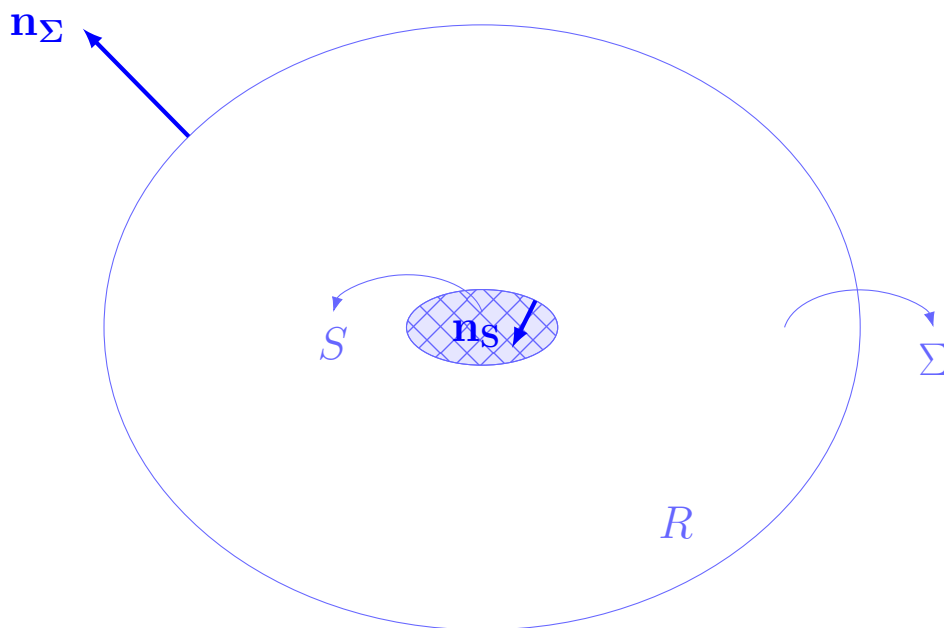
$$\nabla^2 \left[\mathbf{e}_\varphi \frac{1}{r \sin \theta} \psi(r, \theta, t) \right] = \mathbf{0}$$

expanding:

$$\Rightarrow \frac{\partial^2 \psi}{\partial r^2} + \frac{\sin \theta}{r^2} \frac{\partial}{\partial \theta} \left(\frac{1}{\sin \theta} \frac{\partial \psi}{\partial \theta} \right) = 0 \quad (4.6)$$

4.1.2 Far-Field Boundary Conditions

Consider the flow field generated by a solid body moving in an otherwise *quiescent fluid*



Neumann Extension Problem

From divergence theorem

$$\iiint_R \nabla^2 \Phi dV = \iint_S \mathbf{n}_S \cdot \nabla \Phi dS + \iint_\Sigma \mathbf{n}_\Sigma \cdot \nabla \Phi d\Sigma$$

CHAPTER 4. GENERAL PROPERTIES OF IRROTATIONAL MOTION IN AN IDEAL FLUID (POTENTIAL FLOWS)

$$\begin{aligned} \Rightarrow \iint_{\Sigma} \mathbf{n}_{\Sigma} \cdot \nabla \Phi d\Sigma &= - \iint_S \underbrace{\mathbf{n}_S \cdot \nabla \Phi}_{=\mathbf{n}_S \cdot \mathbf{v}_S^{\dagger}} dS \\ &= - \iint_S \mathbf{n}_S \cdot \mathbf{v}_S dS = -\mathbf{v}_S \cdot \iint_S \mathbf{n}_S dS \end{aligned}$$

i.e.,

$$\iint_{\Sigma} \mathbf{n}_{\Sigma} \cdot \nabla \Phi d\Sigma = 0$$

Choose Σ to be a large sphere with radius $r \rightarrow \infty$, then

$$\begin{aligned} \iint_{\Sigma} \underbrace{\mathbf{n}_Z}_{=\mathbf{e}_r} \cdot \nabla \Phi \underbrace{d\Sigma}_{=r^2 \sin \theta d\theta d\varphi} &= 0 \Rightarrow \iint_{\Sigma} \underbrace{\frac{\partial \Phi}{\partial r}}_{=v_r} r^2 \sin \theta d\theta d\varphi = 0 \\ &= \frac{\partial \Phi}{\partial r} = v_r \end{aligned}$$

$$\Rightarrow v_r \equiv \frac{\partial \Phi}{\partial r} \sim o\left(\frac{1}{r^2}\right) \quad (4.7)$$

Similarly, from divergence theorem[‡]

$$\begin{aligned} \iiint_R \nabla \times \nabla \Phi^0 dV &= \iint_S \mathbf{n}_S \times \nabla \Phi dS + \iint_{\Sigma} \mathbf{n}_{\Sigma} \times \nabla \Phi d\Sigma = 0 \\ \Rightarrow \iint_{\Sigma} \mathbf{n}_{\Sigma} \times \nabla \Phi d\Sigma &= - \iint_S \mathbf{n}_S \times \nabla \Phi dS = - \iint_S \mathbf{n}_S \times \mathbf{v} dS \\ &\stackrel{[2]}{=} 0 \end{aligned}$$

p. 294-p. 296
for circulation-free flow

[†]no penetration on S

[‡] $\int_{\partial \mathcal{R}} (\cdot) n_i dA = \int_{\mathcal{R}} (\cdot)_{,i} dV$

4.1. BASIC FORMULATIONS OF POTENTIAL FLOWS

choose Σ to be large sphere radius $r \rightarrow \infty$

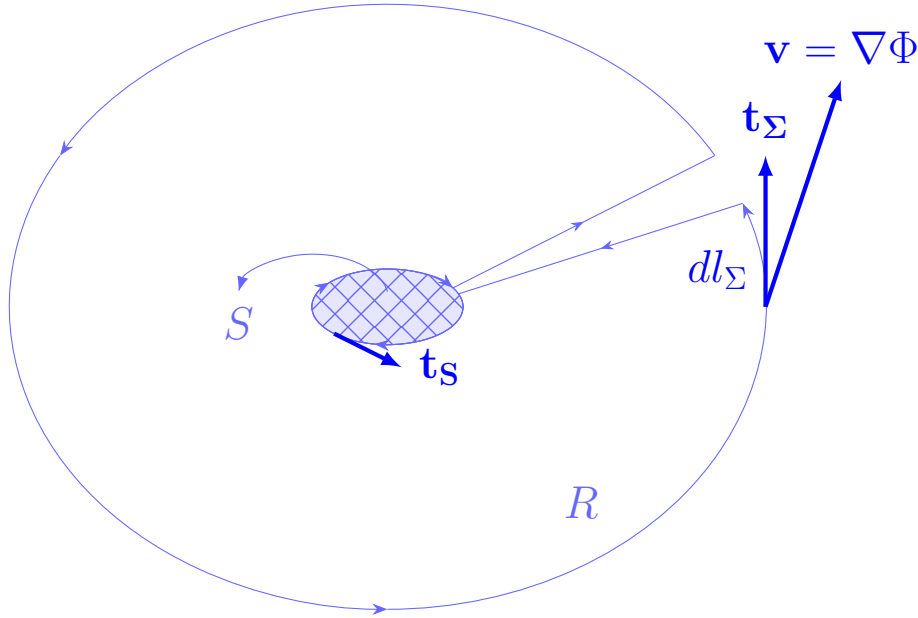
$$\begin{aligned}
 & \iint_{\Sigma} \mathbf{n}_{\Sigma} \times \nabla \Phi d\Sigma \\
 &= \iint_{\Sigma} \mathbf{e}_r \times \left\{ \cancel{\mathbf{e}_r \frac{\partial \Phi}{\partial r}}^0 + \mathbf{e}_{\theta} \frac{1}{r} \frac{\partial \Phi}{\partial \theta} + \mathbf{e}_{\varphi} \frac{1}{r \sin \theta} \frac{\partial \Phi}{\partial \varphi} \right\} r^2 \sin \theta d\theta d\varphi = 0 \\
 &\implies \underbrace{\frac{1}{r} \frac{\partial \Phi}{\partial \theta}}_{v_{\theta}}, \underbrace{\frac{1}{r \sin \theta} \frac{\partial \Phi}{\partial \varphi}}_{v_{\varphi}} \sim o\left(\frac{1}{r^2}\right) \text{ as } r \rightarrow \infty \quad (4.8)
 \end{aligned}$$

Exercise. Show that for 2D flow (i.e. moving cylinder)
B.C.'s at infinity:

$$\begin{aligned}
 v_r &\sim o\left(\frac{1}{r}\right) \\
 v_{\theta} &\sim o\left(\frac{1}{r}\right) \text{ if } \Gamma \neq 0 \\
 &\sim \mathcal{O}\left(\frac{1}{r}\right) \text{ if } \Gamma = 0
 \end{aligned}$$

Proof.

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Consider circulation Γ around solid cylinder, from Stokes' theorem

$$\oint_{\Sigma} \mathbf{t}_{\Sigma} \cdot \nabla \Phi dl_{\Sigma} = \iint_R \mathbf{n}_z \cdot (\nabla \times \nabla \Phi) dA_R - \oint_S \mathbf{t}_S \cdot \nabla \Phi dl_S = 0$$

$$\implies \oint_{\Sigma} \mathbf{t}_{\Sigma} \cdot \nabla \Phi dl_{\Sigma} = \oint_S \mathbf{t}_S \cdot \nabla \Phi dl_S \equiv \Gamma$$

- if $\Gamma = 0$

$$\oint_{\Sigma} \underbrace{\mathbf{t}_{\Sigma}}_{\mathbf{e}_{\theta}} \cdot \nabla \Phi dl_{\Sigma} = 0 \implies \oint_{\Sigma} \underbrace{\frac{1}{r} \frac{\partial \Phi}{\partial \theta}}_{v_{\theta}} r d\theta = 0$$

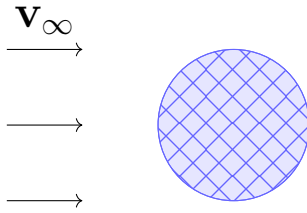
$$\implies v_{\theta} \equiv \frac{1}{r} \frac{\partial \Phi}{\partial \theta} \sim o\left(\frac{1}{r}\right) \text{ as } r \rightarrow \infty$$

- if $\Gamma \neq 0$

4.1. BASIC FORMULATIONS OF POTENTIAL FLOWS

$$\begin{aligned} &\implies \oint_{\Sigma} \frac{1}{r} \frac{\partial \Phi}{\partial \theta} r d\theta = \Gamma \\ &\implies v_{\theta} \equiv \frac{1}{r} \frac{\partial \Phi}{\partial \theta} \sim \mathcal{O}\left(\frac{1}{r}\right) \text{ as } r \rightarrow \infty \end{aligned}$$

Note. For uniform stream past a fixed solid body



use coordinate transformation

$$3\text{D}: (v_r - v_{\infty r}), (v_{\theta} - v_{\infty \theta}), (v_{\varphi} - v_{\infty \varphi}) \sim o\left(\frac{1}{r^2}\right) \text{ as } r \rightarrow \infty$$

$$2\text{D}: (v_r - v_{\infty r}) \sim o\left(\frac{1}{r}\right),$$

$$\begin{aligned} (v_{\theta} - v_{\infty \theta}) &\sim o\left(\frac{1}{r}\right) \text{ for } \Gamma = 0 \\ &\sim \mathcal{O}\left(\frac{1}{r}\right) \text{ for } \Gamma \neq 0 \end{aligned} \quad \text{as } r \rightarrow \infty$$

□

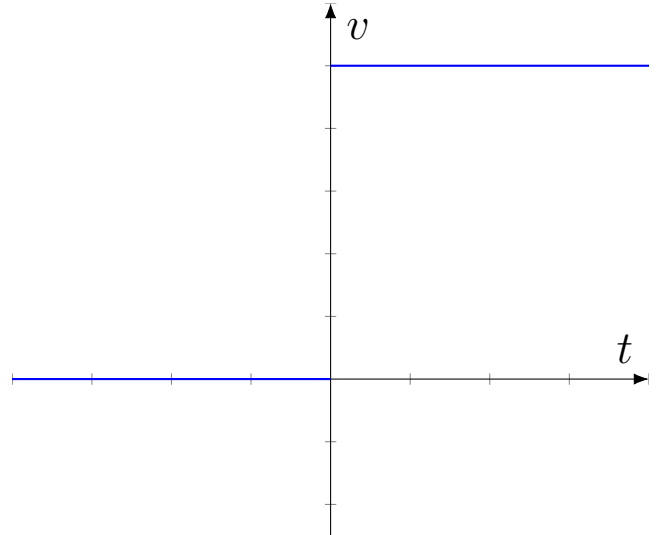
4.1.3 Physical Interpretation of the Velocity Potential Function Φ

Consider an irrotational flow generated by a solid body which is set into motion *impulsively from rest*, i.e.

CHAPTER 4. GENERAL PROPERTIES OF IRROTATIONAL MOTION IN AN IDEAL FLUID (POTENTIAL FLOWS)

$$\mathbf{v} = \mathbf{0} \text{ at } t = 0$$

$$\mathbf{v} = \text{finite at } t = 0^+$$



then

$$\mathbf{v} = \lim_{t \rightarrow 0} \int_0^t \frac{\partial \mathbf{v}}{\partial t} dt$$

From Bernoulli's equation

$$\frac{\partial \mathbf{v}}{\partial t} + \underbrace{\nabla \left(\frac{v^2}{2} + \frac{p}{\rho} + U \right)}_H = 0$$

hence

$$\mathbf{v} = \lim_{t \rightarrow 0} - \int_0^t \nabla H dt \implies \nabla \Phi = \lim_{t \rightarrow 0} - \nabla \int_0^t H dt$$

$$\implies \Phi = \lim_{t \rightarrow 0} - \int_0^t H dt$$

$$\text{for finite } \Phi \implies H \rightarrow \infty \text{ at } t = 0^+ \left[H = \frac{v^2}{2} \overset{\text{finite at } t = 0^+}{\nearrow} + \frac{p}{\rho} + U \right]$$

4.2. FUNDAMENTAL SOLUTION AND GREEN'S THEOREM

Hence

$$\Phi(\mathbf{x}, t) = -\lim_{t \rightarrow 0} \int_0^t \frac{p}{\rho} dt \text{ with } p \rightarrow \infty \text{ as } t = 0^+$$

(*pressure impulse*)

4.2 Fundamental Solution and Green's Theorem

4.2.1 Fundamental Solution (of Laplace equation)

- The Dirac Delta function $\delta(x)$ (unit impulse function)

Definition 4.2.1.

$\delta(x) = 0$ for $x \neq 0$, $\delta(x) \rightarrow \infty$ at $x = 0$

$$\int_{-\infty}^{\infty} \delta(x) = 1 \quad (4.9)$$

It's not an ordinary function, can approximate $\delta(x)$ as the limit of continuous function

$$\textcircled{1} \quad d_n(x) = \frac{n}{\pi(1+n^2x^2)} \text{ as } n \rightarrow \infty$$

$$\textcircled{2} \quad \frac{\epsilon}{\pi(x^2 + \epsilon^2)} \text{ as } \epsilon \rightarrow \infty$$

$$\textcircled{3} \quad \frac{1}{2\sqrt{\pi\epsilon}} e^{-\frac{x^2}{4\epsilon}} \text{ as } \epsilon \rightarrow 0$$

CHAPTER 4. GENERAL PROPERTIES OF IRROTATIONAL MOTION IN AN IDEAL FLUID (POTENTIAL FLOWS)

$$\textcircled{4} \quad \frac{1}{2(\pi)^{3/2} \epsilon^3} e^{-\left(\frac{r}{\epsilon}\right)^2} \text{ as } \epsilon \rightarrow 0$$

i.e. $\delta(r) = 0$ for $r \neq 0$, $\delta(r) \rightarrow \infty$ at $r = 0$, and

$$\iiint_{V_\infty} \delta(r) dV = 1$$

Operational definition of delta function

$$\int_{-\infty}^{\infty} \delta(x) f(x) dx = f(0), \text{ or } \int_{-\infty}^{\infty} \delta(x - y) f(x) dx = f(y)$$

$\delta(x)$: sampling function, distribution, functional

- Fundamental Solution (Green's function)

If a function w satisfy a linear differential equation

$$Lu(x) = \delta(x - y) \quad (4.10)$$

where L denotes a linear differential operators of form,

e.g.

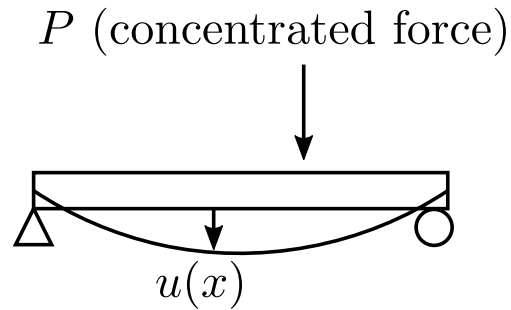
$$a_n \frac{d^n}{dx^n} + a_{n-1} \frac{d^{n-1}}{dx^{n-1}} + \cdots + a_1 \frac{d}{dx} + a_0$$

or

$$\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}$$

e.g.

4.2. FUNDAMENTAL SOLUTION AND GREEN'S THEOREM



i.e.

$$Lw(x; y) = \delta(x - y)$$

$w(x, y)$ is called the fundamental solution to the operator L

Consider an inhomogeneous linear differential equation

$$Lu(x) = f(x) \tag{4.11}$$

The solution to (4.12) is

$$u(x) = \int_{-\infty}^{\infty} \underbrace{w(x; y)}_{\substack{\text{Kernal function} \\ \text{transfer function} \\ \text{influence function}}} f(y) dy \tag{4.12}$$

Proof.

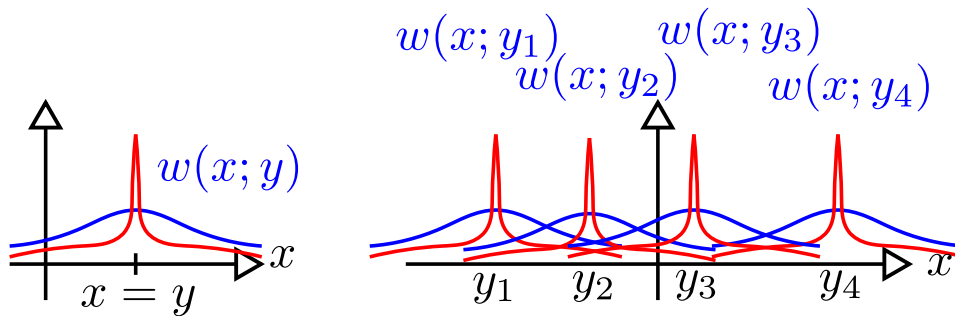
CHAPTER 4. GENERAL PROPERTIES OF IRROTATIONAL MOTION IN AN IDEAL FLUID (POTENTIAL FLOWS)

from (4.12)

$$\begin{aligned}
 Lu(x) &= L \int_{-\infty}^{\infty} w(x; y) f(y) dy \\
 &= \int_{-\infty}^{\infty} [Lw(x; y)] f(y) dy \\
 &= \int_{-\infty}^{\infty} \delta(x - y) f(y) dy = f(x)
 \end{aligned}$$

$$\implies Lw(x; y) = \delta(x - y)$$

$w(x, y)$ is the *response* of the system L to an impulse input at $x = y$



□

- Fundamental Solution to the Laplace equation
Consider

$$\nabla^2 \phi = \delta(r) \tag{4.13}$$

In spherical coordinate with spherical symmetry

$$\nabla^2 \equiv \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right)$$

4.2. FUNDAMENTAL SOLUTION AND GREEN'S THEOREM

Guess $\phi = \frac{A}{r}$, $\nabla^2 \phi = 0$ everywhere except at $r = 0$ where ϕ is *undefined* at $r = 0$

Use integral definition of divergence

$$\nabla^2 \left(\frac{A}{r} \right)_{r=0} = \nabla \cdot \left(\nabla \frac{A}{r} \right)_{r=0} = \lim_{\Delta V \rightarrow 0} \frac{1}{\Delta V} \oiint_{\Delta S} \left(\nabla \frac{A}{r} \right) \cdot \mathbf{n} dS$$

choose ΔV to be a sphere with radius $\epsilon \rightarrow 0$, then

$$\nabla \frac{A}{r} = -\frac{A}{r^2} \mathbf{e}_r$$

and

$$\begin{aligned} \nabla^2 \left(\frac{A}{r} \right)_{r=0} &= \lim_{\Delta V \rightarrow 0} \frac{1}{\Delta V} \oiint_{\Delta S} -\frac{A}{r^2} \mathbf{e}_r \cdot \mathbf{n} dS \\ &= \lim_{\epsilon \rightarrow 0} \frac{-A}{\frac{4}{3}\pi\epsilon^3} \oiint_{\Delta S} \frac{1}{r^2} \underbrace{dS}_{=r^2 \sin \theta d\theta d\varphi} \\ &= \lim_{\epsilon \rightarrow 0} \frac{-A}{\frac{4}{3}\pi\epsilon^3} \oiint_{\Delta \Omega} \frac{1}{r^2} \underbrace{r^2 d\Omega}_{dS} \\ &= \lim_{\epsilon \rightarrow 0} \frac{-4\pi A}{\frac{4}{3}\pi\epsilon^3} \rightarrow \infty \end{aligned}$$

i.e., $\nabla^2 \left(\frac{A}{r} \right)$ behaves like $\delta(r)$.

What is A ?

Recall

$$\iiint_{V(\infty)} \delta(r) dV \equiv 1, \quad \iiint_{V(\epsilon)} \delta(r) dV = 1$$

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but

$$\oint\!\!\!\oint_{V(\infty)} \nabla^2 \left(\frac{A}{r} \right) dV \equiv \oint\!\!\!\oint_{V(\epsilon)} \nabla^2 \left(\frac{A}{r} \right) dV = \nabla^2 \left(\frac{A}{r} \right)_{r=0} \underbrace{V(\epsilon)}_{\frac{4}{3}\pi\epsilon^3} = 1$$

$$\implies A = -\frac{1}{4\pi}$$

i.e. solution to (4.13) is

$$\boxed{\phi = -\frac{1}{4\pi r}} \text{ (fundamental solution)} \quad (4.14)$$

Exercise. Show that 2D fundamental solution to the Laplace equation is

$$\boxed{\phi = \frac{1}{2\pi} \ln r}$$

4.2.2 Green's Theorem (Green's Identities) (Green's Formula)

- Green's 2nd identity

$$\oint\!\!\!\oint_V \left(\phi' \nabla^2 \phi - \phi \nabla^2 \phi' \right) dV = \oint\!\!\!\oint_S \left(\phi' \frac{\partial \phi}{\partial n} - \phi \frac{\partial \phi'}{\partial n} \right) dS \quad (4.15)$$

both ϕ and ϕ' are continuous and differentiable scalar function.

4.2. FUNDAMENTAL SOLUTION AND GREEN'S THEOREM

- Application of Green's Theorem

Use Green's theorem to express ϕ in V , which satisfies $\nabla^2\phi = 0$, in terms of boundary data on S . (This is equivalent to solve $\nabla^2\phi = 0$ in V with B.C.'s, $\frac{\partial\phi}{\partial n}, \phi$ on S .)
 Let $\phi' = 1/r$, the $\nabla^2\phi' = 0$ everywhere except at $r = 0$.

Hence

$$\iiint_V \phi' \nabla^2 \phi dV \stackrel{\text{if } \nabla^2 \phi = 0}{=} 0,$$

$$\iiint_V \phi \nabla^2 \phi' \equiv \iiint_V \phi \nabla^2 \left(\frac{1}{r} \right) dV \neq 0 \text{ if the point } r = 0 \text{ is inside } V$$

Recall that

$$\underbrace{\iiint_V \nabla^2 \left(-\frac{1}{4\pi r} \right) dV}_{\sim \delta(r)} \equiv 1$$

$$\therefore \iiint_V \phi \nabla^2 \left(\frac{1}{r} \right) dV = -4\pi \underbrace{\iiint_V \phi \nabla^2 \left(-\frac{1}{4\pi r} \right) dV}_{\sim \delta(r)} = -4\pi \phi_0$$

where ϕ_0 is the value of ϕ at $r = 0$

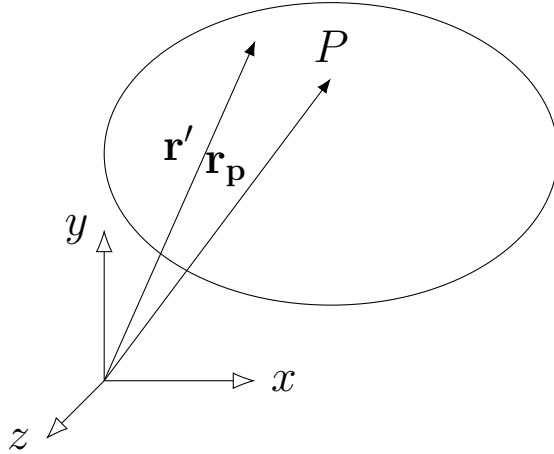
and

$$(4.15) \implies \iint_S \left[\frac{1}{r} \frac{\partial \phi}{\partial n} - \phi \frac{\partial}{\partial n} \left(\frac{1}{r} \right) \right] dS \text{ (Green's formula)} \tag{4.16}$$

Given $\frac{\partial \phi}{\partial n}, \phi$ on S , (4.16) gives ϕ_0 at any point in V .

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In practice, the use of (4.16) is



$$4\pi\phi_P = \oiint_S \left[\frac{1}{r} \frac{\partial\phi}{\partial n} - \phi \frac{\partial}{\partial n} \left(\frac{1}{r} \right) \right] dS \quad (4.17)$$

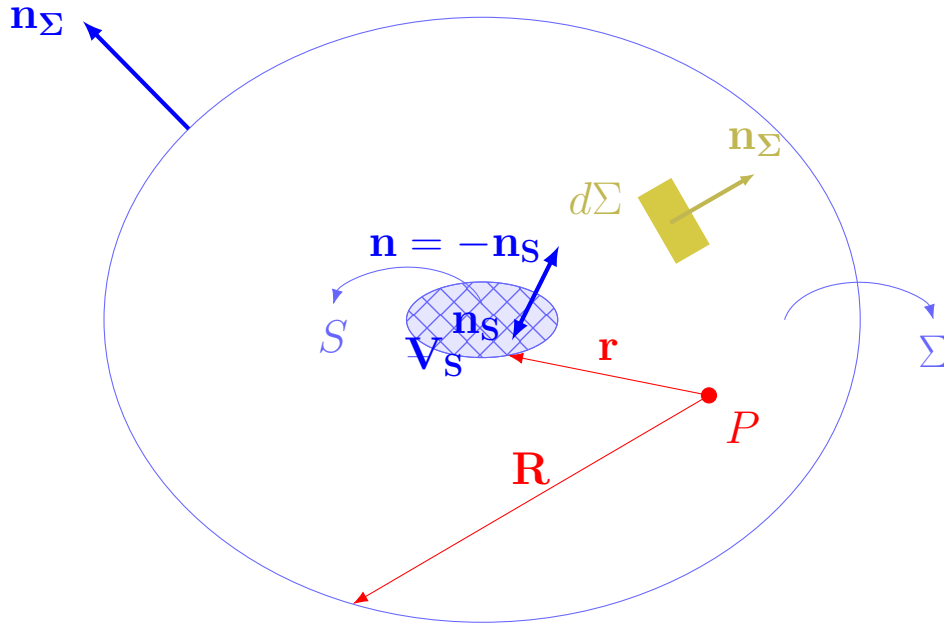
where $r = \frac{1}{|\mathbf{r}' - \mathbf{r}_p|}$

Exercise. For 2D case

$$-2\pi\phi_P = \oint_C \left[\ln \frac{\partial\phi}{\partial n} - \phi \frac{\partial}{\partial n} (\ln r) \right] dl \quad (4.18)$$

4.2. FUNDAMENTAL SOLUTION AND GREEN'S THEOREM

Modification for the problem involving a solid-body moving in infinite domain (Neumann Exterior Boundary)



Consider $\nabla^2 \phi = 0$ and $\phi \rightarrow \phi_\infty$ as $R \rightarrow \infty$ or
 $\nabla \phi \rightarrow 0 \left(\nabla \phi \sim o\left(\frac{1}{R^2}\right) \right)$

(4.17) is modified to

$$4\pi\phi_P = \iint_S \left[\frac{1}{r} \frac{\partial \phi}{\partial n_S} - \phi \frac{\partial}{\partial n_S} \left(\frac{1}{r} \right) \right] dS + \iint_\Sigma \left[\frac{1}{R} \frac{\partial \phi}{\partial n_\Sigma} - \phi \frac{\partial}{\partial n_\Sigma} \left(\frac{1}{R} \right) \right] d\Sigma \quad (*)$$

Choose Σ to be a large sphere with radius $R \rightarrow \infty$

$$\iint_\Sigma \frac{1}{R} \frac{\partial \phi}{\partial n_\Sigma} d\Sigma = \iint \frac{1}{R} \frac{\partial \phi}{\partial R} \underbrace{R^2 \sin \theta d\theta d\varphi}_{d\Sigma} \rightarrow 0 \text{ as } R \rightarrow \infty$$

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$$\begin{aligned}\iint_{\Sigma} \phi \frac{\partial}{\partial n_{\Sigma}} \left(\frac{1}{R} \right) d\Sigma &= \iint_{\Sigma} \phi \frac{\partial}{\partial R} \left(\frac{1}{R} \right) d\Sigma \\ &= - \iint_{\Sigma} \frac{\phi}{R^2} \underbrace{d\Sigma}_{=R^2 d\Omega} = - \iint_{\Omega} \phi d\Omega = -4\pi \phi_{\infty}\end{aligned}$$

Hence

$$\begin{aligned} (*) \implies 4\pi (\phi_P - \phi_{\infty}) &= \iint_S \left[\frac{1}{r} \frac{\partial \phi}{\partial n_S} - \phi \frac{\partial}{\partial n_S} \left(\frac{1}{r} \right) \right] dS \\ &= \iint_S \left[-\frac{1}{r} \frac{\partial \phi}{\partial n} + \phi \frac{\partial}{\partial n} \left(\frac{1}{r} \right) \right] dS \end{aligned} \quad (4.19)$$

Exercise. For 2D case

$$-2\pi (\phi_P - \phi_{\infty}) = \oint_C \left[-\ln r \frac{\partial \phi}{\partial n} + \phi \frac{\partial}{\partial n} (\ln r) \right] dl \quad (4.20)$$

- Interpretation of (4.19)

Let us identify ϕ with the velocity potential function Φ

$$\nabla^2 \Phi = 0$$

Imagine the solid body is replaced by another potential flow with velocity potential $\hat{\Phi}$

$$\begin{aligned} &\boxed{\nabla^2 \hat{\Phi} = 0} \\ &\iint_S \left[\frac{1}{r} \frac{\partial \hat{\Phi}}{\partial n} - \hat{\Phi} \frac{\partial}{\partial n} \left(\frac{1}{r} \right) \right] dS \\ &= \iiint_{V_S} \left[\cancel{\frac{1}{r} \nabla^2 \hat{\Phi}}^0 - \hat{\Phi} \cancel{\nabla^2} \left(\frac{1}{r} \right)^0 \right] dV \end{aligned} \quad (4.21)$$

4.2. FUNDAMENTAL SOLUTION AND GREEN'S THEOREM

From (4.19) + (4.21)

$$\begin{aligned}
 &\Rightarrow 4\pi (\Phi_P - \Phi_\infty) \\
 &= \oint\oint_S \left[\left(\Phi - \hat{\Phi} \right) \frac{\partial}{\partial n} \left(\frac{1}{r} \right) + \frac{1}{r} \left(\frac{\partial \hat{\Phi}}{\partial n} - \frac{\partial \Phi}{\partial n} \right) \right] dS
 \end{aligned} \tag{4.22}$$

Since $\hat{\Phi}$ can be any potential flow, in 2 choice

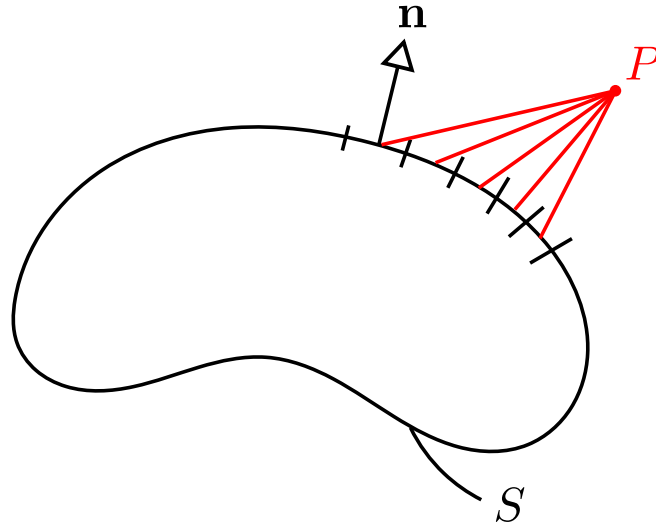
(a) If $\hat{\Phi} = \Phi$ on S (but $\frac{\partial \hat{\Phi}}{\partial n} \neq \frac{\partial \Phi}{\partial n}$ on S)
 (4.21) $\Rightarrow$

$$\begin{aligned}
 4\pi (\Phi_P - \Phi_\infty) &= \oint\oint_S \frac{\left(\frac{\partial \hat{\Phi}}{\partial n} - \frac{\partial \Phi}{\partial n} \right)}{r} dS \\
 &= \oint\oint_S \frac{A(S)}{r} dS
 \end{aligned} \tag{4.23}$$

$A(S)dS \equiv$ strength of the distributed source/sink
 over dS

e.g. Panel method

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(b) If $\frac{\partial \hat{\Phi}}{\partial n} = \frac{\partial \Phi}{\partial n}$ on S
 (4.21) $\implies$

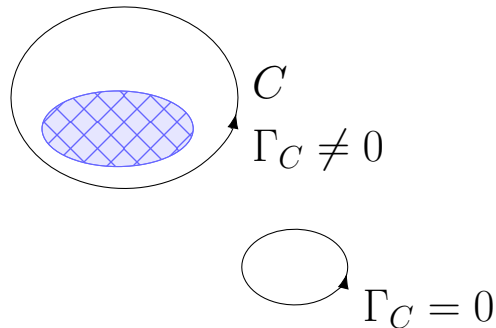
$$\begin{aligned} 4\pi (\Phi_P - \Phi_\infty) &= \oiint_S \left(\Phi - \hat{\Phi} \right) \frac{\partial}{\partial n} \left(\frac{1}{r} \right) dS \\ &= \oiint_S B(S) \frac{\partial}{\partial n} \left(\frac{1}{r} \right) dS \end{aligned} \quad (4.24)$$

4.2.3 Irrotational Motion and Circulation

Irrotational (i.e. $\nabla \times \mathbf{v} = \mathbf{0} \implies \Gamma_C (\equiv \oint_C \mathbf{v} \cdot \mathbf{t} dl)^\S = 0$ in R everywhere)

[§]around any circuit C in R

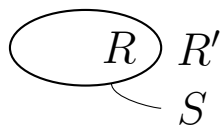
4.2. FUNDAMENTAL SOLUTION AND GREEN'S THEOREM



Some Topological Notions

- Connectivity

A region R is said to be *connected* if any two points in R can be connected by a continuous curve that does not leave R .



R : connected

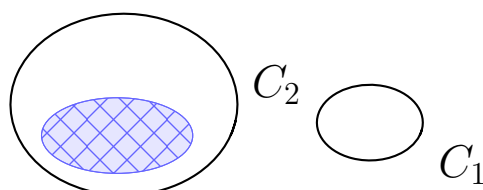
R' : connected

$R \cup R' \setminus S$: not a connected region

- Reducible and irreducible circuits

If a circuit can be continuously contracted to a point without ever leaving R , the circuit is said to be reducible, otherwise, irreducible.

In 3D

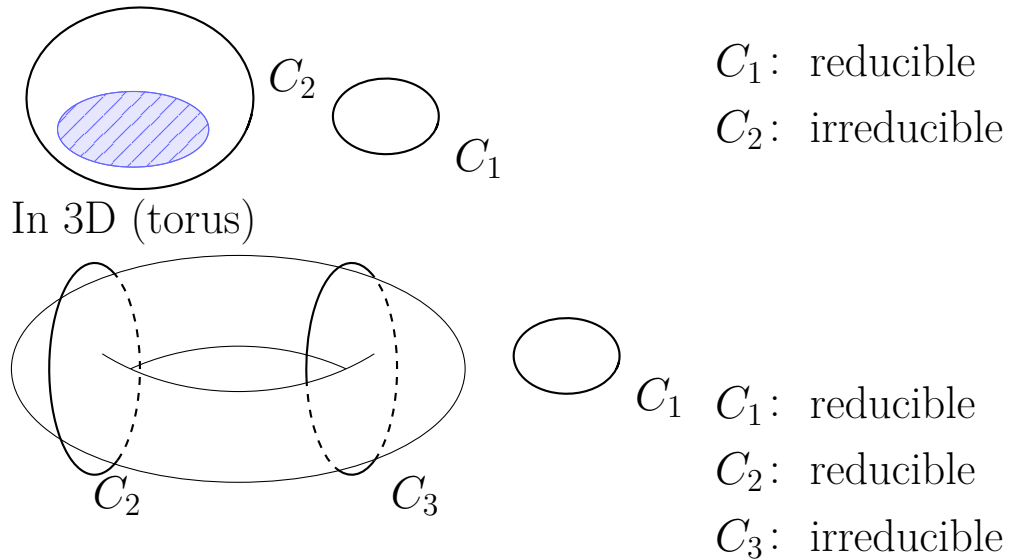


C_1 : reducible

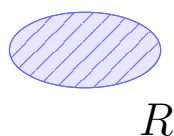
C_2 : reducible

In 2D

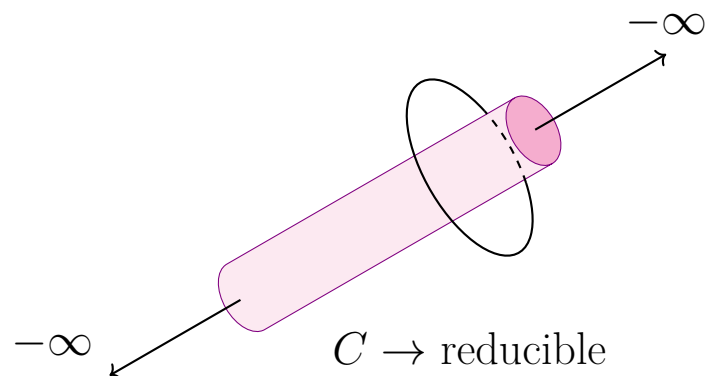
CHAPTER 4. GENERAL PROPERTIES OF IRROTATIONAL MOTION IN AN IDEAL FLUID (POTENTIAL FLOWS)



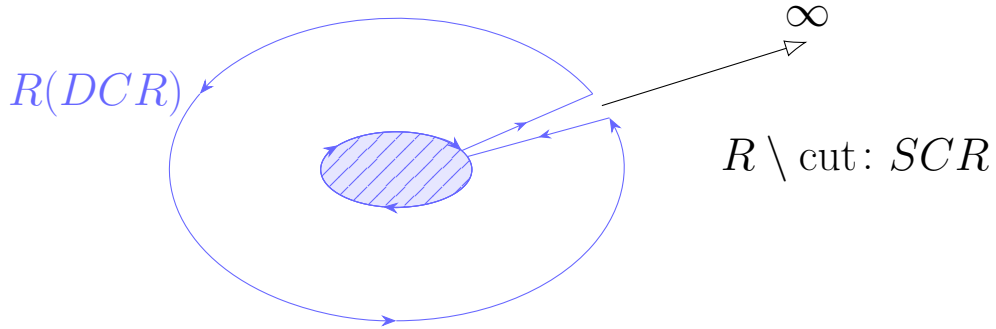
- Simply-Connected Region (SCR)
A connected region in which all circuit are reducible.
- Doubly-Connected Region (DCR)



Can render DCR into SCR



4.2. FUNDAMENTAL SOLUTION AND GREEN'S THEOREM



Irrotational Motion and Circulation

- In a SCR

For any circuit C in SCR R ,

$$\Gamma_C = \oint_C \mathbf{v} \cdot \mathbf{t} dl \stackrel[\text{Thm.}]{\text{Stokes'}} \iint_S \boldsymbol{\xi} \cdot \mathbf{n} dS = 0 \text{ since } \boldsymbol{\xi} = \mathbf{0}$$

hence irrotational flow ($\boldsymbol{\xi} = \mathbf{0}$ everywhere in R)

$$\Longleftrightarrow \Gamma = 0 \text{ around any circuit in } R$$

Further since $\mathbf{v} = \nabla \Phi$

$$\Gamma_C = \oint_C \mathbf{v} \cdot \mathbf{t} dl = \oint_C \underbrace{d\mathbf{t}}_{d\mathbf{r}} \cdot \nabla \Phi = \oint_C d\Phi = \Phi_{P'} - \Phi_P = 0$$

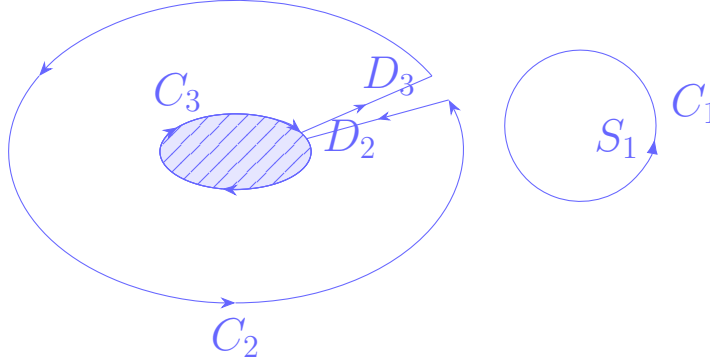
$\implies \Phi$ is a single-valued function.

Discussion.

The irrotational motion of a fluid in a SCR is also a motion without any circulation. Also, the velocity potential Φ is single-valued.

CHAPTER 4. GENERAL PROPERTIES OF IRROTATIONAL MOTION IN AN IDEAL FLUID (POTENTIAL FLOWS)

- In a DCR



- (a) For all reducible circuit in a DCR, like C_1

$$\Gamma_{C_1} = \oint_{C_1} \mathbf{v} \cdot \mathbf{t} dl = \iiint_{S_1} \boldsymbol{\xi} \cdot \mathbf{n} dS = 0 \text{ since } \boldsymbol{\xi} = \mathbf{0} \text{ in } S_1$$

- (b) For irreducible circuit (like C_2)

$$\begin{aligned} \Gamma_{C_2 D_2 C_3 D_3} &= \oint_{C_2} \mathbf{v} \cdot \mathbf{t} dl + \cancel{\int_{D_2} \mathbf{v} \cdot \mathbf{t} dl} + \oint_{C_3} \mathbf{v} \cdot \mathbf{t} dl + \cancel{\int_{D_3} \mathbf{v} \cdot \mathbf{t} dl} \\ &= \oint_{C_2} \mathbf{v} \cdot \mathbf{t} dl - \oint_{C_3} \mathbf{v} \cdot \mathbf{t} dl = 0 \end{aligned}$$

$$\Rightarrow \oint_{C_2} \mathbf{v} \cdot \mathbf{t} dl = \oint_{C_3} \mathbf{v} \cdot \mathbf{t} dl \text{ i.e. } \Gamma_{C_2} = \Gamma_{C_3}$$

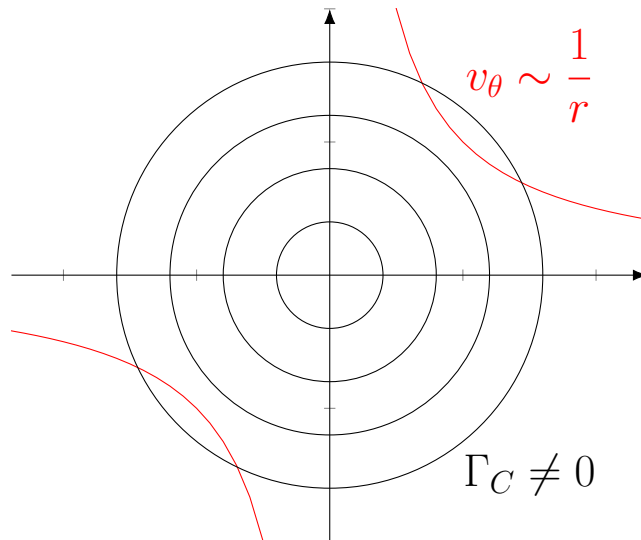
(may or may not be zero)

where C_2 and C_3 are reducible circuit

$$\begin{aligned} \Gamma &= \oint_C \mathbf{v} \cdot \mathbf{t} dl \\ &= \oint_C \nabla \Phi \cdot \underbrace{\mathbf{t} dl}_{d\mathbf{r}} = \oint_C d\Phi = \Phi'_P - \Phi_P \neq 0 \end{aligned}$$

4.2. FUNDAMENTAL SOLUTION AND GREEN'S THEOREM

Example. Free-vortex flow



$\nabla \times \mathbf{v} = \mathbf{0}$ everywhere except at $r = 0$ and

$$\Phi = C\theta$$

i.e., Φ is not a single-valued function, is a multi-valued function.

4.2.4 Some General Properties of Irrotational Motion in an Ideal Fluid (Potential Flow)

For SCR

Let $\nabla^2 \Phi = 0$ at every point in R . From $\mathbf{v} = \nabla \Phi$, $\Gamma_C = 0$ for any circuit C on R .

(1) Mean-Value Theorem

CHAPTER 4. GENERAL PROPERTIES OF IRROTATIONAL MOTION IN AN IDEAL FLUID (POTENTIAL FLOWS)

Define average value of Φ , $\bar{\Phi}_P$ over a spherical surface with radius r centred at P

$$\bar{\Phi}_P = \iint_S \Phi dS / \iint_S dS \quad (*)$$

$$\begin{aligned} &= \iint_S \Phi dS / \oint_{\Omega} r^2 d\Omega \\ &= \oint_{\Omega} \Phi r^2 d\Omega / 4\pi r^2 = \frac{1}{4\pi} \oint_{\Omega} \Phi d\Omega \quad (**) \end{aligned}$$

Now

$$\begin{aligned} \frac{d\bar{\Phi}_P}{dr} &= \frac{1}{4\pi} \oint_{\Omega} \frac{\partial \Phi}{\partial r} d\Omega \\ &= \frac{1}{4\pi r^2} \oint_{\Omega} \frac{\partial \Phi}{\partial r} \underbrace{r^2 d\Omega}_{dS} = \frac{1}{4\pi r^2} \iint_S \underbrace{\frac{\partial \Phi}{\partial n}}_{\mathbf{n} \cdot \nabla \Phi} dS \\ &= \frac{1}{4\pi r^2} \iiint_V \underbrace{\nabla \cdot (\nabla \Phi)}_{\nabla^2 \Phi = 0} dV \\ &= 0 \end{aligned}$$

$\implies \bar{\Phi}_P$ is independent of r , i.e. $\bar{\Phi}_P = \Phi_P$, value of Φ at point P .

(2) Maximum and Minimum Principles

The potential Φ can neither be a local maximum nor minimum in the interior of region R .

Proof.

4.2. FUNDAMENTAL SOLUTION AND GREEN'S THEOREM

From (1)

$$\bar{\Phi}_P = \Phi_P \text{ independent of } r$$

Now if Φ has a local maximum (minimum) at P , then all Φ on S are smaller (larger) than Φ_P , i.e.

$$\Phi > \bar{\Phi}_P = \frac{1}{4\pi} \oint_{\Omega} \Phi d\Omega \text{ contradict } \bar{\Phi}_P = \Phi_P$$

□

- (3) Each (Cartesian) component of $\nabla\Phi$ satisfies a Laplace equation.

Proof.

$$\because \nabla^2\Phi = 0 \therefore \frac{\partial (\nabla^2\Phi)}{\partial x(y)} = 0 \implies \nabla^2 \underbrace{\left(\frac{\partial\Phi}{\partial x(y)} \right)}_{v_x(v_y)} = 0$$

□

- (4) The component of $\nabla\Phi(\equiv \mathbf{v})$ can neither be a local maximum nor minimum in the interior of R .
- (5) The magnitude of the velocity $|\mathbf{v}|$ cannot be a local maximum in the interior of R .
- (6) Minimum pressure cannot occur in the interior of R .

CHAPTER 4. GENERAL PROPERTIES OF IRROTATIONAL MOTION IN AN IDEAL FLUID (POTENTIAL FLOWS)

Proof.

From (5) + Bernoulli's equation □

(7)

$$\underbrace{\iiint_R |\nabla\Phi|^2 dV}_{=\iiint_R |\mathbf{v}|^2 dV \propto \text{KE}} = \iiint_R (\nabla\Phi \cdot \nabla\Phi) dV$$

$$= \iiint_R \left[\underbrace{\nabla \cdot (\Phi \nabla\Phi)}_{=\mathbf{n} \cdot (\Phi \nabla\Phi) dS} - \cancel{\Phi \nabla^2 \Phi} \right] dV$$

Hence if

$$\Phi = 0 \quad \text{or} \quad \frac{\partial\Phi}{\partial n} \quad \text{on } S$$

$$\implies \iiint_R |\mathbf{v}|^2 dV = 0 \implies \mathbf{v} = \mathbf{0} \text{ everywhere, } \Phi = \text{constant in } R$$

(8)

$$\iiint_R \underbrace{\nabla^2 \Phi}_{=\nabla \cdot (\nabla\Phi)} = \iint_S \mathbf{n} \cdot \nabla\Phi dS = \iint_S \frac{\partial\Phi}{\partial n} dS = 0 \quad (4.25)$$

i.e. the natural B.C. (Neumann problem) $\frac{\partial\Phi}{\partial n}$ on S cannot be arbitrary specified.

4.2. FUNDAMENTAL SOLUTION AND GREEN'S THEOREM

(9) Minimum Kinetic Energy

Among all possible velocity fields satisfying continuity equation and the specified B.C.'s on S , the irrotational one has the minimum KE in R .

Proof.

Let $\mathbf{v}$ be an irrotational flow field in R , then $\mathbf{v} = \nabla\Phi$ and $\nabla \cdot \mathbf{v} = 0$. Let $\mathbf{v}_1 = \mathbf{v} + \mathbf{v}'$ be another velocity field which also satisfies $\nabla \cdot \mathbf{v}_1 = 0$ and $\mathbf{v}_1 \equiv \mathbf{v}$ (i.e. $\mathbf{v}' = 0$) on the boundary S , but $\mathbf{v}_1$ is not irrotational.

Let

$$T = \frac{\rho}{2} \iiint_R v^2 dV, \quad T_1 = \frac{\rho}{2} \iiint_R v_1^2 dV$$

$$\begin{aligned} T_1 - T &= \frac{\rho}{2} \iiint_R (v_1^2 - v^2) dV \\ &= \frac{\rho}{2} \iiint_R [(\mathbf{v} + \mathbf{v}') \cdot (\mathbf{v} + \mathbf{v}') - v^2] dV \\ &= \frac{\rho}{2} \iiint_R (2\mathbf{v} \cdot \mathbf{v}' + v'^2) \end{aligned}$$

but

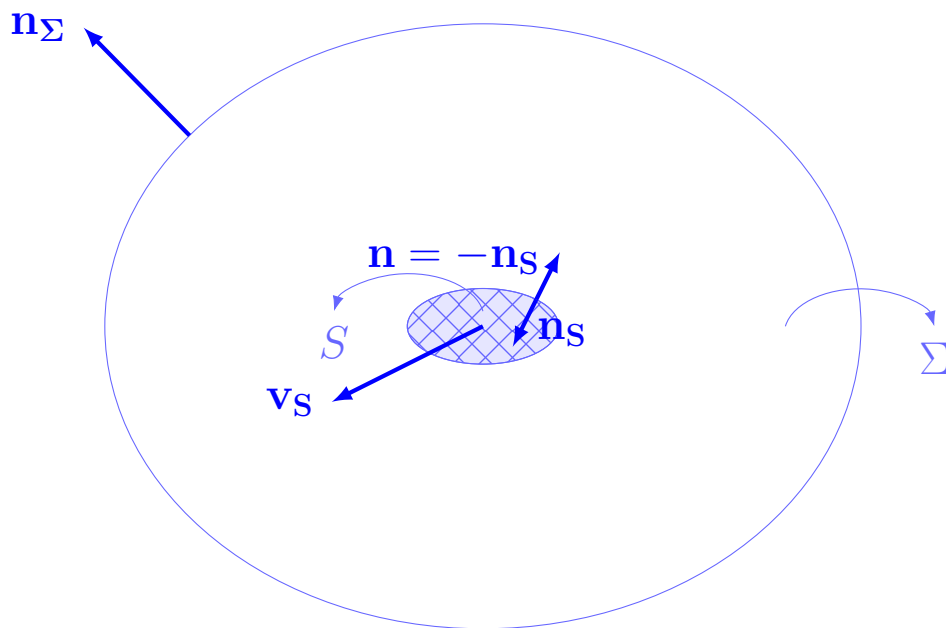
$$\begin{aligned} \iiint_R \mathbf{v} \cdot \mathbf{v}' dV &= \iiint_R \mathbf{v}' \cdot \nabla\Phi dV \\ &= \iiint_R [\nabla \cdot (\Phi\mathbf{v}') - \Phi\nabla \cdot \mathbf{v}'] \\ &= \iint_S \Phi (\mathbf{n} \cdot \mathbf{v}') dS = 0 \text{ since } \mathbf{v}' = \mathbf{0} \text{ on } S \end{aligned}$$

CHAPTER 4. GENERAL PROPERTIES OF IRROTATIONAL MOTION IN AN IDEAL FLUID (POTENTIAL FLOWS)

$$\therefore T_1 - T = \iiint_R v'^2 dV \geq 0$$

□

- (10) The solution of the Neumann exterior problem in a SCR is *unique* up to an *additive constant*.



Proof.

Let Φ_1 and Φ_2 be two solutions to $\nabla^2 \Phi = 0$ and
 $\frac{\partial \Phi_1}{\partial n} = \frac{\partial \Phi_2}{\partial n} = \mathbf{n} \cdot \mathbf{v}_s$ on S .
 Also

$$\nabla \Phi_1, \nabla \Phi_2 \sim o\left(\frac{1}{r^2}\right) \text{ as } r \rightarrow \infty \text{ (i.e. on } \Sigma)$$

Let $\Phi' = \Phi_2 - \Phi_1$, then $\nabla^2 \Phi' = 0$, $\frac{\partial \Phi'}{\partial n} = 0$ on S ,

4.2. FUNDAMENTAL SOLUTION AND GREEN'S THEOREM

$$\frac{\partial \Phi'}{\partial n_\Sigma} \sim o\left(\frac{1}{r^2}\right) \text{ on } \Sigma$$

$$\begin{aligned} \iiint_R |\nabla \Phi'|^2 dV &= \iiint_R \underbrace{(\nabla \Phi' \cdot \nabla \Phi')}_{=\nabla \cdot (\Phi' \nabla \Phi') - \Phi' \nabla^2 \Phi'} dV \\ &= \iint_S \Phi' \mathbf{n}_S \cdot \nabla \Phi' dS + \underbrace{\iint_\Sigma \Phi' \underbrace{\mathbf{n}_\Sigma \cdot \nabla \Phi'}_{=\frac{\partial \Phi'}{\partial n_\Sigma} \sim o\left(\frac{1}{r^2}\right)} \underbrace{d\Sigma}_{\sim \mathcal{O}\left(\frac{1}{r^2}\right)}}_{\rightarrow 0} \\ &= \iint_S \Phi' \underbrace{\frac{\partial \Phi'}{\partial n_S}}_{=-\frac{\partial \Phi'}{\partial n}=0} dS = 0 \end{aligned}$$

$$\implies \nabla \Phi' = 0 \implies \Phi' = \text{constant} \implies \Phi_2 - \Phi_1 = \text{constant}$$

□

The conclusion is not *generally true* for a DCR.

For DCR

- All (1)–(6) and (8) are also valid for DCR

(10) Uniqueness property of solution



4.2. FUNDAMENTAL SOLUTION AND GREEN'S THEOREM

$$\frac{\partial \Phi'}{\partial n_\Sigma} \sim o\left(\frac{1}{r}\right) \text{ on } \Sigma$$

$$\begin{aligned}
 \iint_R |\nabla \Phi'|^2 dA_R &= \iint_R \underbrace{(\nabla \Phi' \cdot \nabla \Phi')}_{=\nabla \cdot (\Phi' \nabla \Phi') - \Phi' \nabla^2 \Phi'} dA_R \\
 &= \oint_\Sigma \underbrace{\Phi' \frac{\partial \Phi'}{\partial n_\Sigma}}_{\sim o(\frac{1}{r})} \underbrace{dl}_{\sim O(r)} + \oint_S \cancel{\Phi' \frac{\partial \Phi'}{\partial n_S}} dl \\
 &\quad + \int_A^B \Phi'_+ \left(\frac{\partial \Phi'}{\partial n_+} \right)_+ dl + \int_C^D \Phi'_- \left(\frac{\partial \Phi'}{\partial n_-} \right)_- dl \\
 &= \int_A^B \Phi'_+ \left(\frac{\partial \Phi'}{\partial n_+} \right)_+ dl + \int_C^D \Phi'_- \left(\frac{\partial \Phi'}{\partial n_-} \right)_- dl \\
 &= \int_{\text{cut}} (\Phi'_+ - \Phi'_-) \frac{\partial \Phi'}{\partial n_C} dl \\
 &= \int_{\text{cut}} [(\Phi_2 - \Phi_1)_+ - (\Phi_2 - \Phi_1)_-] \frac{\partial \Phi'}{\partial n_C} dl \\
 &= \int_{\text{cut}} \left\{ \underbrace{[(\Phi_2)_+ - (\Phi_2)_-]}_{-\Gamma_2} - \underbrace{[(\Phi_1)_+ - (\Phi_1)_-]}_{-\Gamma_1} \right\} \frac{\partial \Phi'}{\partial n_C} dl \\
 &= (\Gamma_1 - \Gamma_2) \int_{\text{cut}} \frac{\partial \Phi'}{\partial n_C} dl
 \end{aligned}$$

(i) If $\Gamma_1 = \Gamma_2$, then

$$\iint_R (\nabla \Phi') \cdot (\nabla \Phi') dA_R = 0 \implies \nabla \Phi' = 0$$

CHAPTER 4. GENERAL PROPERTIES OF IRROTATIONAL MOTION IN AN IDEAL FLUID (POTENTIAL FLOWS)

$$\implies \Phi' = \text{constant}$$

$$\implies \Phi_2 = \Phi_1 + \text{constant}$$

(ii) If $\Gamma_1 \neq \Gamma_2$, then

$$\Phi_2 \neq \Phi_1 + \text{in general}$$

Discussion.

In a DCR, the solution to the Neumann exterior problem is unique determined only when the *circulation is specified*. Different circulations gives different solutions.

4.2.5 Flow Induced by Vorticity

Recall vorticity that $\boldsymbol{\xi} = \nabla \times \mathbf{v}$

$$\text{If } \mathbf{v}(\mathbf{r}, t) \text{ is known } \xRightarrow{\nabla \times \mathbf{v}} \boldsymbol{\xi}(\mathbf{r}, t)$$

$$\text{If } \boldsymbol{\xi}(\mathbf{r}, t) \text{ is known } \xRightarrow{?} \mathbf{v}(\mathbf{r}, t)$$

Note that $\mathbf{v}(\mathbf{r}, t)$ thus determined is not unique!

Biot-Savart Law

(Calculation of the velocity field induced by vorticity)

$$\mathbf{v} = \nabla \times \boldsymbol{\Psi} \text{ (from } \nabla \cdot \mathbf{v} = 0)$$

Now

$$\boldsymbol{\xi} = \nabla \times \mathbf{v} = \nabla \times (\nabla \times \boldsymbol{\Psi}) = -\nabla^2 \boldsymbol{\Psi} + \nabla (\nabla \cdot \boldsymbol{\Psi})$$

4.2. FUNDAMENTAL SOLUTION AND GREEN'S THEOREM

As before, without loss of generality can set $\nabla \cdot \Psi = 0$.

Hence

$$\boxed{\nabla^2 \Psi = -\xi} \quad (4.26a)$$

In component form

$$\nabla^2 \Psi_i = -\xi_i \quad (4.26b)$$

Recall

$$Lu(x) = f(x)$$

$$\underbrace{u(x)}_{w(x;y)} = \delta(x - y)$$

$$u(x) = \int w(x; y) f(y) dy$$

then the solution to the (4.26b) is

Recall

$$\nabla^2 \Psi_i = \delta(|\mathbf{r} - \mathbf{r}'|)$$

solution is

$$\Psi_{i,\text{FS}}(\mathbf{r}; \mathbf{r}') = -\frac{1}{4\pi|\mathbf{r} - \mathbf{r}'|}$$

$$\begin{aligned} \Psi_i(\mathbf{r}, t) &= \iiint_{V_{r'}} -\xi_i(\mathbf{r}', t) \Psi_{i,\text{FS}}(\mathbf{r}; \mathbf{r}') dV_{r'} \\ &= \frac{1}{4\pi} \iiint_{V_{r'}} \frac{\xi(\mathbf{r}', t)}{|\mathbf{r} - \mathbf{r}'|} dV_{r'} \end{aligned}$$

Sum up all 3 components

$$\Psi(\mathbf{r}, t) = \frac{1}{4\pi} \iiint_{V_{r'}} \frac{\boldsymbol{\xi}(\mathbf{r}', t)}{|\mathbf{r} - \mathbf{r}'|} dV_{r'} \quad (4.27)$$

solution to the inhomogeneous (4.26a)

CHAPTER 4. GENERAL PROPERTIES OF IRROTATIONAL MOTION IN AN IDEAL FLUID (POTENTIAL FLOWS)

The induced velocity field $\mathbf{v}(\mathbf{r}, t)$ is then

$$\begin{aligned}
 \mathbf{v}(\mathbf{r}, t) &= \nabla \times \Psi \\
 &= \frac{1}{4\pi} \nabla_r \P \times \iiint_{V_{r'}} \frac{\boldsymbol{\xi}(\mathbf{r}', t)}{|\mathbf{r} - \mathbf{r}'|} dV_{r'} \\
 &= \frac{1}{4\pi} \iiint_{V_{r'}} \left[\nabla_r \left(\frac{1}{|\mathbf{r} - \mathbf{r}'|} \right) \times \boldsymbol{\xi}(\mathbf{r}', t) \right] dV_{r'} \\
 &= \frac{1}{4\pi} \iiint_{V_{r'}} \frac{\boldsymbol{\xi}(\mathbf{r}', t) \times \mathbf{R}}{R^3} dV_{r'} \tag{4.28}
 \end{aligned}$$

where $\mathbf{R} = \mathbf{r} - \mathbf{r}'$ and note that

$$\nabla_r \times (f \boldsymbol{\xi}(\mathbf{r}', t)) = \nabla_r f \times \boldsymbol{\xi} + f \cancel{\nabla_r \times \boldsymbol{\xi}(\mathbf{r}', t)}$$

Exercise. Show that

$$\nabla_r \left(\frac{1}{|\mathbf{r} - \mathbf{r}'|} \right) = -\frac{\mathbf{r} - \mathbf{r}'}{|\mathbf{r} - \mathbf{r}'|^3} = -\frac{\mathbf{R}}{R^3}$$

Proof.

Let $\mathbf{R} = x_i \mathbf{e}_i$ and $R = (x_i x_i)^{1/2}$, then

$$\begin{aligned}
 \nabla (R^n) &= \nabla (x_i x_i)^{n/2} = \mathbf{e}_i \frac{n}{2} (x_j x_j)^{n/2-1} 2 \underbrace{x_{k,i} x_k}_{=\delta_{ki}} \\
 &= n \mathbf{e}_i (x_j x_j)^{\frac{n-2}{2}} x_i = n R^{n-2} \mathbf{R}
 \end{aligned}$$

[¶]Note that the subscript r on the curl to emphasize that the curl is to be taken with respect to the coordinate of the point $\mathbf{r}$.

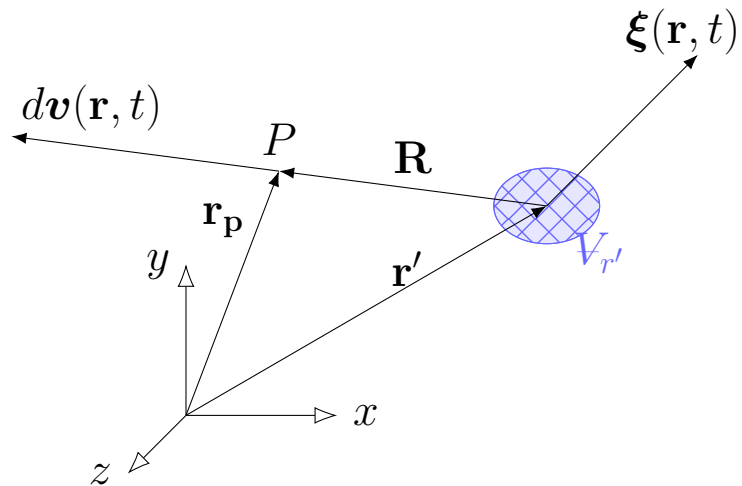
4.2. FUNDAMENTAL SOLUTION AND GREEN'S THEOREM

□

In elemental form

$$d\Psi(\mathbf{r}, t) = \frac{1}{4\pi} \frac{\boldsymbol{\xi}(\mathbf{r}', t) dV_{r'}}{|\mathbf{r} - \mathbf{r}'|} \quad (4.29)$$

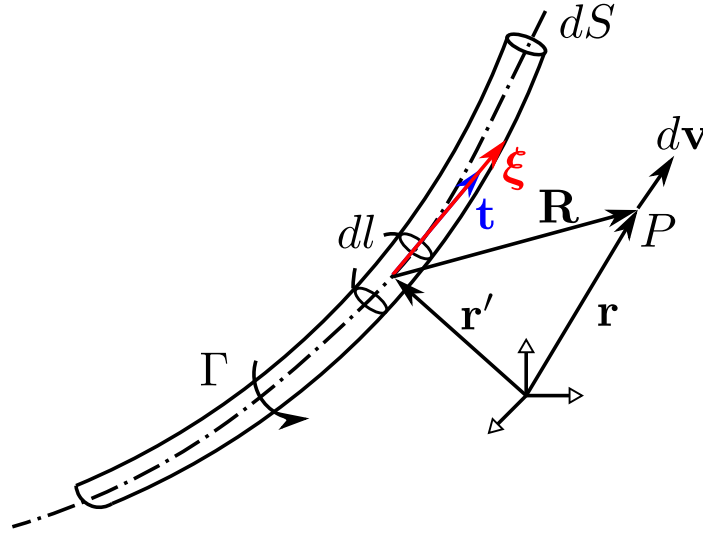
$$d\mathbf{v}(\mathbf{r}, t) = \frac{1}{4\pi} \frac{\boldsymbol{\xi}(\mathbf{r}', t) dV_{r'} \times \mathbf{R}}{|\mathbf{r} - \mathbf{r}'|} \quad (4.30)$$



Velocity Induced by Concentrated Vorticity

- Single Vortex Filament

CHAPTER 4. GENERAL PROPERTIES OF IRROTATIONAL MOTION IN AN IDEAL FLUID (POTENTIAL FLOWS)



From Helmholtz Theorem

$$\boldsymbol{\xi} \cdot \mathbf{t} dS = \Gamma = \text{constant}$$

From (4.30)

$$\Rightarrow d\mathbf{v}(\mathbf{r}) = \frac{1}{4\pi} \frac{\boldsymbol{\xi}(\mathbf{r}') dV_{r'} \times \mathbf{R}}{R^3} \parallel \frac{1}{4\pi} \frac{\Gamma dl \mathbf{t} \times \mathbf{R}}{R^3} \quad (4.31)$$

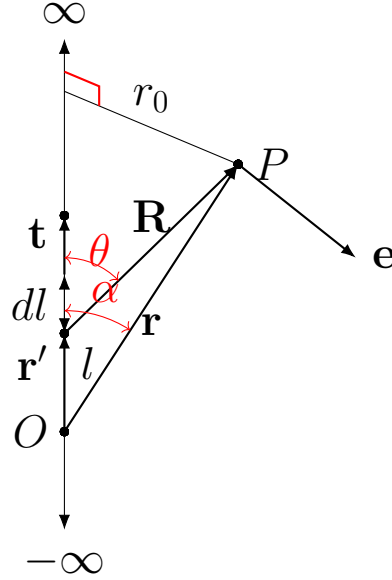
Since Γ is constant

$$\mathbf{v}(\mathbf{r}) = \frac{\Gamma}{4\pi} \int \frac{\mathbf{t} \times \mathbf{R}}{R^3} dl \quad (4.32)$$

- Straight Infinite-long vortex filament

$$\parallel \boldsymbol{\xi}(\mathbf{r}') dV_{r'} = \boldsymbol{\xi} (\mathbf{t} dS \cdot d\mathbf{l}) = \boldsymbol{\xi} \left(\mathbf{t} dS \cdot \frac{\boldsymbol{\xi}}{\xi} dl \right) = \boldsymbol{\xi} \frac{\Gamma}{\xi} dl = \Gamma \mathbf{t} dl$$

4.2. FUNDAMENTAL SOLUTION AND GREEN'S THEOREM



$$\mathbf{t} \times \mathbf{R} = R \sin \theta \mathbf{e} \quad (4.33)$$

where $\mathbf{e}$ is unit vector into board.

From (4.32)

$$\begin{aligned} \mathbf{v}(\mathbf{r}) &= \frac{\Gamma}{4\pi} \int_{-\infty}^{\infty} \mathbf{e} \frac{\sin \theta}{R^2} dl & (*) \\ &\stackrel{\text{in this case}}{=} \mathbf{e} \frac{\Gamma}{4\pi} \int_{-\infty}^{\infty} \frac{\sin \theta}{R^2} dl \stackrel{**}{=} \mathbf{e} \frac{\Gamma}{4\pi} \int_0^\pi \frac{\sin \theta}{r_0} d\theta \\ &= \frac{\Gamma}{2\pi r_0} \mathbf{e} & (4.34) \end{aligned}$$

Note.

**From geometry, $\frac{r \cos \alpha - l}{r_0} = \cot \theta \implies dl = \frac{r_0 d\theta}{\sin^2 \theta}$ and $R^2 = \frac{r_0^2}{\sin^2 \theta}$

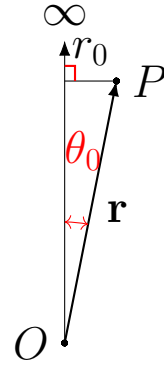
CHAPTER 4. GENERAL PROPERTIES OF IRROTATIONAL MOTION IN AN IDEAL FLUID (POTENTIAL FLOWS)

- ① For semi-infinite long vortex filament

$$\mathbf{v}(\mathbf{r}) = \frac{\Gamma}{4\pi r_0} (1 + \cos \theta_0) \mathbf{e}$$

(4.35)

$$\left(\int_{\theta=0}^{\theta=\pi} (\dots) d\theta \longrightarrow \int_{\theta=\theta_0}^{\theta=\pi} (\dots) d\theta \right)$$

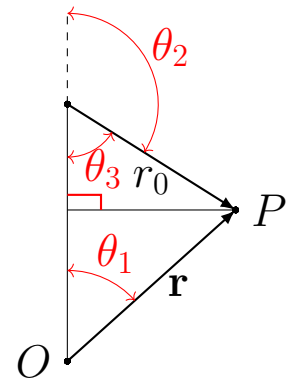


- ② For a segment of vortex filament

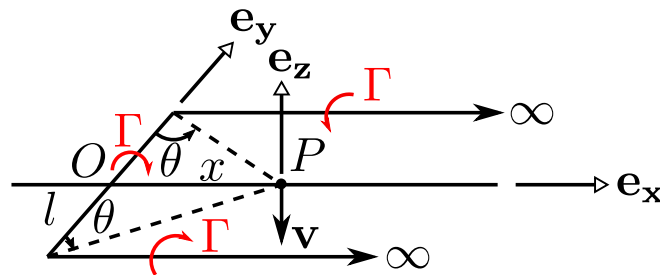
$$\mathbf{v}(\mathbf{r}) = \frac{\Gamma}{4\pi r_0} (\cos \theta_1 - \cos \theta_2)$$

$$= \frac{\Gamma}{4\pi r_0} (\cos \theta_1 + \cos \theta_3)$$

(4.36)



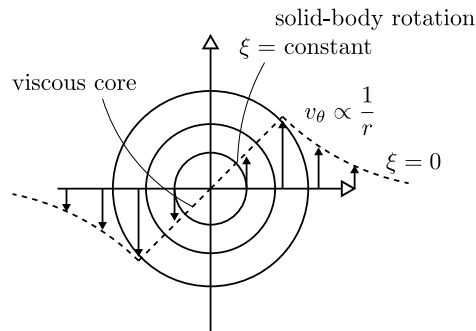
- (Concentrated) Horseshoe Vortex filament



$$\mathbf{v} = -\mathbf{e}_z \left[\frac{\Gamma}{4\pi x} (\cos \theta + \cos \theta) + 2 \frac{\Gamma}{4\pi l} (1 + \sin \theta) \right]$$

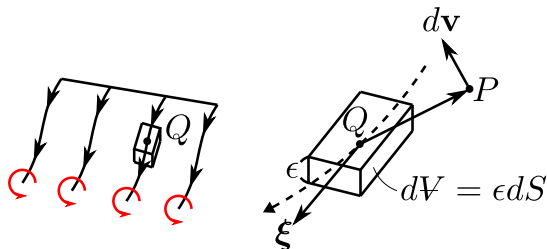
CHAPTER 4. GENERAL PROPERTIES OF IRROTATIONAL MOTION IN AN IDEAL FLUID (POTENTIAL FLOWS)

– In real viscous flow



which is Rankine vortex (or Circular vortex).

Velocity Induced by Concentrated Vorticity- Concentrated Vortex Sheet



Biot-Savart law:

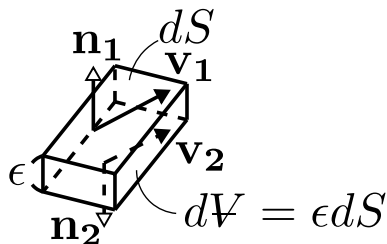
$$\boldsymbol{\xi} dV = \boldsymbol{\xi} \epsilon dS$$

as $\epsilon \rightarrow 0$, let $|\boldsymbol{\xi}| \rightarrow \infty$

so that $|\boldsymbol{\xi} \epsilon| = |\boldsymbol{\gamma}| = \text{constant at point } Q$

$$d\mathbf{v}(\mathbf{r}) = \frac{1}{4\pi} \frac{\boldsymbol{\gamma}(\mathbf{r}') dS \times \mathbf{R}}{R^3} \quad (4.38)$$

What is $|\boldsymbol{\gamma}|$?



4.2. FUNDAMENTAL SOLUTION AND GREEN'S THEOREM

Now

$$\boldsymbol{\xi} dV = \nabla \times \mathbf{v} dV \stackrel{\dagger\dagger}{=} \oint\!\!\!\oint_{\Delta S} \mathbf{n} \times \mathbf{v} dS$$

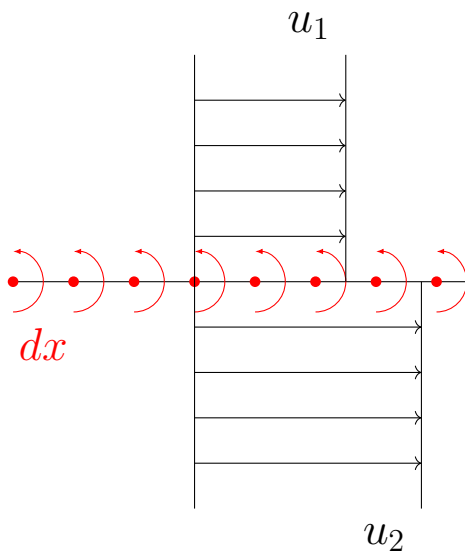
Hence

$$\begin{aligned} \boldsymbol{\xi} dV &= \lim_{\epsilon \rightarrow 0} \boldsymbol{\xi} \epsilon dS \\ &= \lim_{\epsilon \rightarrow 0} \oint\!\!\!\oint_{\Delta S} \mathbf{n} \times \mathbf{v} dS \\ &= (\mathbf{n}_1 \times \mathbf{v}_1 + \mathbf{n}_2 \times \mathbf{v}_2) dS \\ &\implies \boldsymbol{\gamma} = \mathbf{n} \times (\mathbf{v}_1 - \mathbf{v}_2) \end{aligned}$$

but $(\mathbf{v}_1 - \mathbf{v}_2) \perp \mathbf{n}$

hence

$$|\boldsymbol{\gamma}| = |\mathbf{v}_1 - \mathbf{v}_2| = |u_1 - u_2| \quad (4.39)$$

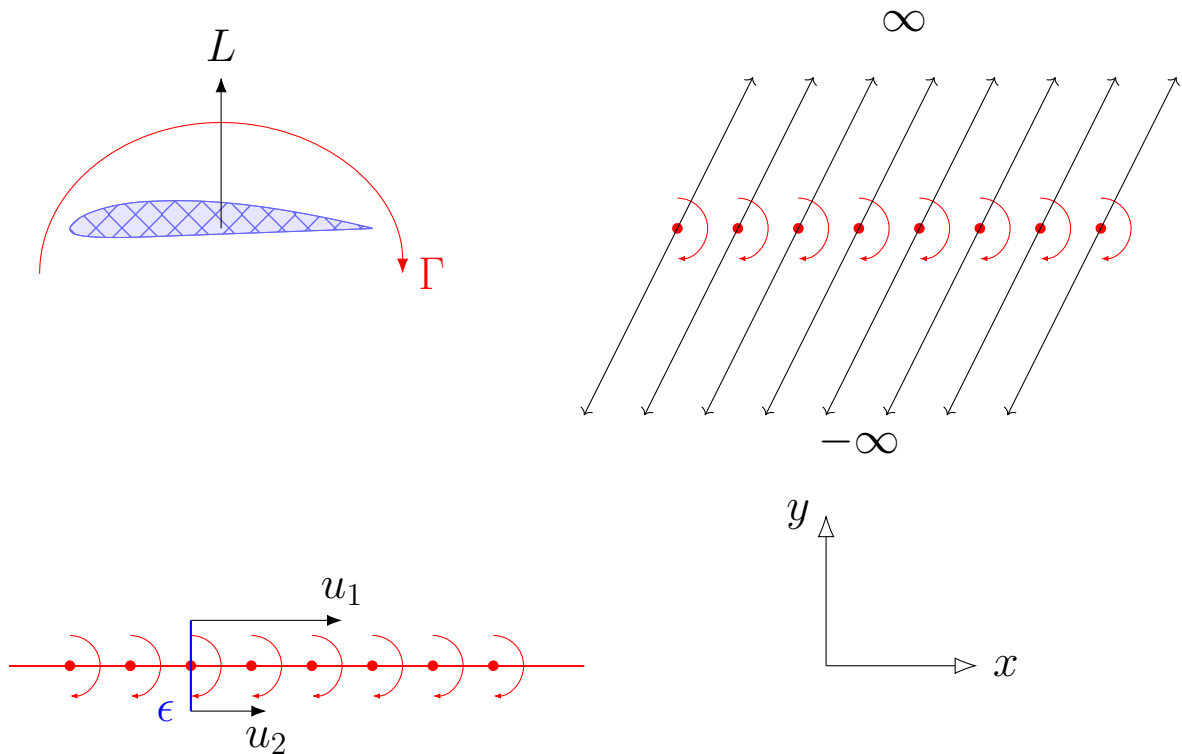


$$d\Gamma = (u_2 - u_1) dx = \gamma dx$$

^{††}integral definition of $\nabla \times \mathbf{v}$

CHAPTER 4. GENERAL PROPERTIES OF IRROTATIONAL MOTION IN AN IDEAL FLUID (POTENTIAL FLOWS)

- Vortex Sheet of Straight, Infinite-long Vortex Filament

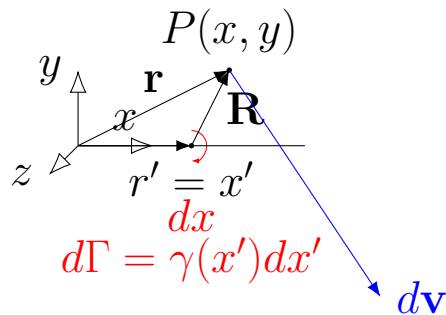


$$d\Gamma = (u_1 - u_2) dx = \gamma dx$$

in general

$$\gamma(x) = u_1(x) - u_2(x) \quad (4.40)$$

4.2. FUNDAMENTAL SOLUTION AND GREEN'S THEOREM



Recall, from (4.34)

$\mathbf{v}(\mathbf{r}) = \frac{\Gamma}{2\pi R} \mathbf{e}$ for a single concentrated vortex filament

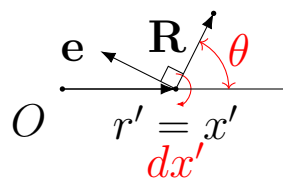
$$\Rightarrow d\mathbf{v} = \frac{d\Gamma}{2\pi R} \mathbf{e} = \frac{\gamma(x')dx'}{2\pi R} \mathbf{e}$$

$$\mathbf{v}(x, y) = \frac{1}{2\pi} \int \left[\frac{\gamma(x')dx'}{R(x')} \mathbf{e} \right] \quad (4.41)$$

Exercise.

Show that if $\gamma(x) \equiv \gamma = \text{constant}$ along $-\infty < x < \infty$, then

$$\begin{aligned} \mathbf{v}(x, y) &= \frac{\gamma}{2} \mathbf{e}_x \text{ at } + \text{ side} \\ &= -\frac{\gamma}{2} \mathbf{e}_x \text{ at } - \text{ side} \end{aligned}$$



$$\begin{aligned} u &= \frac{\gamma}{2} \int_{-\infty}^{\infty} \frac{-\cos(\pi/2 - \theta)}{\sqrt{(x - x')^2 + y^2}} dx' = \frac{\gamma}{2\pi} \underbrace{\frac{y}{(x - x')^2 + y^2}}_{=\pi} dx' \\ &= \frac{\gamma}{2} \end{aligned}$$

Chapter 5

Two-Dimensional Potential Flows

$$\left. \begin{array}{l} \text{Incompressible: } \nabla \cdot \mathbf{v} = 0 \\ \text{Irrotational: } \nabla \times \mathbf{v} = 0 \end{array} \right\} \implies \nabla^2 \Phi = 0 \quad (5.1)$$

Equation (5.1) can be solved via using Complex-Variable Theory.

5.1 Brief Review of Complex Theory

- A complex variable: $z = x + iy = re^{i\theta}$, $i = \sqrt{-1}$,
 $r = \sqrt{x^2 + y^2}$, $\theta = \arctan(y/x)$
- A complex function

$$f(z) = \xi(x, y) + i\eta(x, y)$$

$$\xi \equiv \Re \{f\}, \quad \eta \equiv \Im \{f\}$$

5.1. BRIEF REVIEW OF COMPLEX THEORY

- Complex conjugate (C-C)

$$\bar{z} = x - iy, \quad z, \bar{z}: \text{C-C pair}$$

$$\overline{f(z)}: \text{C-C of } f(z)$$

Note that $\overline{f(z)} \neq \bar{f}(z) \neq f(\bar{z})$, e.g.

$$\begin{aligned} \text{if } f(z) &= (2 + 3i)z + e^{4iz}, \\ \overline{f(z)} &= (2 - 3i)\bar{z} + e^{-4i\bar{z}}, \quad \bar{f}(z) = (2 - 3i)z + e^{-4iz}, \\ f(\bar{z}) &= (2 + 3i)\bar{z} + e^{4i\bar{z}} \end{aligned}$$

- The conjugate z and $\bar{z}$

$$\text{Since } \begin{cases} z = x + iy \\ \bar{z} = x - iy \end{cases} \implies \begin{cases} x = \frac{1}{2}(z + \bar{z}) \\ y = -\frac{i}{2}(z - \bar{z}) \end{cases}$$

$$(x, y) \iff (z, \bar{z})$$

$$\begin{cases} \frac{\partial}{\partial z} = \frac{\partial x}{\partial z} \frac{\partial}{\partial x} + \frac{\partial y}{\partial z} \frac{\partial}{\partial y} = \frac{1}{2} \left(\frac{\partial}{\partial x} - i \frac{\partial}{\partial y} \right) \\ \frac{\partial}{\partial \bar{z}} = \frac{\partial x}{\partial \bar{z}} \frac{\partial}{\partial x} + \frac{\partial y}{\partial \bar{z}} \frac{\partial}{\partial y} = \frac{1}{2} \left(\frac{\partial}{\partial x} + i \frac{\partial}{\partial y} \right) \end{cases}$$

$$\implies \begin{cases} \frac{\partial}{\partial x} = \frac{\partial}{\partial z} + \frac{\partial}{\partial \bar{z}} \\ \frac{\partial}{\partial y} = i \frac{\partial}{\partial z} - i \frac{\partial}{\partial \bar{z}} \end{cases}$$

Example.

CHAPTER 5. TWO-DIMENSIONAL POTENTIAL FLOWS

$$\psi(x, y) \implies \psi(z, \bar{z})$$

$$u = \frac{\partial \psi}{\partial y} = i \frac{\partial \psi}{\partial z} - i \frac{\partial \psi}{\partial \bar{z}}, \quad v = -\frac{\partial \psi}{\partial x} = -\frac{\partial \psi}{\partial z} - \frac{\partial \psi}{\partial \bar{z}}$$

$$\implies u - iv = 2i \frac{\partial \psi}{\partial z}, \quad u + iv = -2i \frac{\partial \psi}{\partial \bar{z}}$$

2D vorticity

$$\begin{aligned} \zeta &= \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} = - \left(\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} \right) \\ &= - \underbrace{\left(\frac{\partial}{\partial x} - i \frac{\partial}{\partial y} \right)}_{=2\frac{\partial}{\partial z}} \underbrace{\left(\frac{\partial}{\partial x} + i \frac{\partial}{\partial y} \right)}_{=2\frac{\partial}{\partial \bar{z}}} \psi \\ &\implies -\zeta = \nabla^2 \psi = 4 \frac{\partial^2 \psi}{\partial z \partial \bar{z}} \end{aligned}$$

5.1.1 Analytical Functions

- Differentiation of a complex function

$$f'(z) = \lim_{\Delta z \rightarrow 0} \frac{f(z + \Delta z) - f(z)}{\Delta z} \text{ exists and is single-valued.}$$

- Analytic function

For a closed C on a complex plane, a function $f(z)$ is said to be *analytic* within C if

- ① $f(z)$ exists and is finite
- ② $f'(z)$ exists and is single-valued

5.1. BRIEF REVIEW OF COMPLEX THEORY

- Functions z^n ($n \geq 0, n$: integer), e^z , $\sin z$, $\cos z$, $\cosh z \cdots$ are analytic in $|z| < \infty$
- The functions z^{-n} ($n > 0$) is analytic for $|z| > 0$
- The function $\ln z$ is analytic in any finite region which does not enclose the origin.

$$\begin{aligned}\ln z &= \ln r e^{i(\theta+2k\pi)}, k = 0, \pm 1, \pm 2, \cdots \\ &= \ln r + \underbrace{i(\theta + 2k\pi)}_{\text{multi-valued function}}\end{aligned}$$

5.1.2 Cauchy-Riemann Conditions

$$f(z) = f(x + iy) = \xi(x, y) + i\eta(x, y)$$

$$f'(z) = \lim_{\Delta z \rightarrow 0} \frac{f(z + \Delta z) - f(z)}{\Delta z}$$

Choose $\Delta z \equiv \Delta x$, then

$$\begin{aligned}f'(z) &= \lim_{\Delta x \rightarrow 0} \frac{f(z + \Delta x) - f(z)}{\Delta x} \\ &= \lim_{\Delta x \rightarrow 0} \left[\frac{\xi(x + \Delta x, y) - \xi(x, y)}{\Delta x} + i \frac{\eta(x + \Delta x, y) - \eta(x, y)}{\Delta x} \right] \\ &= \frac{\partial \xi}{\partial x} + i \frac{\partial \eta}{\partial x}\end{aligned}$$

(*)

CHAPTER 5. TWO-DIMENSIONAL POTENTIAL FLOWS

Similarly, choose $\Delta z = i\Delta y$

$$f'(z) = -i\frac{\partial \xi}{\partial y} + \frac{\partial \eta}{\partial y} \quad (**)$$

For differentiable $\implies (*) = (**)$

$$\implies \boxed{\frac{\partial \xi}{\partial x} = \frac{\partial \eta}{\partial y}, \frac{\partial \xi}{\partial y} = -\frac{\partial \eta}{\partial x}} \text{ (C-R-C)} \quad (5.2)$$

From (5.2)

$$\begin{aligned} \frac{\partial^2 \xi}{\partial x^2} &= \frac{\partial^2 \eta}{\partial x \partial y} = -\frac{\partial^2 \xi}{\partial y^2} \\ \implies \frac{\partial^2 \xi}{\partial x^2} + \frac{\partial^2 \xi}{\partial y^2} &= 0 \implies \nabla^2 \xi = 0 \end{aligned} \quad (5.3a)$$

Similarly,

$$\nabla^2 \eta = 0 \quad (5.3b)$$

Moreover

$$\nabla \xi \cdot \nabla \eta = \frac{\partial \xi}{\partial x} \frac{\partial \eta}{\partial x} + \frac{\partial \xi}{\partial y} \frac{\partial \eta}{\partial y} \stackrel{\text{use C-R-C}}{=} 0$$

i.e. curves of $\xi(x, y) = \text{constant}$ and curves of $\eta(x, y) = \text{constant}$ are perpendicular.

So any analytic complex function

$$f(z) = \underbrace{\xi(x, y)}_{=\Phi(x, y)} + i \underbrace{\eta(x, y)}_{=\Psi(x, y)}$$

to be the velocity potential function and stream function of 2D potential flow.

5.1.3 Integration of an Analytic Functions

- $$\oint_C f(z)dz = 0 \text{ Cauchy Integral Theorem (CIT)} \quad (5.4)$$

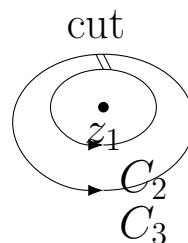
- If $f(z)$ is analytic everywhere except at certain isolated points, $z_1, z_2, \dots, z_n$.

$$\textcircled{1} \quad \oint_{C_1} f(z) dz = 0 \quad (5.5) \quad \begin{array}{c} \bullet \\ \circlearrowright \\ z_1 \quad C_1 \end{array}$$

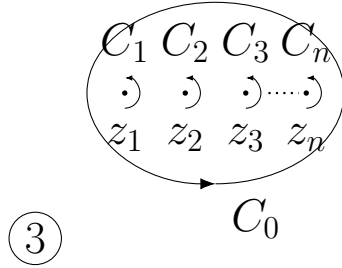
$$\oint_{C_2} f(z)dz = \oint_{C_3} f(z)dz^* = 0 \quad (5.6)$$

$$\neq 0 \quad (5.7)$$

C_2, C_3 : reducible circuits



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$$\oint_{C_0} f(z)dz = \oint_{C_1} f(z)dz + \oint_{C_2} f(z)dz + \cdots + \oint_{C_n} f(z)dz \quad (5.8)$$

(extension of CIT)

Example. Consider $f(z) = z^n$, n is integer

(a) for $n \geq 0$

$f(z) = z^n$ is analytic in $|z| < \infty$

$\therefore \oint_C z^n dz \stackrel{\text{CIT}}{=} 0$ or $\oint_C z^n dz = \frac{1}{n+1} \oint_C d(z^{n+1}) = 0$ for every finite circuit C

(b) for $n \leq -2$, let $m = -n$, i.e. $m \geq 2$

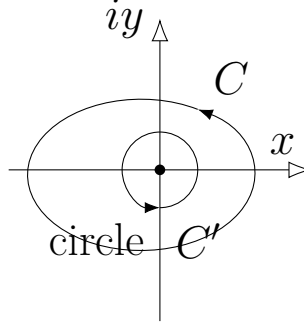
$f(z) = \frac{1}{z^m}$ is analytic everywhere except at $z = 0$

- for every closed curve C not enclosing $z = 0$

$$\oint_C \frac{1}{z^m} dz = \frac{1}{-m+1} \oint_C d(z^{-m+1}) \stackrel{\text{CIT}}{=} 0$$

- for any closed curve C enclosing $z = 0$

5.1. BRIEF REVIEW OF COMPLEX THEORY



$$\oint_C \frac{1}{z^m} dz \stackrel{C, C' \text{ are reducible}}{=} \oint_{C'} \frac{1}{(re^{i\theta})^m} d(re^{i\theta}) = \frac{i}{r^{m-1}} \int_{\theta=0}^{\theta=2\pi} e^{-im\theta} e^{i\theta} d\theta$$

$$= \frac{1}{(1-m)r^{m-1}} e^{i(1-m)\theta} \Big|_0^{2\pi} = 0$$

for any $m \neq 1$ (i.e., $n \neq -1$)

(c) for $n = -1$ (i.e. $m = 1$)

$f(z) = \frac{1}{z}$: singular at $z = 0$

- for any closed curve C not enclosing $z = 0$

$$\oint_C \frac{1}{z} dz = \oint_C d(\ln z) = \ln z \Big|_{z_0}^{z'_0} = \ln re^{i\theta} \Big|_{r_0 e^{i\theta'_0}}^{r_0 e^{i\theta_0}} = 0$$

- for any closed curve C enclosing $z = 0$

$$\oint_C \frac{1}{z} dz = \oint_C d(\ln z) = \ln z \Big|_{z_0}^{z'_0} = \ln re^{i\theta} \Big|_{r_0 e^{i\theta_0}}^{r_0 e^{i(\theta_0+2\pi)}} = 2\pi i$$

- Similarly for any function $f(z) = (z - z_0)^n$

CHAPTER 5. TWO-DIMENSIONAL POTENTIAL FLOWS

(a) $\oint_C (z - z_0)^n dz = 0$ for every circuit C when $n \geq 0$
or $n \leq -2$ (n : integer)

(b) for $n = -1$

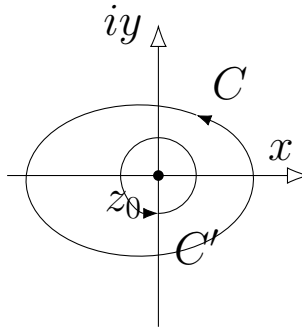
$$\oint_C \frac{1}{z - z_0} dz = 0 \text{ (for every circuit } C \text{ not enclosing } z = z_0)$$

$$= 2\pi i \text{ (for every circuit } C \text{ enclosing } z = z_0)$$

5.1.4 Cauchy Integral Formula (CIF)

- If $f(z)$ is analytic within a closed curve C , and point z_0 inside C , then

$$\frac{1}{2\pi i} \oint_{C \rightarrow C'} \frac{f(z)}{z - z_0} dz = f(z_0) \quad (5.9)$$



- In (5.9), let z_0 be any point inside C and be denoted by ζ , then

$$f(\zeta) = \frac{1}{2\pi i} \oint_C \frac{f(z)}{z - \zeta} dz$$

$$\zeta \Leftrightarrow z \implies f(z) = \frac{1}{2\pi i} \oint_C \frac{f(\zeta)}{\zeta - z} d\zeta \text{ (CIF)} \quad (5.10)$$

5.1. BRIEF REVIEW OF COMPLEX THEORY

- CIF for higher-derivatives

If $f(z)$ is analytic (i.e. $f'(z)$ exists) within C , then

$$f^{(n)}(z) = \frac{n!}{2\pi i} \oint_C \frac{f(\zeta)}{(\zeta - z)^{n+1}} d\zeta \quad (5.11)$$

5.1.5 Taylor Series and Laurent Series

- Taylor Series

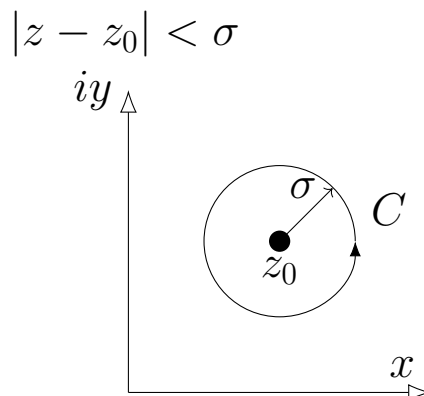
Let $f(z)$ be an analytic function within a circle C centred at z_0 , then

$$f(z) = \sum_{n=0}^{\infty} A_n (z - z_0)^n \quad (5.12)$$

where

$$A_n = \frac{1}{2\pi i} \oint_C \frac{f(\zeta)}{(\zeta - z_0)^{n+1}} d\zeta = \frac{1}{n!} f^{(n)}(z_0)$$

The radius of convergence of the power-series is σ , i.e.



- Laurent Series

CHAPTER 5. TWO-DIMENSIONAL POTENTIAL FLOWS

Let $f(z)$ be analytic in an annulus between C_1 and C_2 , both centred at z_0 , then

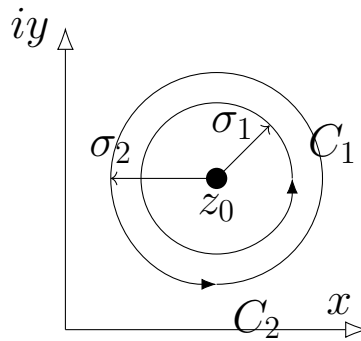
$$f(z) = \sum_{-\infty}^{\infty} A_n (z - z_0)^n \quad (5.13)$$

where

$$A_n = \frac{1}{2\pi i} \oint_C \frac{f(\zeta)}{(\zeta - z_0)^{n+1}} d\zeta \left(\neq \frac{1}{n!} f^{(n)}(z_0) \right)$$

radii of convergence

$$\sigma_1 < |z - z_0| < \sigma_2$$



5.1.6 The Residue Theorem

- Consider the integral $I = \oint_C f(z) dz$, $f(z)$ is analytic within C except an isolated singularity $z = z_0$, then

$$I = 2\pi A_{-1} \quad (5.14)$$

5.1. BRIEF REVIEW OF COMPLEX THEORY

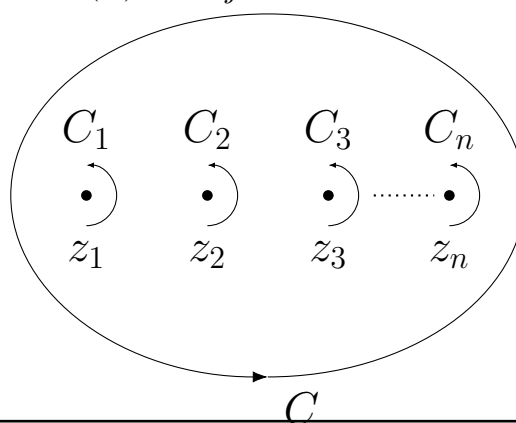
where A_{-1} (coefficient of $\frac{1}{z - z_0}$) is called the residue of $f(z)$ at z_0 .

$$\begin{aligned} \int f(z) dz &= \sum_{-\infty}^{\infty} \int A_n (z - z_0)^n dz \\ &= \int \frac{A_{-2}}{(z - z_0)^2} dz + \int \frac{A_{-1}}{(z - z_0)} dz + \int A_0 dz \\ &\quad + \int A_1 (z - z_0) dz + \dots \end{aligned}$$

If $f(z)$ has singularities at $z_1, z_2, \dots, z_n$ within C , then

$$\begin{aligned} I &= \oint_{C_1} f(z) dz + \oint_{C_2} f(z) dz + \dots + \oint_{C_n} f(z) dz \\ &= I_1 + I_2 + \dots + I_n \\ &= \sum_{j=1}^n I_j = 2\pi i \sum_{j=1}^n A_{-1}^{(j)} \end{aligned}$$

where $A_{-1}^{(j)}$ is the residue of $f(z)$ at z_j .



CHAPTER 5. TWO-DIMENSIONAL POTENTIAL FLOWS

- Classification of Singularities

$f(z)$ has a singular point at $z = z_0$, then

$$\begin{aligned} f(z) &= \sum_{n=-\infty}^{\infty} A_n (z - z_0)^n \\ &= \cdots + \frac{A_{-m}}{(z - z_0)^m} + \frac{A_{-m+1}}{(z - z_0)^{m+1}} + \cdots + A_0 \\ &\quad + A_1 (z - z_0) + A_2 (z - z_0)^2 + \cdots \end{aligned}$$

- ① If Laurent series about z_0 of $f(z)$ terminates at $\frac{1}{(z - z_0)^m}$, i.e. $A_{-m-1} = A_{-m-2} = A_{-m-3} = \cdots = 0$ but $A_{-m} \neq 0$
 $\implies f(z)$ has an m^{th} -order pole at z_0 .
- ② If the Laurent series contains infinite number of negative-power terms
 $\implies$ essential singularity at z_0

Note.

if $m = 1 \longrightarrow$ simple pole

In ①, if $m = 2 \longrightarrow$ double pole.

Example.

For essential singularity:

$f(z) = e^{-1/z^2} = 1 - \frac{1}{z^2} + \frac{1}{2!z^4} - \cdots$, $f(z)$ has an essential singularity at $z = 0$

5.1. BRIEF REVIEW OF COMPLEX THEORY

①

$$\frac{z^2 + 3i}{(z - 2)(z + i)}, \quad z = 2 \longrightarrow \text{simple pole}$$
$$z = -i \longrightarrow \text{simple pole}$$

②

$$\frac{\sin z}{z^2(z^2 + 4)}, \quad z = \pm 2i \longrightarrow \text{simple pole}$$
$$z = 0 \longrightarrow \text{simple pole (removable singularity)}$$

- Calculation of Residues

- If $f(z)$ has a simple pole at $z = z_0$, then

$$A_{-1} = \lim_{z \rightarrow z_0} [(z - z_0) f(z)] \quad (5.15)$$

- If $f(z)$ has a pole of order m at $z = z_0$, then

$$A_{-1} = \lim_{z \rightarrow z_0} \frac{1}{(m-1)!} \frac{d^{m-1}}{dz^{m-1}} [(z - z_0)^m f(z)] \quad (5.16)$$

Example. $f(z) = \frac{\sin z}{z(z - 2i)}$ has a simple pole at $z = 2i$

$$A_{-1} = \lim_{z \rightarrow 2i} \left[(z - 2i) \frac{\sin z}{z(z - 2i)} \right] = \frac{\sin(2i)}{2i}$$

Example. $f(z) = \frac{z}{(z + 1)(z - i)^2}$ has a double-pole at $z = i$

$$A_{-1} = \lim_{z \rightarrow i} \frac{1}{1!} \frac{d}{dz} \left[(z - i)^2 \frac{z}{(z + 1)(z - i)^2} \right]$$
$$= \lim_{z \rightarrow i} \frac{1}{(z + 1)^2} = \frac{1}{(1 + i)^2}$$

CHAPTER 5. TWO-DIMENSIONAL POTENTIAL FLOWS

5.2 (Complex) Potential Flows

- Complex potential

Any analytic function $F(z) = \xi(x, y) + i\eta(x, y)$

$$\text{CRC} \implies \frac{\partial \xi}{\partial x} = \frac{\partial \eta}{\partial y}, \frac{\partial \xi}{\partial y} = -\frac{\partial \eta}{\partial x}$$

$$\implies \begin{cases} \frac{\partial^2 \xi}{\partial x^2} + \frac{\partial^2 \xi}{\partial y^2} = 0 \\ \frac{\partial^2 \eta}{\partial x^2} + \frac{\partial^2 \eta}{\partial y^2} = 0 \end{cases} \quad (*)$$

Consider a 2D irrotational and incompressible flow

$$\text{irrotational: } u = \frac{\partial \Phi}{\partial x}, v = \frac{\partial \Phi}{\partial y}$$

$$\text{incompressible: } u = \frac{\partial \Psi}{\partial y}, v = \frac{\partial \Psi}{\partial x}$$

$$\implies \frac{\partial \Phi}{\partial x} = \frac{\partial \Psi}{\partial y}, \frac{\partial \Phi}{\partial y} = -\frac{\partial \Psi}{\partial x} \implies \begin{cases} \frac{\partial^2 \Phi}{\partial x^2} + \frac{\partial^2 \Phi}{\partial y^2} = 0 \\ \frac{\partial^2 \Psi}{\partial x^2} + \frac{\partial^2 \Psi}{\partial y^2} = 0 \end{cases} \quad (**)$$

<cf> (*) and (**)

Can identify

$$\xi = \Phi, \quad \eta = \Psi \implies F(z) = \Phi(x, y) + i\Psi(x, y) \quad (5.17)$$

where $F(z)$ is Complex Potential Function.

5.2. (COMPLEX) POTENTIAL FLOWS

- Velocity

$$\begin{aligned} W(z) &= \frac{dF}{dz} = \frac{\partial F}{\partial x} = \frac{\partial \Phi}{\partial x} + i \frac{\partial \Psi}{\partial x} = u - iv \\ &= \frac{\partial F}{i \partial y} = \frac{1}{i} \frac{\partial \Phi}{\partial y} + \frac{\partial \Psi}{\partial y} = u - iv \end{aligned}$$

$W(z) = u - iv$ is Complex Velocity

$$W\bar{W} = u^2 + v^2 = |\mathbf{v}|^2$$

5.2.1 Some Elementary Flows

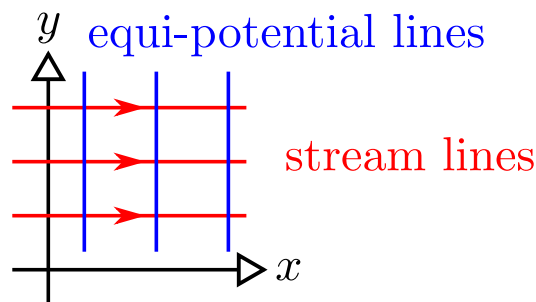
- Uniform flow

$$F(z) = Uz \tag{5.18}$$

where U is real number $U > 0$

$$W(z) = \frac{dF}{dz} = U$$

$$\implies \Phi = Ux, \quad \Psi = Uy$$

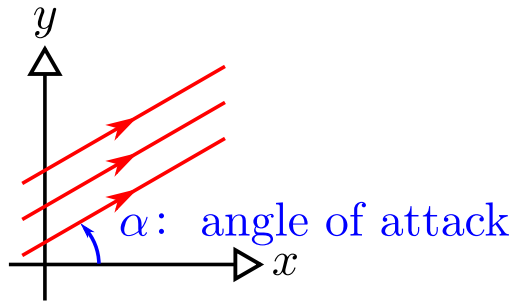


or

$$F(z) = V e^{-i\alpha} z \tag{5.19}$$

CHAPTER 5. TWO-DIMENSIONAL POTENTIAL FLOWS

$$\begin{aligned} W(z) &= \frac{dF}{dz} = V e^{-i\alpha} \\ &= V \cos \alpha \mathbf{e}_x - iV \sin \alpha \mathbf{e}_y \end{aligned}$$



- Flow around a corner of arbitrary angle

$$F(z) = Az^n = A \left(r e^{i\theta} \right)^n = Ar^n e^{in\theta} \quad (5.20)$$

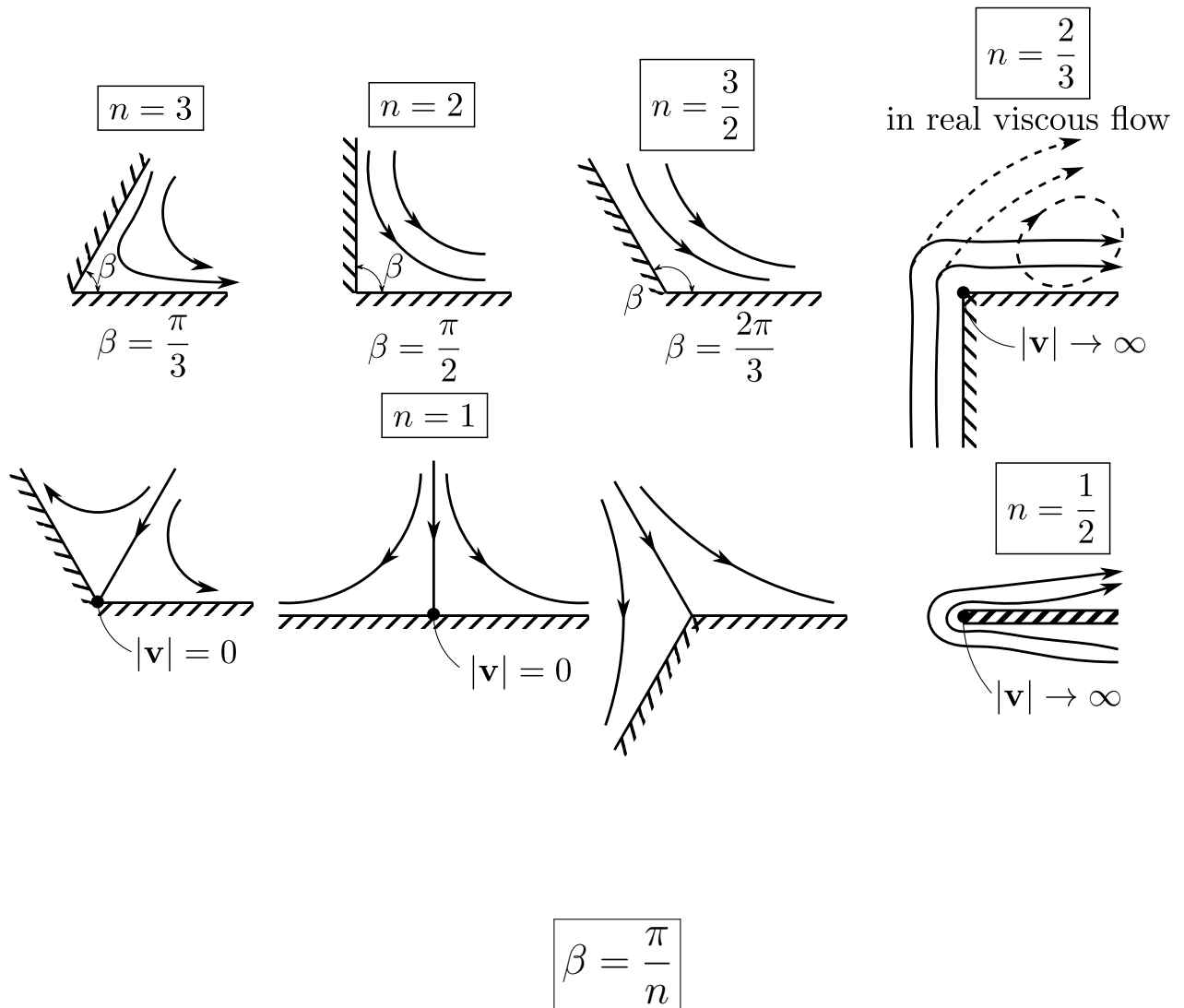
where n is positive and A is real number > 0
i.e.

$$\begin{cases} \Phi = Ar^n \cos n\theta \\ \Psi = Ar^n \sin n\theta \end{cases}$$

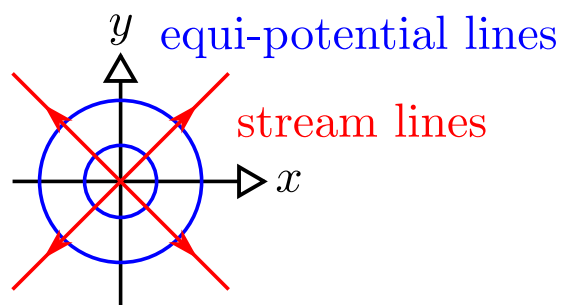
and

$$\begin{cases} v_r = \frac{\partial \Phi}{\partial r} = nAr^{n-1} \cos n\theta & (5.21a) \\ v_\theta = \frac{1}{r} \frac{\partial \Phi}{\partial \theta} = -nAr^{n-1} \sin n\theta & (5.21b) \end{cases}$$

5.2. (COMPLEX) POTENTIAL FLOWS



- Plane (Line) Source/Sink



CHAPTER 5. TWO-DIMENSIONAL POTENTIAL FLOWS

$$\begin{aligned}
 F(z) &= m \ln z \\
 &= m \ln (re^{i\theta}) = m \ln r + im\theta
 \end{aligned}
 \tag{5.22}$$

where m is real number, $m: + \longrightarrow$ source, $m: - \longrightarrow$ sink

$$\begin{cases} \Phi = m \ln r \end{cases} \tag{5.23a}$$

$$\begin{cases} \Psi = m\theta \end{cases} \tag{5.23b}$$

$$\begin{cases} v_r = \frac{m}{r} \end{cases} \tag{5.24a}$$

$$\begin{cases} v_\theta = 0 \end{cases} \tag{5.24b}$$

volume flow rate is

$$Q = \oint_C \mathbf{v} \cdot \mathbf{n} dl = \int_0^{2\pi} v_r r d\theta = \int_0^{2\pi} \frac{m}{r} r d\theta = 2\pi m$$

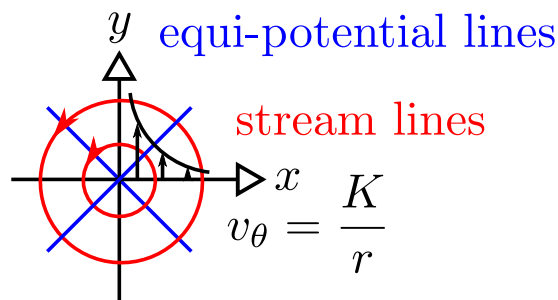
or

$$F(z) = \frac{Q}{2\pi} \ln z \tag{5.25}$$

If the origin of the source/sink is at z_0

$$F(z) = m \ln (z - z_0) = \frac{Q}{2\pi} \ln (z - z_0) \tag{5.26}$$

- Plane (Line) Vortex



5.2. (COMPLEX) POTENTIAL FLOWS

$$\begin{aligned} F(z) &= -iK \ln z \\ &= K\theta - iK \ln r \end{aligned} \quad (5.27)$$

where K is real number.

$$\begin{cases} \Phi = K\theta \\ \Psi = -K \ln r \end{cases} \quad \begin{matrix} (5.28a) \\ (5.28b) \end{matrix}$$

$$\begin{cases} v_r = 0 \\ v_\theta = \frac{K}{r} \end{cases} \quad \begin{matrix} (5.29a) \\ (5.29b) \end{matrix}$$

Circulation is

$$\Gamma = \oint_C \mathbf{v} \cdot \mathbf{t} dl = \int_0^{2\pi} v_\theta r d\theta = 2\pi K \neq 0$$

or

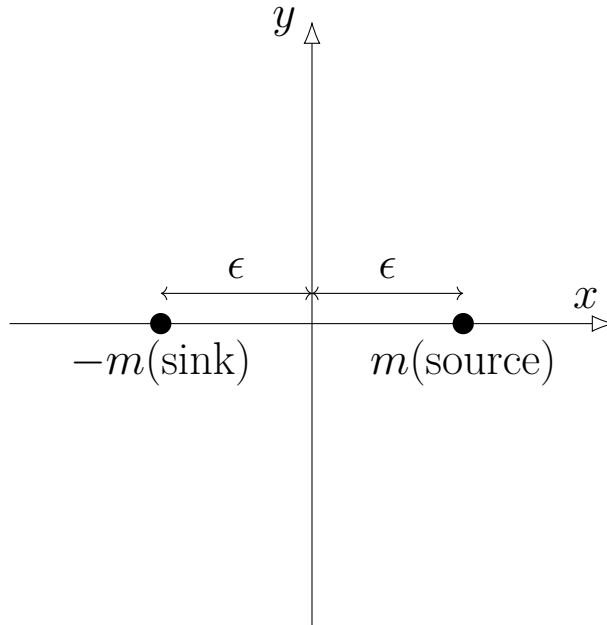
$$F(z) = -i \frac{\Gamma}{2\pi} \ln z \quad (5.30)$$

If the origin of the vortex is at z_0

$$F(z) = -iK \ln(z - z_0) = -i \frac{\Gamma}{2\pi} \ln(z - z_0) \quad (5.31)$$

CHAPTER 5. TWO-DIMENSIONAL POTENTIAL FLOWS

- Plane (Line) Doublet (Dipole)



$$\begin{aligned}
 F(z) &= m \ln(z + \epsilon) - m \ln(z - \epsilon) \\
 &= m \ln \frac{1 + \epsilon/z}{1 - \epsilon/z} \\
 &= m \ln \left[1 + 2\frac{\epsilon}{z} + \mathcal{O}\left(\frac{\epsilon}{z}\right)^2 \right] \\
 &\cong m \left(2\frac{\epsilon}{z} \right)
 \end{aligned}$$

Let $\lim_{\epsilon \rightarrow 0} (2\epsilon m) = \mu$ where μ is real number

$$\begin{aligned}
 \therefore F(z) &= \frac{\mu}{z} \\
 &= \frac{\mu}{re^{i\theta}} = \frac{\mu}{r} \cos \theta - i \frac{\mu}{r} \sin \theta
 \end{aligned} \tag{5.32}$$

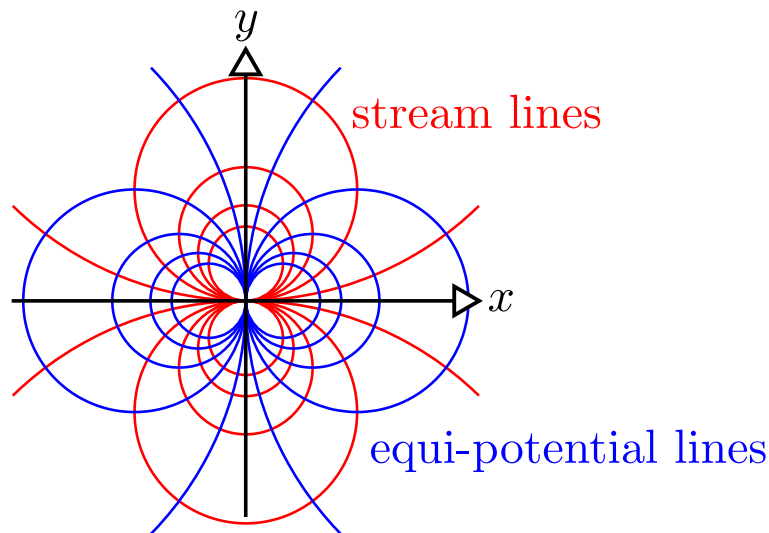
5.2. (COMPLEX) POTENTIAL FLOWS

$$\begin{cases} \Phi = \frac{\mu}{r} \cos \theta \\ \Psi = -\frac{\mu}{r} \sin \theta \end{cases} \quad \begin{matrix} (5.33a) \\ (5.33b) \end{matrix}$$

$$\begin{cases} v_r = -\frac{\mu}{r^2} \cos \theta \\ v_\theta = -\frac{\mu}{r^2} \sin \theta \end{cases} \quad \begin{matrix} (5.34a) \\ (5.34b) \end{matrix}$$

If origin of doublet is at $z - z_0$, then

$$F(z) = \frac{\mu}{z - z_0} \quad (5.35)$$

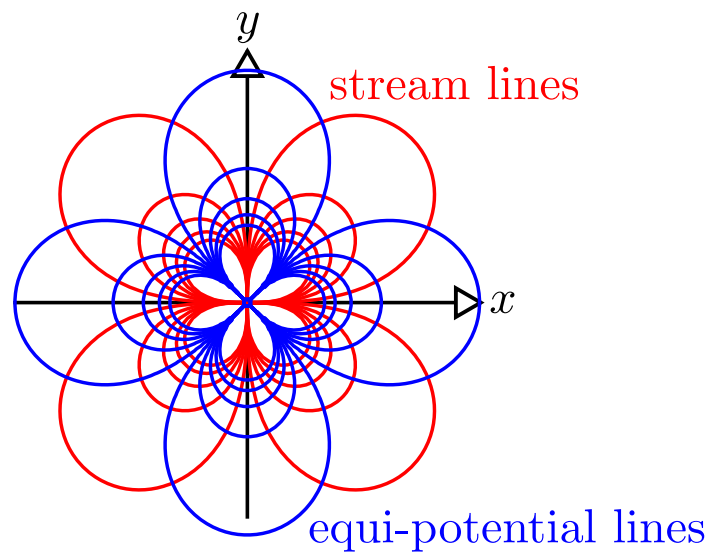


Note.

$$\textcircled{1} \quad \frac{d}{dz} \underbrace{(m \ln z)}_{=\text{source/sink}} = \underbrace{\frac{m}{z}}_{=\text{doublet}}$$

CHAPTER 5. TWO-DIMENSIONAL POTENTIAL FLOWS

$$\textcircled{2} \quad \underbrace{\frac{d}{dz} \left(\frac{m}{z} \right)}_{=\text{doublet}} = \underbrace{-\frac{m}{z^2}}_{=\text{quadruplet}}$$



$$\textcircled{3} \quad \frac{1}{z^2}, \frac{1}{z^3}, \frac{1}{z^4}, \dots : \text{higher-order singularities}$$

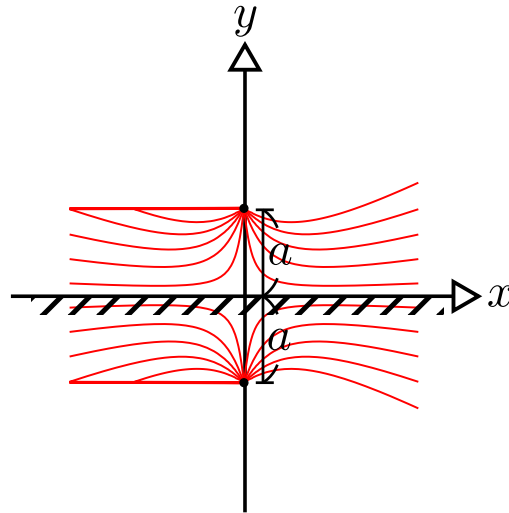
5.2.2 Images-Superposition

- A singularities (source/sink, vortex, doublet, etc...) near an infinite plane wall

– source/sink

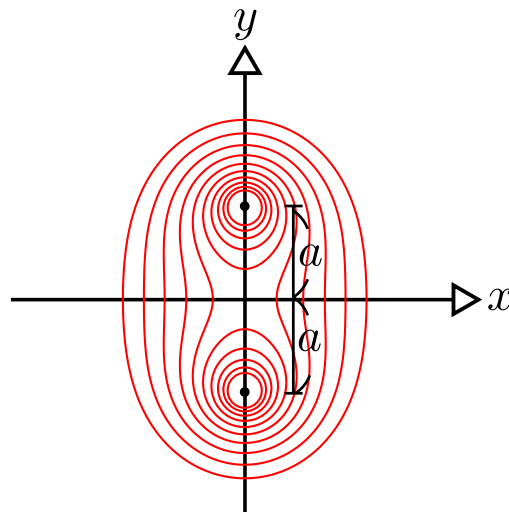
$$\begin{aligned} F(z) &= m \ln(z - ia) + m \ln(z + ia) \\ &= m \ln(z^2 + a^2) \end{aligned}$$

5.2. (COMPLEX) POTENTIAL FLOWS



– vortex

$$\begin{aligned} F(z) &= -\frac{i\Gamma}{2\pi} \ln(z - ia) + \frac{i\Gamma}{2\pi} \ln(z + ia) \\ &= \frac{i\Gamma}{2\pi} \ln\left(\frac{z + ia}{z - ia}\right) \end{aligned}$$

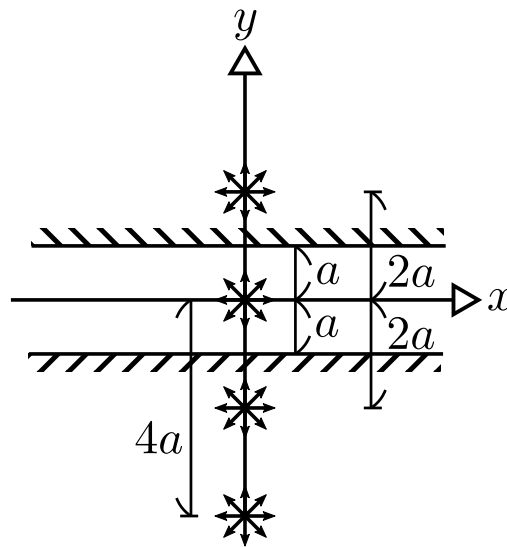


- A singularity confined between two infinite plane walls

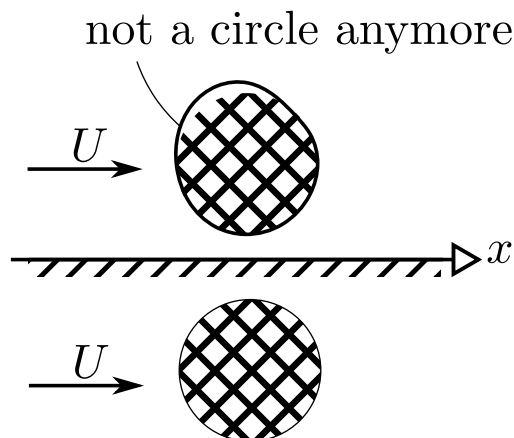
CHAPTER 5. TWO-DIMENSIONAL POTENTIAL FLOWS

An infinite row of images of source

$$F(z) = m \sum_{n=-\infty}^{\infty} \ln(z + i2na)$$



- Distortion due to image



ref: A. B. Basset. On the motion of two spheres in a

5.2. (COMPLEX) POTENTIAL FLOWS

liquid, and allied problems. Proceedings of the London Mathematical Society

5.2.3 Superposition-Flow around a circular cylinder with/without circulation

- without circulation
Uniform stream + doublet:

$$F(z) = Uz + \frac{\mu}{z}$$

require $r = a$ a streamline, then

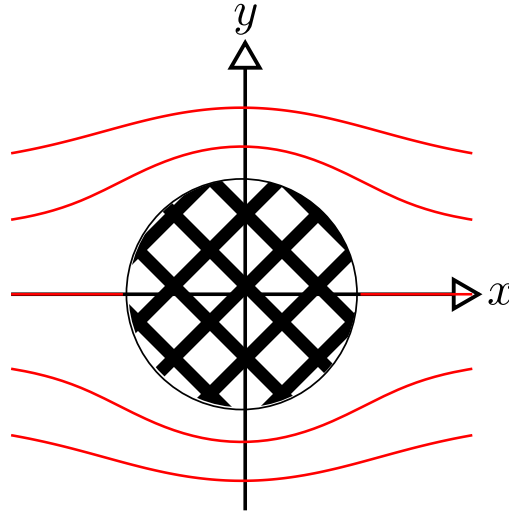
$$F(ae^{i\theta}) = \left(Ua + \frac{\mu}{a}\right) \cos \theta + i \left(Ua - \frac{\mu}{a}\right) \sin \theta$$

$$\Psi = \left(Ua - \frac{\mu}{a}\right) \sin \theta \implies \Psi = 0 \implies \mu = Ua^2$$

thus

$$F(z) = U \left(z + \frac{a^2}{z} \right) \tag{5.36}$$

CHAPTER 5. TWO-DIMENSIONAL POTENTIAL FLOWS



- with circulation Γ

Can add a CW free vortex of strength Γ to the above $F(z)$, i.e. uniform flow + doublet + vortex:

$$F(z) = U \left(z + \frac{a^2}{z} \right) + \frac{i\Gamma}{2\pi} \ln z + C$$

require $r = a$ a streamline, then

$$\begin{aligned} F(ae^{i\theta}) &= U \left(ae^{i\theta} + ae^{-i\theta} \right) + \frac{i\Gamma}{2\pi} \ln \left(ae^{i\theta} \right) + C \\ &= 2Ua \cos \theta - \frac{\Gamma\theta}{2\pi} + \frac{i\Gamma}{2\pi} \ln a + C \end{aligned}$$

choose $C = -\frac{i\Gamma}{2\pi} \ln a$, i.e.

$$F(z) = U \left(z + \frac{a^2}{z} \right) + \frac{i\Gamma}{2\pi} \ln \frac{z}{a} \quad (5.37)$$

5.2. (COMPLEX) POTENTIAL FLOWS

$$\left\{ \begin{array}{l} \Phi = U \cos \theta \left(r - \frac{a^2}{r} \right) - \frac{\Gamma \theta}{2\pi} \end{array} \right. \quad (5.38a)$$

$$\left\{ \begin{array}{l} \Psi = U \sin \theta \left(r - \frac{a^2}{r} \right) + \frac{\Gamma}{2\pi} \ln \frac{r}{a} \end{array} \right. \quad (5.38b)$$

$$\left\{ \begin{array}{l} v_r = U \cos \theta \left(1 - \frac{a^2}{r^2} \right) \end{array} \right. \quad (5.39a)$$

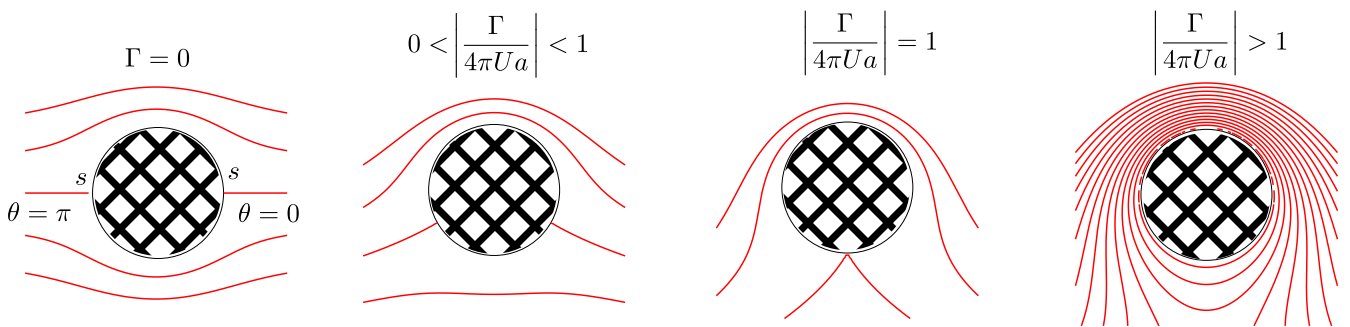
$$\left\{ \begin{array}{l} v_\theta = -U \sin \theta \left(1 + \frac{a^2}{r^2} \right) - \frac{\Gamma}{2\pi r} \end{array} \right. \quad (5.39b)$$

On the surface of the circular cylinder ($r = a$)

$$\Rightarrow \left\{ \begin{array}{l} v_r = 0 \\ v_\theta = -2U \sin \theta - \frac{\Gamma}{2\pi a} \end{array} \right.$$

Stagnation points

$$\Rightarrow \left\{ \begin{array}{l} v_r = 0 \\ v_\theta = 0 \end{array} \right. \Rightarrow \sin \theta_s = -\frac{\Gamma}{4\pi U a} \quad (5.40)$$



CHAPTER 5. TWO-DIMENSIONAL POTENTIAL FLOWS

5.2.4 The Circle Theorem (First)

- Let $F(z)$ be the complex potential of a potential flow in the absence of any rigid boundaries and let all singularities of $F(z)$ be outside the circle $|z| = a$.

If a circular cylinder $|z| = a$ is introduced into the base flow $F(z)$, then the resulting complex potential $f(z)$ is given by

$$f(z) = F(z) + \overline{F}\left(\frac{a^2}{z}\right) \text{ (circle theorem)} \quad (5.41)$$

Proof.

- ① All singularities in $F(z)$ will lie inside the circle $|z| = a$ in $\overline{F}\left(\frac{a^2}{z}\right)$
- ② If z is on the circle $|z| = a$, then $a^2/z = \bar{z}$, then

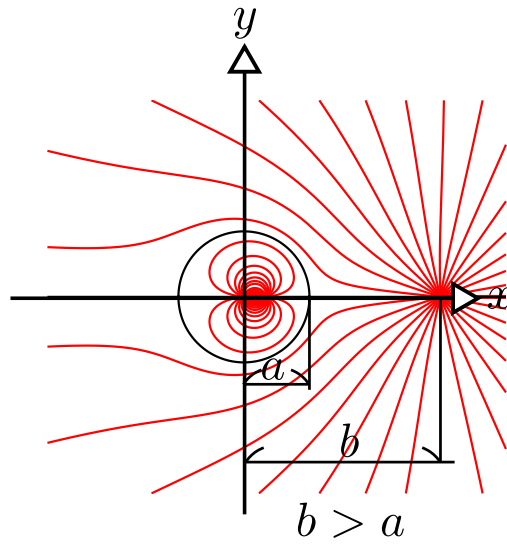
$$f(z) = F(z) + \overline{F}(\bar{z}) = \text{real function}$$

$$\implies \Psi = 0 \text{ on } |z| = a, \text{ i.e. } |z| = a \text{ is a streamline.}$$

□

Example.

5.2. (COMPLEX) POTENTIAL FLOWS



$$F(z) = m \ln(z - b)$$

$$\Rightarrow f(z) = m \ln(z - b) + m \ln\left(\frac{a^2}{z} - b\right)$$

CHAPTER 5. TWO-DIMENSIONAL POTENTIAL FLOWS

5.2.5 Uniform Flow Past a Cylinder of Arbitrary Cross-Section

- In general

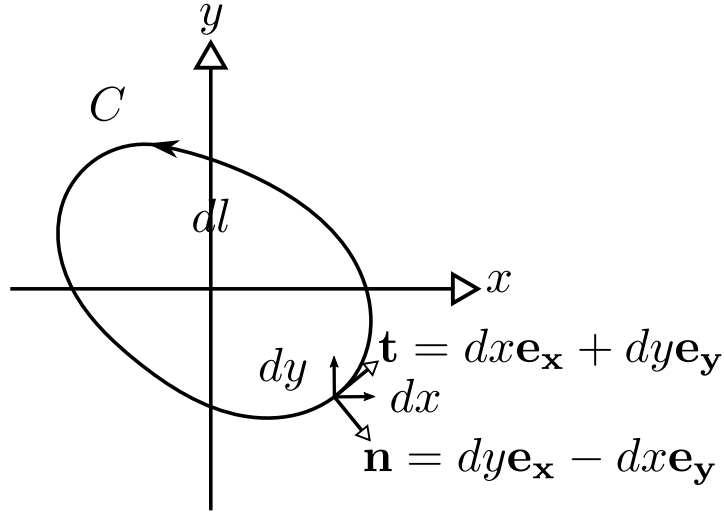
$$\begin{aligned}
 u - iv \equiv W(z) &\equiv \frac{dF(z)}{dz} = A_0 + \frac{A_1}{z} + \frac{A_2}{z^2} + \cdots \quad (\text{Laurent Series}) \\
 &= \sum_{n=0}^{\infty} \frac{A_n}{z^n}
 \end{aligned}
 \tag{5.42}$$

$$\begin{aligned}
 \Rightarrow F(z) &= \underbrace{A_0 z}_{\text{uniform stream}} + \underbrace{A_1 \ln z}_{\substack{\text{source/sink} \\ \text{or} \\ \text{vortex}}} - \sum_{n=2}^{\infty} \frac{A_n}{n-1} \frac{1}{z^{n-1}} + \text{const}
 \end{aligned}
 \tag{5.43}$$

An $n \geq 2$ determined by geometry

- About A_0 and A_1 in (5.43)

5.2. (COMPLEX) POTENTIAL FLOWS



Note that

$$\Gamma = \oint_C \mathbf{v} \cdot \mathbf{t} dl = \oint_C (u dx + v dy) = \oint_C d\Phi$$

$$Q = \oint_C \mathbf{v} \cdot \mathbf{n} dl = \oint_C (u dy - v dx) = \oint_C d\Psi$$

Hence

$$\begin{aligned} \Gamma + iQ &= \oint_C (d\Phi + i d\Psi) \\ &= \oint_C dF = \oint_C W(z) dz \end{aligned} \quad (5.44)$$

In our present case, uniform free stream U at $z \rightarrow \infty$.

Hence $A_0 = U$ and

$$\begin{aligned} \Gamma + iQ &= \oint_C W(z) dz \\ &= 0 \text{ if } C \text{ does not enclose the cylinder} \\ &= 2\pi i A_1 \text{ for all circuits enclosing the cylinder} \end{aligned}$$

CHAPTER 5. TWO-DIMENSIONAL POTENTIAL FLOWS

Let $A_1 = a_1 + ib_1$

$$\Rightarrow \begin{cases} \Gamma = -2\pi b_1 \Rightarrow b_1 = -\frac{\Gamma}{2\pi} \\ Q = 0 = 2\pi i a_1 \Rightarrow a_1 = 0 \end{cases}$$

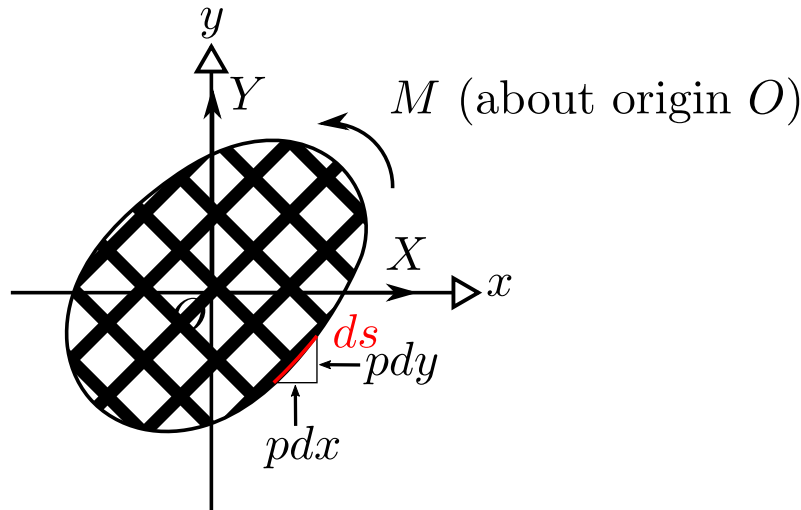
for cylinder

$$W(z) = U - \frac{i\Gamma}{2\pi z} + \sum_{n=2}^{\infty} \frac{A_n}{z^n} \quad (5.45)$$

$$F(z) = Uz - \frac{i\Gamma}{2\pi} \ln z + \sum_{n=2}^{\infty} \frac{A_n}{n-1} \frac{1}{z^{n-1}} + \mathcal{O}^0 \quad (5.46)$$

5.2.6 Force and Moment on a Cylinder

- Blasius Theorem



$$dX = -pdy, \quad dY = pdx, \quad dM = p(xdx + ydy)$$

5.2. (COMPLEX) POTENTIAL FLOWS

pressure: $p = p_\infty - \frac{\rho}{2}|\mathbf{v}|^2$ can set $p_\infty = 0$ without loss of generality.

Thus

$$d(X - iY) = \frac{\rho}{2}|\mathbf{v}|^2(dy + idx), \quad dM = -\frac{\rho}{2}|\mathbf{v}|^2(xdx + ydy)$$

Now

$$dy + idx = id\bar{z}, \quad xdx + ydy = \Re\{z d\bar{z}\}$$

and

$$|\mathbf{v}|^2 = W\overline{W} = \frac{dF}{dz} \frac{d\overline{F}}{d\bar{z}}$$

Thus

$$d(X - iY) = \frac{\rho}{2}i \frac{dF}{dz} d\overline{F}, \quad dM = \Re\left\{-\frac{\rho}{2}z \frac{dF}{dz} d\overline{F}\right\} \text{ along } C$$

On the cylinder, $\Psi = \text{constant}$, $d\Psi = 0$

therefore

$$d\overline{F} = d(\Phi - i\Psi) = d(\Phi + i\Psi) = dF = \frac{dF}{dz} dz$$

along the cylinder surface

hence

$$d(X - iY) = \frac{\rho}{2}i \left(\frac{dF}{dz}\right)^2 dz, \quad dM = \Re\left\{-\frac{\rho}{2}z \left(\frac{dF}{dz}\right)^2 dz\right\} \text{ along } C$$

CHAPTER 5. TWO-DIMENSIONAL POTENTIAL FLOWS

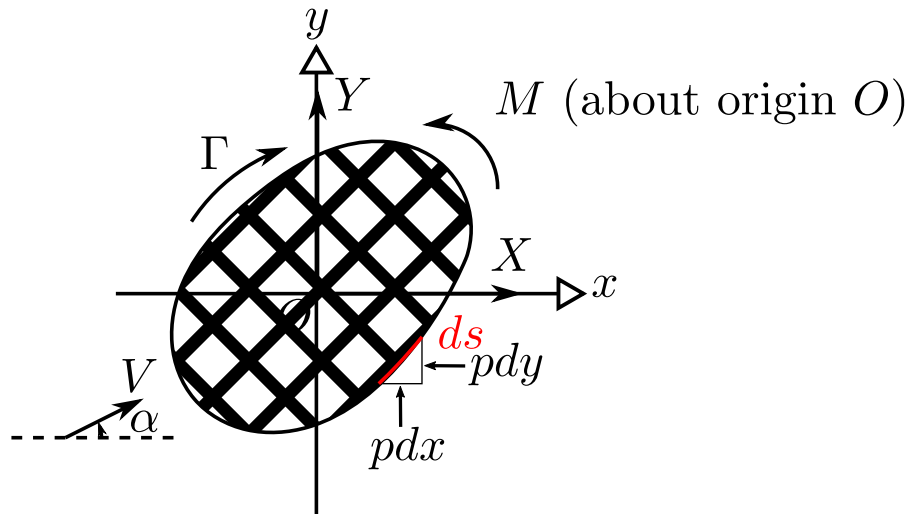
i.e.

$$\left\{ \begin{array}{l} X - iY = \frac{i\rho}{2} \oint_C \left(\frac{dF}{dz} \right)^2 dz \end{array} \right. \quad (5.47a)$$

$$\left\{ \begin{array}{l} M = \Re \left\{ -\frac{\rho}{2} \oint_C z \left(\frac{dF}{dz} \right)^2 dz \right\} \end{array} \right. \quad (5.47b)$$

C could be any closed curve enclosing the cylinder

5.2.7 Kutta-Joukowski Theorem



- Consider a cylinder (of arbitrary cross-section) placed in a uniform stream of speed V at an angle-of-attack α with CW circulation of strength Γ .

$$F(z) = V e^{-i\alpha} z + \frac{i\Gamma}{2\pi} \ln z - \sum_{n=2}^{\infty} \frac{A_n}{n-1} \frac{1}{z^{n-1}} \quad (5.48)$$

5.2. (COMPLEX) POTENTIAL FLOWS

$$\Rightarrow W(z) = \frac{dF}{dz} = Ve^{-i\alpha} + \frac{i\Gamma}{2\pi z} + \sum_{n=2}^{\infty} \frac{A_n}{z^n}$$

$$\therefore W^2 = \underbrace{V^2 e^{-2i\alpha}}_{B_0} + \underbrace{\frac{i\Gamma V e^{-i\alpha}}{\pi}}_{B_{-1}} \frac{1}{z} + \underbrace{\frac{-\Gamma^2 + 8\pi^2 A_2 V e^{-i\alpha}}{4\pi^2}}_{B_{-2}} \frac{1}{z^2} + \dots$$

From Blasius Theorem

$$\begin{aligned} X - iY &= \frac{i\rho}{2} \oint_C W^2(z) dz = \frac{i\rho}{2} 2\pi i [B_{-1}] \\ &= \frac{i\rho}{2} 2\pi i \left(\frac{i\Gamma V e^{-i\alpha}}{\pi} \right) \\ &= -i\rho V e^{-i\alpha} \Gamma \end{aligned} \quad (5.49)$$

or

$$X + iY = \overline{X - iY} = i\rho V e^{-i\alpha} \Gamma \quad (5.50)$$

In Aerodynamic:

Lift force L is $\perp$ to free stream
 Drag force L is $\parallel$ to free stream

$$D + iL = (X + iY) e^{-i\alpha} = i\rho V \Gamma \quad (5.51)$$

$$\left\{ \begin{array}{l} L = \rho V \Gamma \text{ (Kutta-Joukowski Theorem)} \end{array} \right. \quad (5.52a)$$

$$\left\{ \begin{array}{l} D = 0 \text{ (d'Alembert paradox)} \end{array} \right. \quad (5.52b)$$

CHAPTER 5. TWO-DIMENSIONAL POTENTIAL FLOWS

- Moment on the cylinder

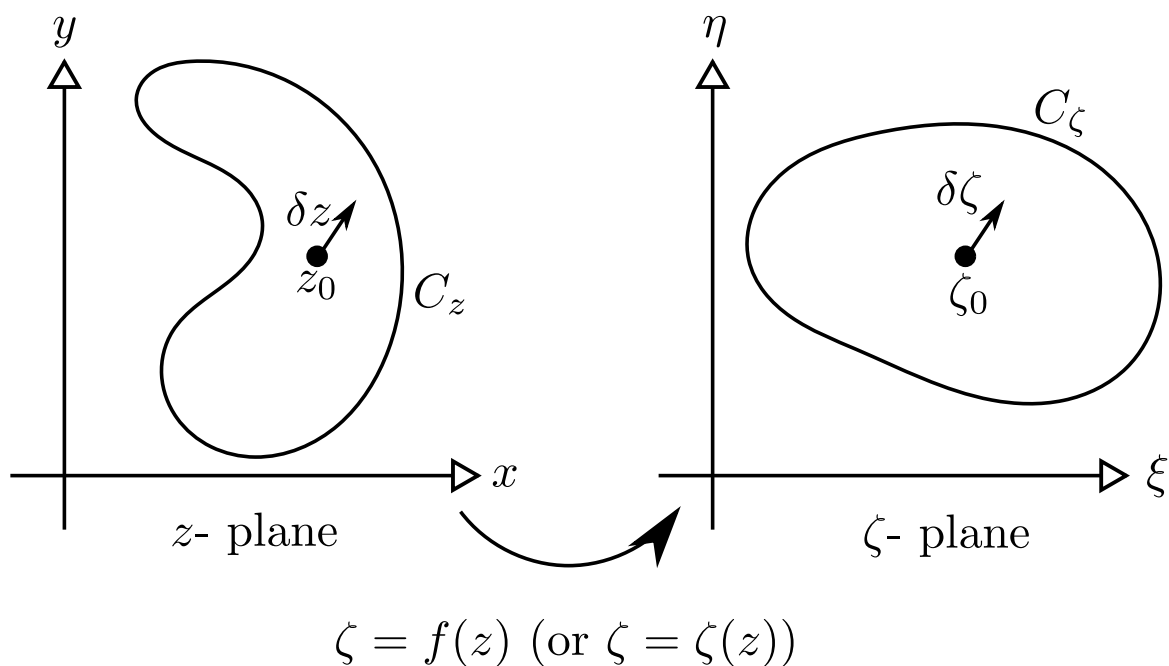
$$\begin{aligned} M &= \Re \left\{ -\frac{\rho}{2} \oint_C z W^2(z) dz \right\} = -\frac{\rho}{2} \Re \{ 2\pi i B_{-2} \} = \dots \\ &= -2\pi \rho V \Re \{ i A_2 e^{-i\alpha} \} \end{aligned}$$

5.3 Potential Flows Generated by Transformations

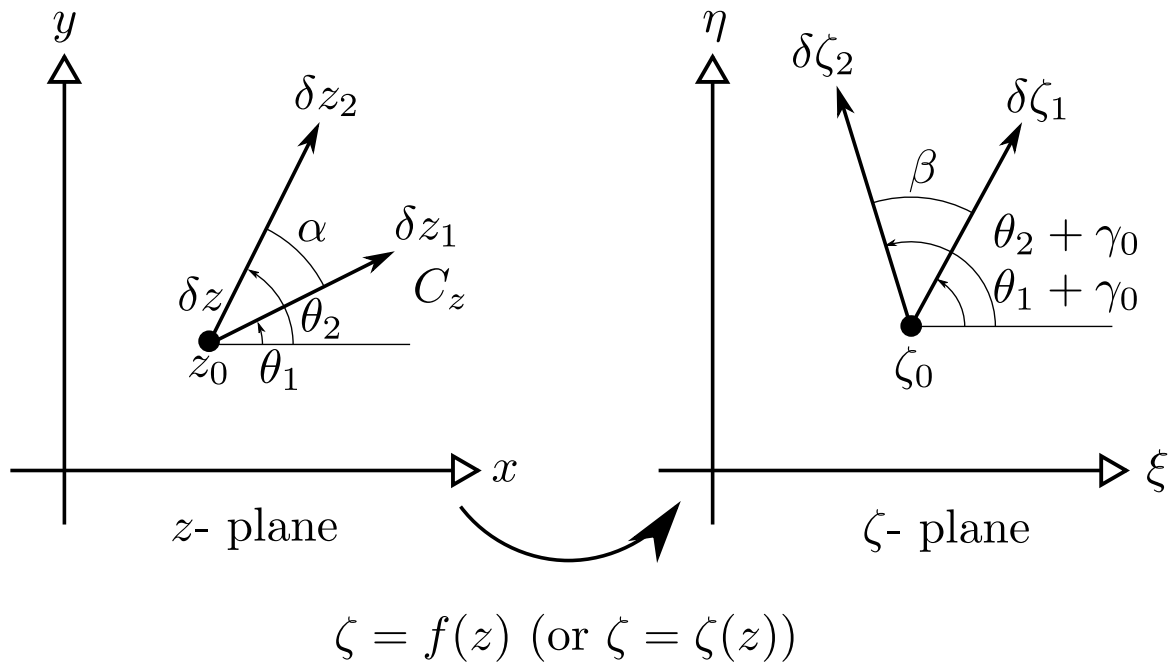
5.3.1 Conformal Transform (Mapping)

General Properties

- Transformation (mapping) by analytic functions



5.3. POTENTIAL FLOWS GENERATED BY TRANSFORMATIONS



Assume $f(z)$ is analytic function and $\frac{d\zeta}{dz} \neq 0$, then

$$\delta\zeta = \left(\frac{d\zeta}{dz}\right)_{z_0} \delta z \implies |\delta\zeta| = a_0 |\delta z|, \arg(\delta\zeta) = \gamma_0 + \arg(\delta z)$$

$$\beta = (\theta_2 + \gamma_0) - (\theta_1 + \gamma_0) = \theta_2 - \theta_1 = \alpha$$

angle is preserved under mapping $\implies$ Conformal mapping

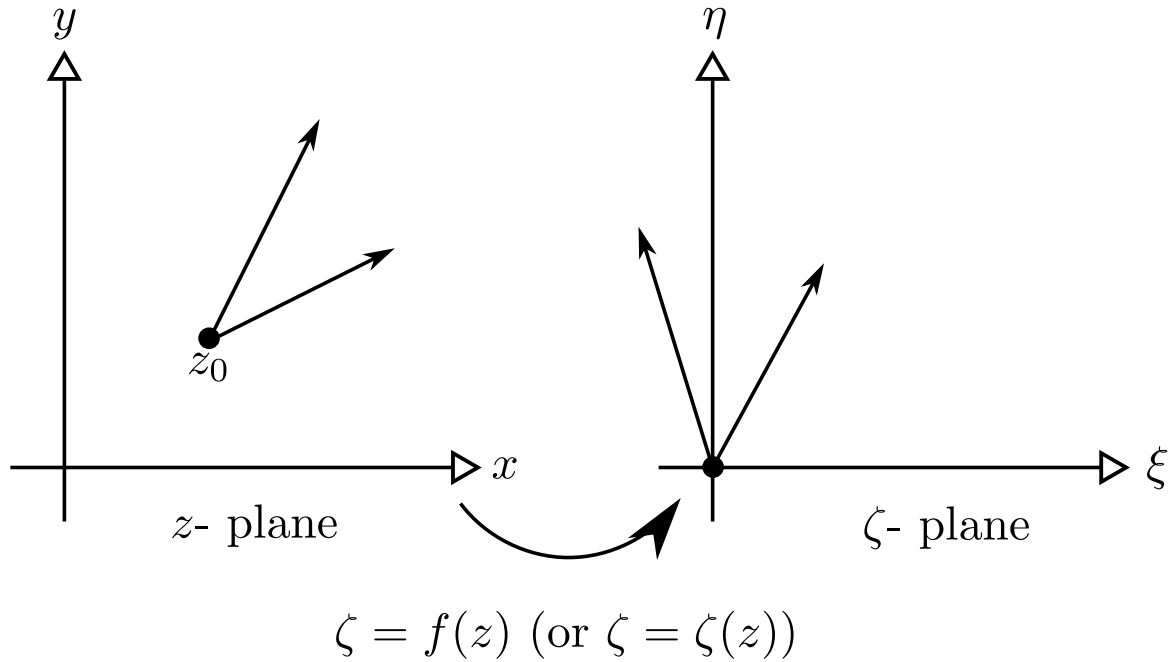
What if $\boxed{\frac{d\zeta}{dz} = 0}$ (critical point)

- Analysis of the critical point of a transformation
Consider the transformation of form:

$$\zeta = (z - z_0)^n \quad (n > 1, n: \text{ integer}), \quad \frac{d\zeta}{dz} = n(z - z_0)^{n-1}$$

hence equal to zero at $z = z_0$

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Let $z - z_0 = re^{i\theta}$, $\zeta = \rho e^{i\phi}$, then

$$\begin{aligned}\zeta &= (z - z_0)^n \implies \rho e^{i\phi} = r^n e^{in\theta} \\ &\implies \rho = r^n, \phi = n\theta \implies \beta = n\alpha\end{aligned}$$

angle is not preserved at critical point under mapping.

Now consider the general transformation $\zeta = \zeta(z)$ and z_0 is a critical point

$$\zeta(z) = \zeta(z_0) + a_1(z - z_0) + a_2(z - z_0)^2 + \cdots$$

where

$$a_k = \frac{1}{k!} \left(\frac{d^k \zeta}{dz^k} \right)_{z=z_0}$$

Let us suppose the first $(n - 1)$ derivatives are zero, i.e.

$$a_1 = a_2 = a_3 = \cdots = a_{n-1} = 0, a_n \neq 0$$

5.3. POTENTIAL FLOWS GENERATED BY TRANSFORMATIONS

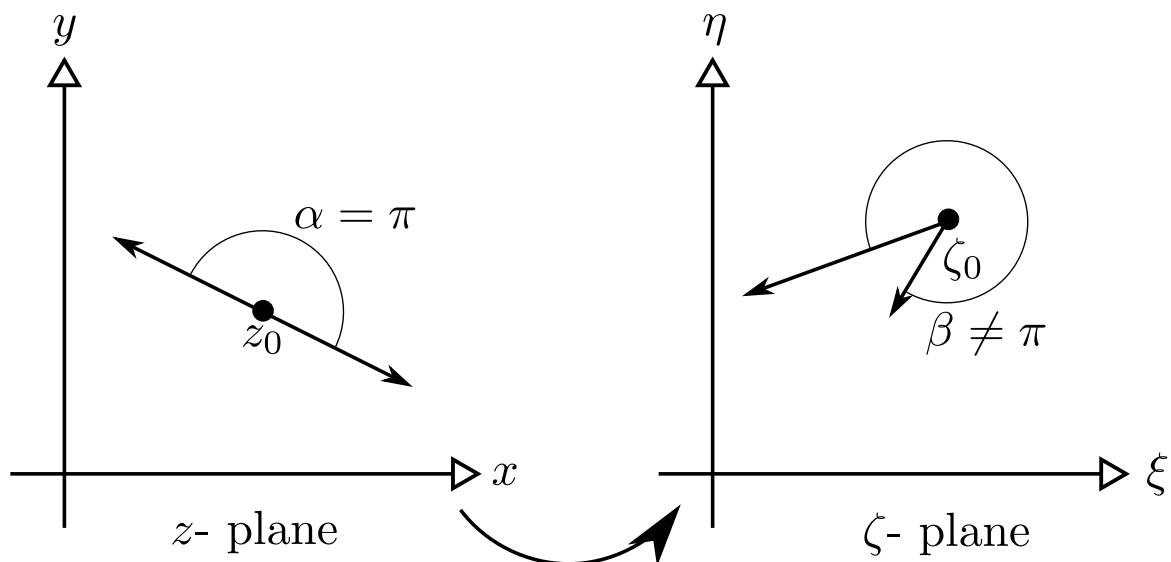
then

$$\zeta = \zeta_0 + a_n (z - z_0)^n \overbrace{\left[1 + \frac{a_{n+1}}{a_n} (z - z_0) + \frac{a_{n+2}}{a_n} (z - z_0)^2 + \cdots \right]}^{f(z-z_0)}$$

$$\implies \zeta - \zeta_0 = (z - z_0)^n f(z - z_0) \text{ as } z \rightarrow z_0,$$

$$f(z - z_0) \approx 1 + O(z - z_0)$$

Example.



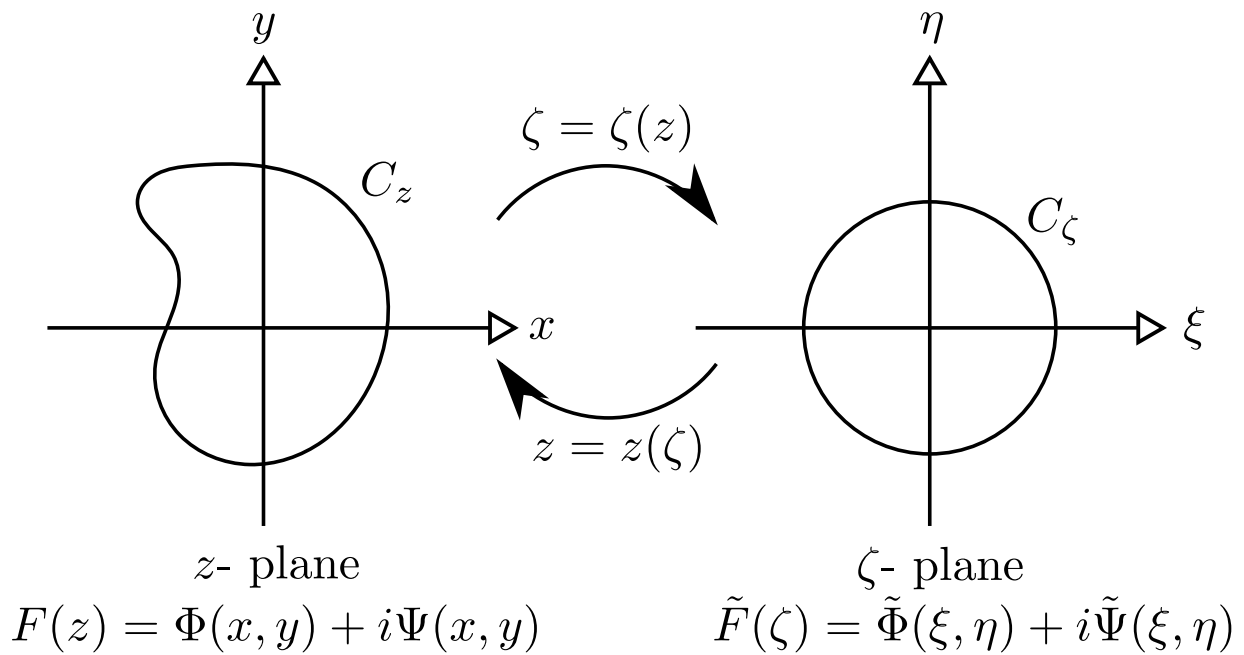
$$\zeta = f(z) \text{ (or } \zeta = \zeta(z))$$

smooth curve will map to a sharp corner at critical point.

- Application to the Potential Flow
 - (a) Consider a potential flow over certain configuration C_z in a z -plane with a complex potential

CHAPTER 5. TWO-DIMENSIONAL POTENTIAL FLOWS

$F(z) = \Phi(x, y) + i\Psi(x, y)$, suppose a conformal transformation $\zeta = \zeta(z)$, $\zeta = \xi + i\eta$, $z = x + iy$, which maps C_z on z - plane into a relatively simple configuration C_ζ (e.g. a circle) on ζ - plane.



$$\tilde{F}(\zeta) = F(z(\zeta)) \quad (5.53)$$

Now

$$\frac{d\tilde{F}}{d\zeta} = \frac{dF}{dz} \frac{dz}{d\zeta} = \frac{dF}{dz} / \frac{d\zeta}{dz}$$

exists at all points where $d\zeta/dz \neq 0$

$\implies \tilde{F}(\zeta)$ is an analytic function except at critical point of transformation.

5.3. POTENTIAL FLOWS GENERATED BY TRANSFORMATIONS

(b) The complex velocity on ζ - plane

$$\tilde{W}(\zeta) = \frac{d\tilde{F}}{d\zeta} = \frac{dF}{dz} / \frac{d\zeta}{dz} = W(z) / \frac{d\zeta}{dz} \quad (5.54)$$

$\tilde{W}(\zeta) \rightarrow \infty$ at the critical point $\frac{d\zeta}{dz} = 0$

(c) Transformation of uniform condition at ∞

$$\frac{dz}{d\zeta} \rightarrow 1 \quad \text{as} \quad \zeta \rightarrow \infty$$

i.e.

$$z \longleftrightarrow \zeta \quad \text{as} \quad \zeta \rightarrow \infty$$

hence the form of the transformation

$$z(\zeta) = \zeta + \frac{c_1}{\zeta} + \frac{c_2}{\zeta^2} + \cdots + = \zeta + \sum_{n=1}^{\infty} \frac{c_n}{\zeta^n}$$

(d)

$$\begin{aligned} \Gamma_{\zeta} + iQ_{\zeta} &= \oint_{C_{\zeta}} \overline{W}(\zeta) d\zeta = \oint_{C_{\zeta}} \frac{d\overline{F}}{d\zeta} d\zeta \\ &= \oint_{C_z} \frac{dF}{dz} \frac{dz}{d\zeta} d\zeta \\ &= \oint_{C_z} W(z) dz = \Gamma_z + iQ_z \end{aligned}$$

Based on (a)~(d)

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Example.

Transformation of a uniform stream past a circular cylinder on ζ - plane into that past an arbitrary cylinder on z - plane

$$z = \zeta + \frac{c_1}{\zeta} + \frac{c_2}{\zeta^2} + \cdots \quad (5.55)$$

an from Blasius Theorem

$$\begin{aligned} X - iY &= \frac{i\rho}{2} \oint_{C_z} W^2(z) dz \\ &= \frac{i\rho}{2} \oint_{C_\zeta} W^2(\zeta) \left(\frac{d\zeta}{dz} \right)^2 dz \\ &= \frac{i\rho}{2} \oint_{C_\zeta} W^2(\zeta) \frac{1}{dz/d\zeta} d\zeta \end{aligned} \quad (5.56)$$

$$\begin{aligned} M &= -\frac{\rho}{2} \Re \left\{ \oint_{C_z} z W^2(z) dz \right\} \\ &= -\frac{\rho}{2} \Re \left\{ \oint_{C_\zeta} z(\zeta) W^2(\zeta) \frac{1}{dz/d\zeta} d\zeta \right\} \end{aligned} \quad (5.57)$$

5.3.2 Joukowski Transformation

- General properties of J-T

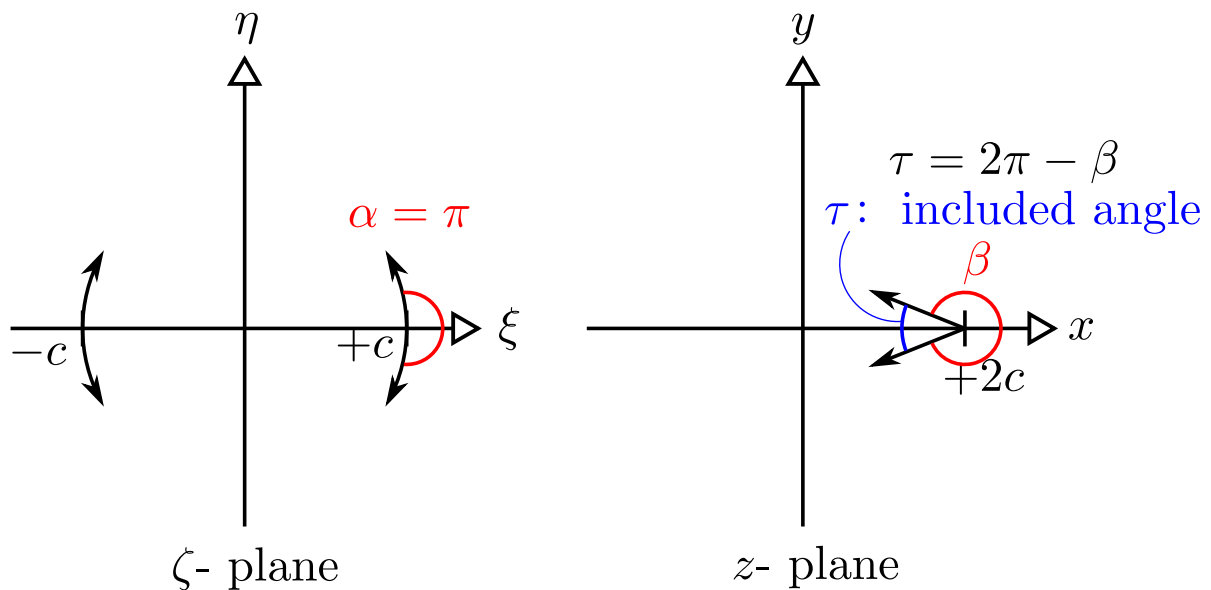
$$z = \zeta + \frac{c^2}{\zeta} \quad (\text{J-T}) \quad (5.58)$$

$$\frac{dz}{d\zeta} = 1 - \frac{c^2}{\zeta^2} \quad (5.59)$$

5.3. POTENTIAL FLOWS GENERATED BY TRANSFORMATIONS

As $\zeta \rightarrow \infty$, $\frac{dz}{d\zeta} \rightarrow 1$, i.e. $z \rightarrow \zeta$

A singular point at $\zeta = 0$; Two critical points at $\zeta = \pm c$



Previously, we have $\boxed{\beta = n\alpha}$

i.e. $2\pi - \tau = n\alpha \implies \beta = n\alpha \implies \tau = 2\pi - n\alpha$ where n is the order of derivatives, i.e., $\frac{d^n z}{d\zeta^n}$ that first becomes non-zero.

For J-T, $n \equiv 2$, $\therefore \tau = 2\pi - n\alpha = 0$

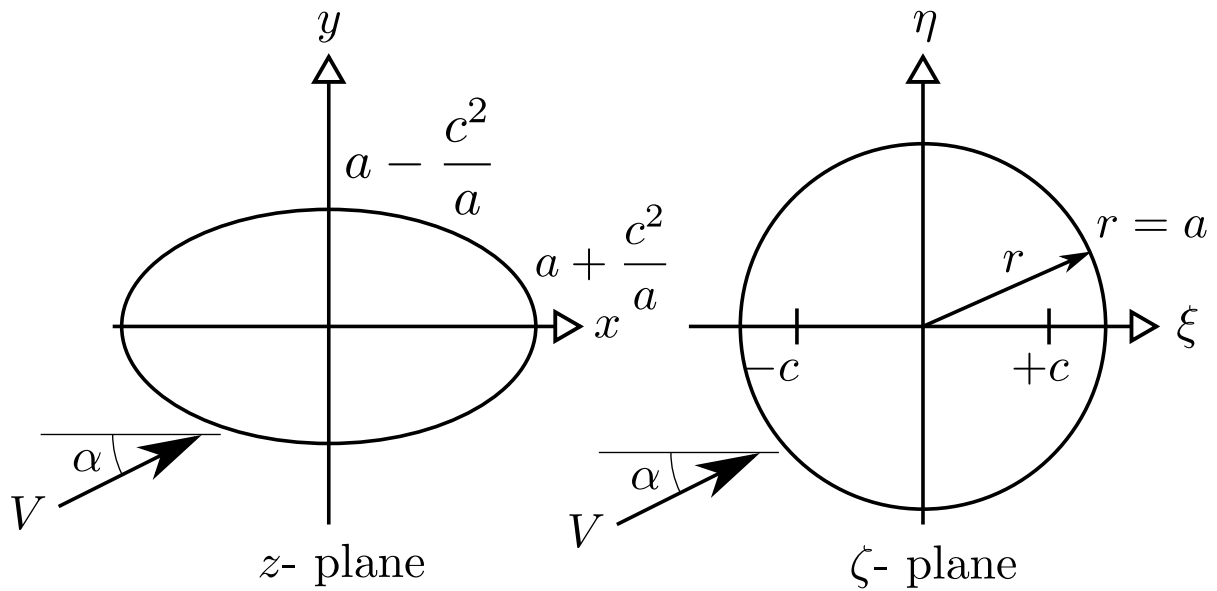
$\tau = 0$
cusps

5.3.3 Families of J-T

(a) Flow around elliptic cylinder

Consider a circle of radius a ($a > c$), centred at origin on ζ - plane

CHAPTER 5. TWO-DIMENSIONAL POTENTIAL FLOWS



J-T

$$z = \zeta + \frac{c^2}{\zeta}, \quad \zeta = ae^{i\theta}$$

$$\begin{aligned} \Rightarrow z &= ae^{i\theta} + \frac{c^2}{ae^{i\theta}} \\ &= \left(a + \frac{c^2}{a}\right) \cos \theta + i \left(a - \frac{c^2}{a}\right) \sin \theta \\ &= x + iy \end{aligned}$$

$$\Rightarrow \begin{cases} x = \left(a + \frac{c^2}{a}\right) \cos \theta \\ y = \left(a - \frac{c^2}{a}\right) \sin \theta \end{cases}$$

5.3. POTENTIAL FLOWS GENERATED BY TRANSFORMATIONS

or

$$\left(\frac{x}{a + c^2/a}\right)^2 - \left(\frac{y}{a - c^2/a}\right)^2 = 1$$

It is an ellipse!

Consider a uniform stream of speed V and angle-of-attack α past the circular cylinder on ζ - plane

$$F(\zeta) = V \left(e^{-i\alpha} \zeta + \frac{a^2}{\zeta} e^{i\alpha} \right) \quad (5.60)$$

From J-T

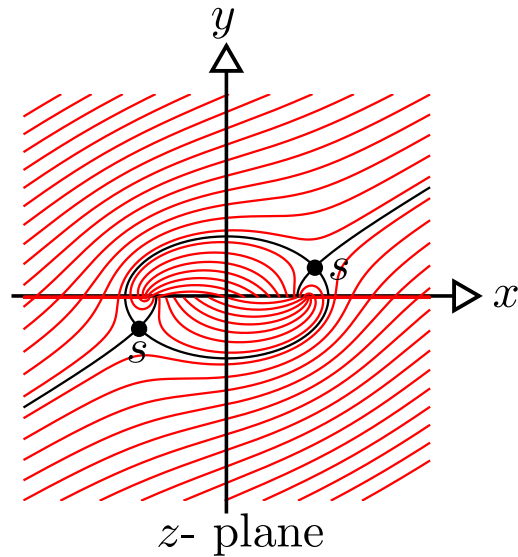
$$\zeta^2 - z\zeta + c^2 = 0 \implies \zeta \overset{\dagger}{=} \frac{z}{2} \pm \sqrt{\left(\frac{z}{2}\right)^2 - c^2}$$

$$\therefore F(z) = F(\zeta(z))$$

$$\begin{aligned} &= V \left\{ \left[\frac{z}{2} + \sqrt{\left(\frac{z}{2}\right)^2 - c^2} \right] e^{-i\alpha} + \frac{a^2 e^{i\alpha}}{\frac{z}{2} + \sqrt{\left(\frac{z}{2}\right)^2 - c^2}} \right\} \\ &= V \left[z e^{-i\alpha} + \left(\frac{a^2}{c^2} e^{i\alpha} - e^{-i\alpha} \right) \left(\frac{z}{2} - \sqrt{\left(\frac{z}{2}\right)^2 - c^2} \right) \right] \end{aligned}$$

[†]discard – since $z \rightarrow \zeta$ as $\zeta \rightarrow \infty$

CHAPTER 5. TWO-DIMENSIONAL POTENTIAL FLOWS



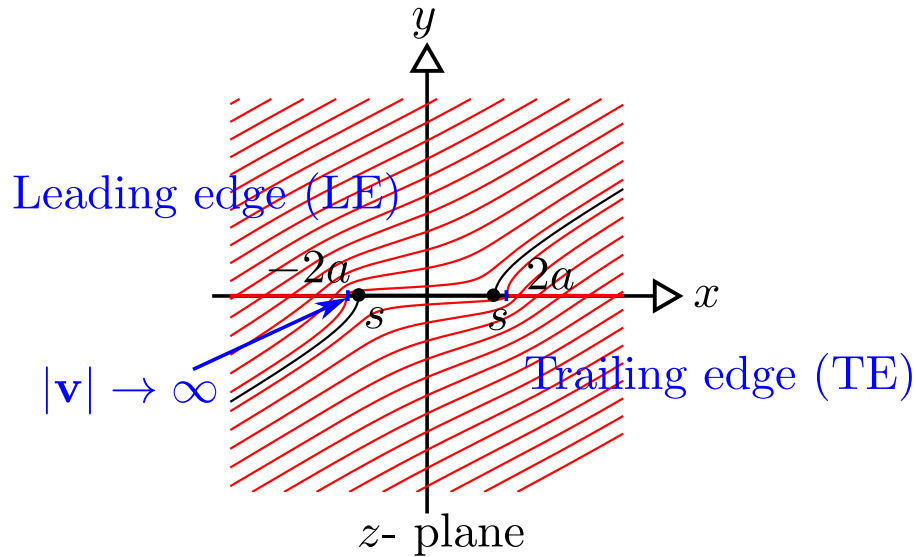
Example. Stagnation points on the ellipse
 $\zeta = \pm ae^{i\alpha}$ subs into J-T

$$z = \zeta + \frac{c^2}{\zeta} \implies z = \pm \left(a + \frac{c^2}{a} \right) \cos \alpha \pm i \left(a - \frac{c^2}{a} \right) \sin \alpha$$

(5.61)

(b) Flow around a flat plate

5.3. POTENTIAL FLOWS GENERATED BY TRANSFORMATIONS



In (a), let $c = a$, the $\zeta = ae^{i\theta} \xrightarrow{\text{maps to}} -2a \leq x \leq 2a$ on z -plane

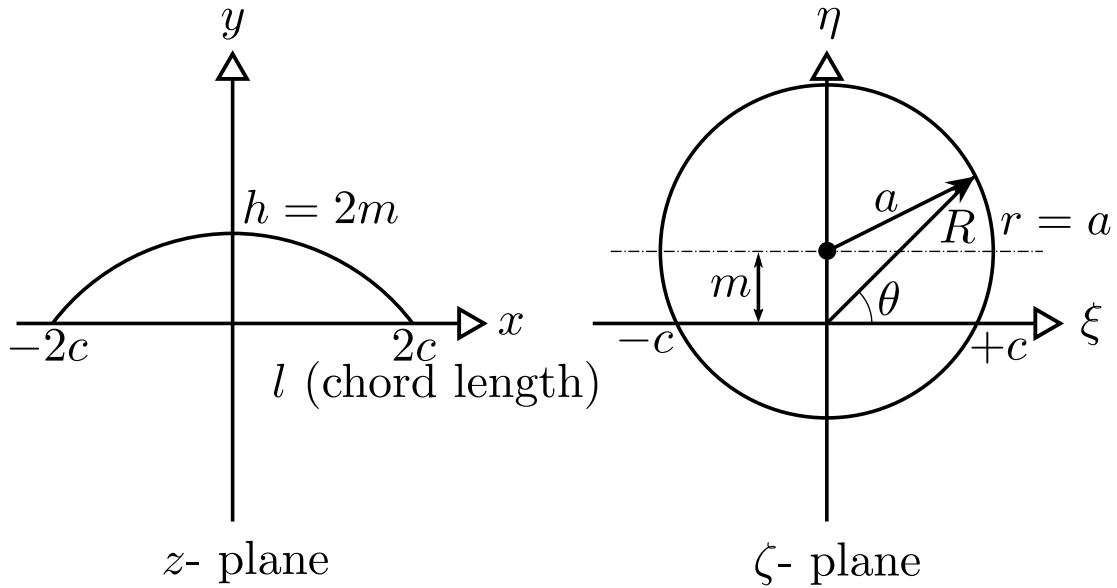
$$F(z) = V \left[ze^{-i\alpha} + (e^{i\alpha} - e^{-i\alpha}) \left(\frac{z}{2} - \sqrt{\left(\frac{z}{2}\right)^2 - a^2} \right) \right] \quad (5.62)$$

Example. Stagnation points on the plate
 $\zeta = \pm ae^{i\alpha}$ subs into J-T

$$z = \zeta + \frac{c^2}{\zeta} \implies z = \pm 2a \cos \alpha \quad (5.63)$$

(c) Flow around a circular arc (camber line)

CHAPTER 5. TWO-DIMENSIONAL POTENTIAL FLOWS



From $\zeta = R(\theta)e^{i\theta}$ and J-T: $z = Re^{i\theta} + \frac{c}{R^2}e^{-i\theta}$

$$\Rightarrow \begin{cases} x = \left(R + \frac{c^2}{R}\right) \cos \theta \\ y = \left(R - \frac{c^2}{R}\right) \sin \theta \end{cases}$$

$$\xrightarrow[\text{R}]{\text{eliminating}} x^2 \sin^2 \theta - y^2 \cos^2 \theta = 4c^2 \sin^2 \theta \cos^2 \theta \quad (5.64)$$

use relation

$$c^2 + m^2 = a^2 = R^2 + m^2 - 2Rm \overbrace{\cos \left(\frac{\pi}{2} - \theta\right)}^{\sin \theta} \quad (*)$$

Example.

5.3. POTENTIAL FLOWS GENERATED BY TRANSFORMATIONS

$$(5.64) \implies x^2 + \left[y + c \left(\frac{c}{m} - \frac{m}{c} \right) \right]^2 = c^2 \left[4 + \left(\frac{c}{m} - \frac{m}{c} \right)^2 \right] \quad (5.65)$$

Solution.

From (*)

$$\begin{aligned} \sin \theta = \frac{R^2 - c^2}{2Rm} \text{ and } \sin \theta = \frac{y}{R - c^2/R} &\implies \sin \theta = \frac{y}{2m \sin \theta} \\ \implies \sin^2 \theta = \frac{y}{2m}, \quad \cos^2 \theta = 1 - \sin^2 \theta = 1 - \frac{y}{2m} \end{aligned}$$

$$(5.65) \implies \begin{cases} l = 4c \\ h = 2m \end{cases} \implies \begin{cases} c = \frac{l}{4} \\ m = \frac{h}{2} \end{cases} \text{ If we let } \frac{m}{c} = \epsilon \ll 1$$

then

$$x^2 + \left(y + \frac{c^2}{m} \right)^2 = c^2 \left(4 + \frac{c^2}{m^2} \right) + \mathcal{O}(\epsilon^2) \quad (5.66a)$$

correct to $\mathcal{O}(\epsilon)$

CHAPTER 5. TWO-DIMENSIONAL POTENTIAL FLOWS

in terms of l and h

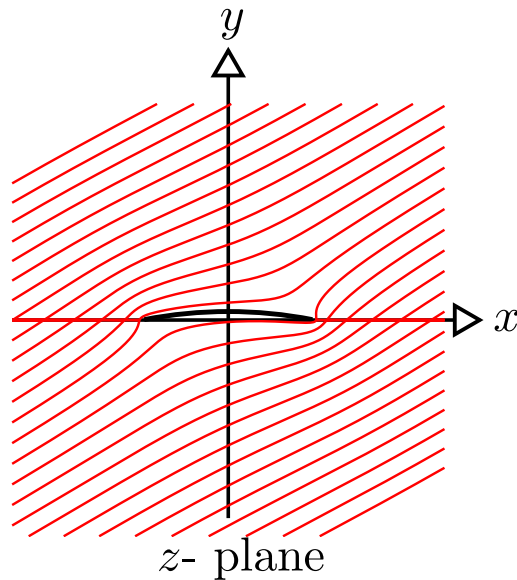
$$x^2 + \left(y + \frac{l^2}{8h}\right)^2 = \frac{l^2}{4} \left(1 + \frac{l^2}{16h^2}\right) \quad (5.66b)$$

The complex potential is

$$F(\zeta) = V \left[(\zeta - im) e^{-i\alpha} + \frac{a^2}{\zeta - im} e^{i\alpha} \right] \quad (5.67)$$

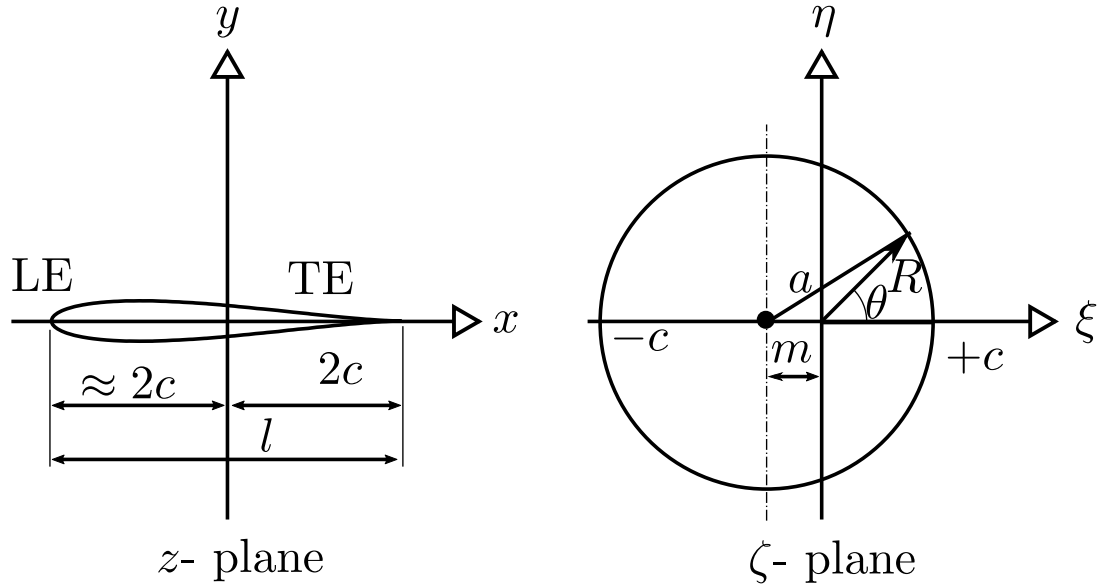
with

$$z = \zeta + \frac{c^2}{\zeta}$$



(d) Flow around a symmetric Joukowski airfoil

5.3. POTENTIAL FLOWS GENERATED BY TRANSFORMATIONS



From J-T and infinitesimal assumption

$$z = \zeta + \frac{c^2}{\zeta}, \quad \epsilon = \frac{m}{c} \ll 1$$

hence $\zeta = c \implies z = 2c$

$$\zeta = -(c + 2m) \implies z = -c(1 + 2\epsilon) - \frac{c}{1 + 2\epsilon} = -2c + \mathcal{O}(\epsilon^2)$$

hence

$$l \approx 4c + \epsilon^\epsilon$$

Exercise. Show that correct to ϵ , the airfoil shape on z -plane is given by

$$z \approx c [2 \cos \theta + i 2 \epsilon (1 - \cos \theta) \sin \theta] + \mathcal{O}(\epsilon^2)$$

i.e.

$$\begin{cases} x \approx 2c \cos \theta \\ y \approx 2c \epsilon (1 - \cos \theta) \sin \theta \end{cases}$$

CHAPTER 5. TWO-DIMENSIONAL POTENTIAL FLOWS

or

$$y \approx \pm 2c\epsilon \left(1 - \frac{x}{2c}\right) \sqrt{1 - \left(\frac{x}{2c}\right)^2} + \mathcal{O}(\epsilon^2) \quad (5.68)$$

Solution.

From cosine theorem

$$\begin{aligned} a^2 &= (c + m)^2 = R^2 \left(1 + \frac{m^2}{R^2} + 2\frac{m}{R} \cos \theta\right) \\ &\stackrel{\ddagger}{=} R^2 \left(1 + 2 \underbrace{\frac{m}{R} \cos \theta}_{\mathcal{O}(\epsilon)} + \mathcal{O}(\epsilon^2)\right) \\ &\implies c + m \approx R \left(1 + \frac{m}{R} \cos \theta\right) \\ &\therefore R \approx c [1 + \epsilon (1 - \cos \theta)] \end{aligned}$$

From J-T

$$\begin{aligned} z &= \zeta + \frac{c^2}{\zeta} \\ &= Re^{i\theta} + \frac{c^2}{R} e^{-i\theta} \\ &= c [1 + \epsilon (1 - \cos \theta)] e^{i\theta} + \frac{ce^{-i\theta}}{1 + \epsilon (1 - \cos \theta)} \\ &\stackrel{\S}{=} c [1 + \epsilon (1 - \cos \theta)] e^{i\theta} + c \left[1 - \epsilon (1 - \cos \theta) + \mathcal{O}(\epsilon^2)\right] e^{-i\theta} \end{aligned}$$

[‡]From $R \geq c$, $m/R \leq m/c$, i.e. $m/R = \mathcal{O}(\epsilon)$

5.3. POTENTIAL FLOWS GENERATED BY TRANSFORMATIONS

Maximum thickness t occurs at $\frac{dy}{dx} = 0$

$$\Rightarrow \begin{cases} x = -c \\ y = \pm \frac{3\sqrt{3}}{2}c\epsilon \end{cases} \Rightarrow t \approx 3\sqrt{3}c\epsilon$$

hence thickness ratio

$$\frac{t}{l} = \frac{3\sqrt{3}}{4}\epsilon \Rightarrow \epsilon = \frac{m}{c} \approx \frac{4}{3\sqrt{3}}\frac{t}{l} \approx 0.77\frac{t}{l}$$

equation for the airfoil in terms of t and l

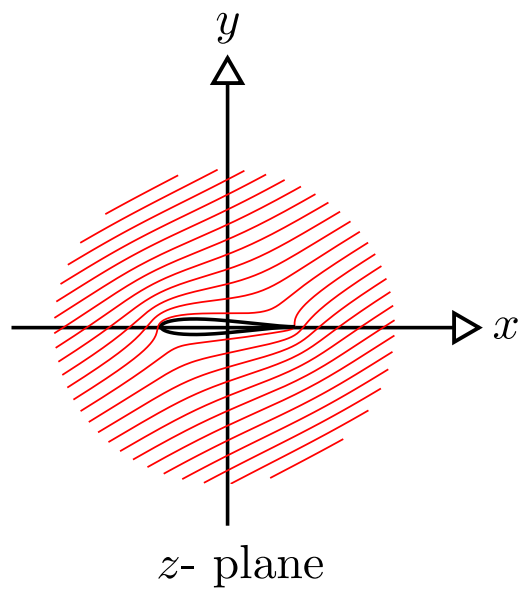
$$\frac{y}{t} \approx \pm 0.385 \left(1 - 2\frac{x}{l}\right) \sqrt{1 - \left(\frac{2x}{l}\right)^2} \quad (5.69)$$

The complex potential is

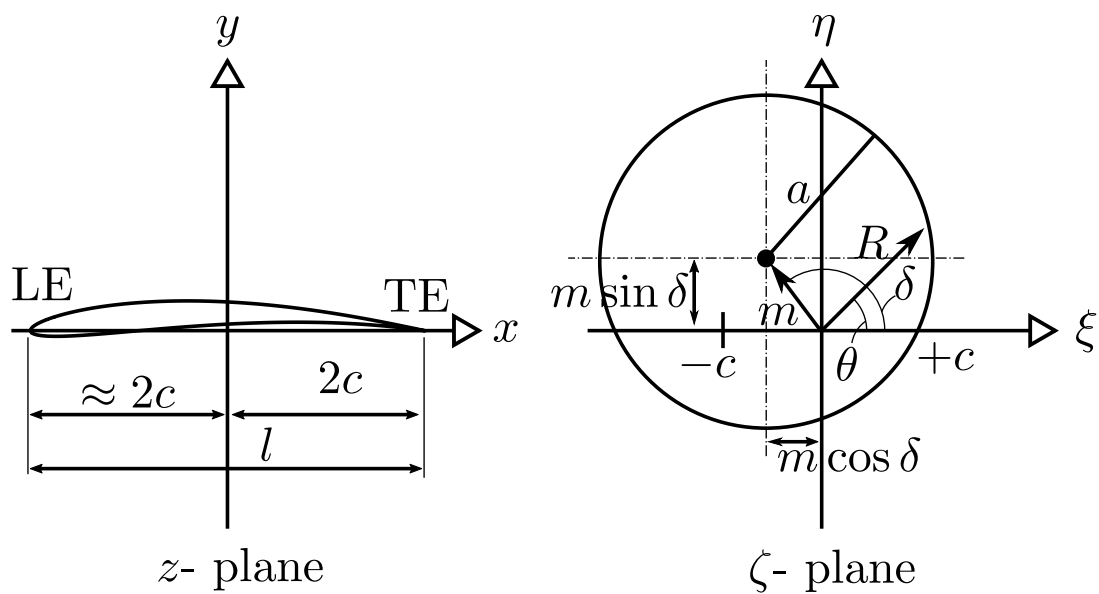
$$F(\zeta) = V \left[(\zeta + m) e^{-i\alpha} + \frac{a^2}{\zeta + m} e^{i\alpha} \right] \quad (5.70)$$

[§] $\frac{ce^{-i\theta}}{1 + \epsilon(1 - \cos \theta)} = ce^{-i\theta} [1 - \epsilon(1 - \cos \theta) + \mathcal{O}(\epsilon^2)]$

CHAPTER 5. TWO-DIMENSIONAL POTENTIAL FLOWS



(e) Flow around a cambered Joukowski airfoil



For $\epsilon = m/c \ll 1$

5.3. POTENTIAL FLOWS GENERATED BY TRANSFORMATIONS

correct to $\mathcal{O}(\epsilon)$, equation of the cambered Joukowski airfoil

$$y = \underbrace{\sqrt{\frac{l^2}{4} \left(1 + \frac{l^2}{16h^2}\right) - x^2} - \frac{l^2}{8h}}_{\substack{\text{equation for camber} \\ (5.66b)}} \pm \underbrace{0.385 \left(1 - 2\frac{x}{l}\right) \sqrt{1 - \left(\frac{2x}{l}\right)^2}}_{\substack{\text{equation for thickness} \\ (5.69)}} + \mathcal{O}\left(\frac{t}{l}\right)^2 \quad (5.71)$$

The complex potential is

$$F(\zeta) = V \left[\left(\zeta - me^{i\delta} \right) e^{-i\alpha} + \frac{a^2 e^{i\alpha}}{\zeta - me^{i\delta}} \right] \quad (5.72)$$

with $z = \zeta + \frac{c^2}{\zeta}$ and

$$m \cos \delta \approx \frac{0.77}{4}t, \quad m \sin \delta \approx \frac{h}{2}, \quad a \approx \frac{l}{4} \left(1 + 0.77\frac{t}{l} \right)$$

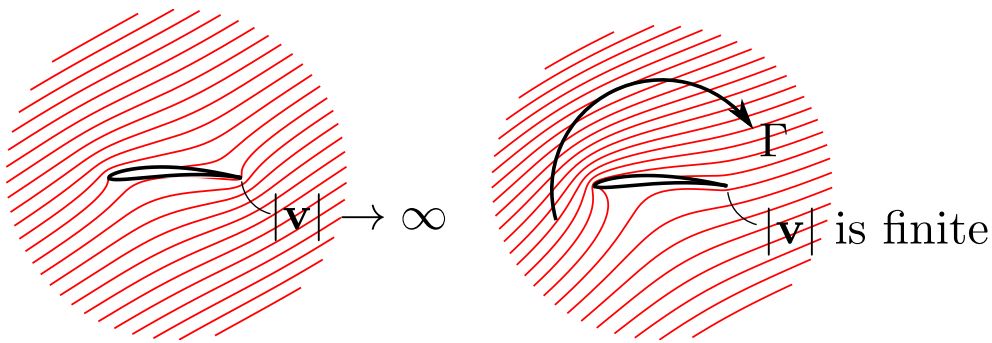
5.3.4 Kutta Condition

Above (b)~(e) cases all involve a singular point at TE where velocity $\rightarrow \infty$. The viscous effect must take action.

- Kutta-Joukowski hypothesis

CHAPTER 5. TWO-DIMENSIONAL POTENTIAL FLOWS

Whenever a flow passing around sharp corner, the viscous effect becomes significant and cannot be neglected therefore,
instead of



These should be a circulation around the airfoil as a manifestation of the viscous effect (flow separation), the magnitude of the circulation being such that to ensure the flow leave the TE smoothly, i.e.

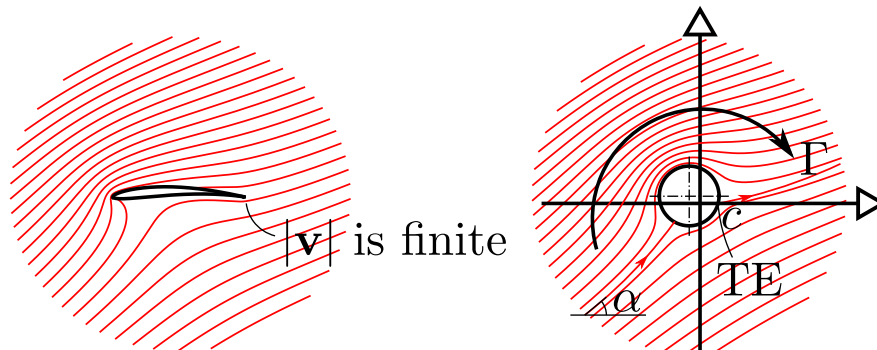
Γ is fixed by the requirement

$$\frac{dF(z)}{dz} = \frac{dF}{d\zeta} / \frac{dz}{d\zeta} \text{ is finite at TE} \quad (5.73)$$

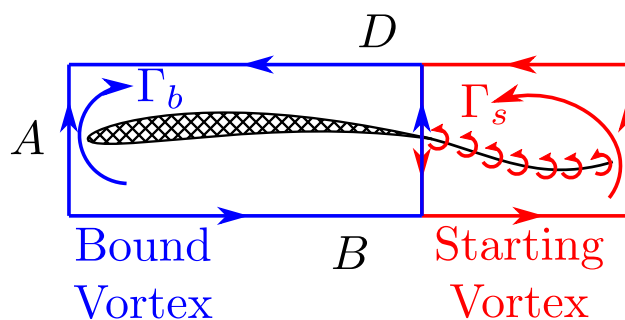
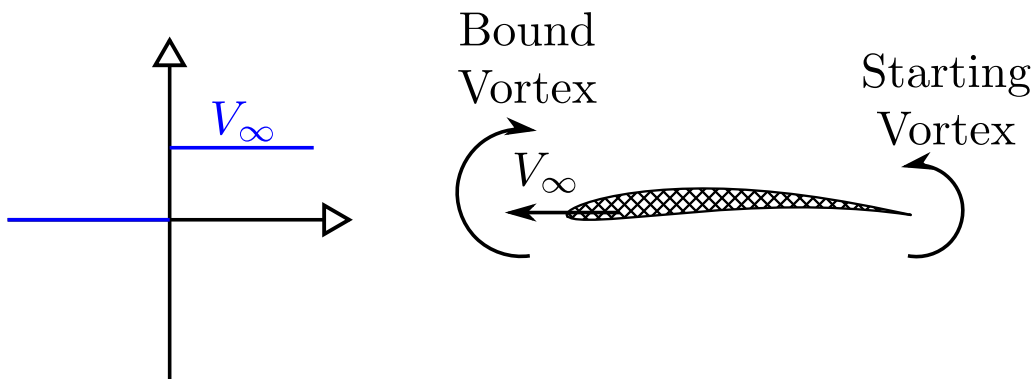
which implies

$$\frac{dF}{d\zeta} = 0 \text{ at TE (at critical point } \zeta = c) \quad (5.74)$$

5.3. POTENTIAL FLOWS GENERATED BY TRANSFORMATIONS



- Generation of Stating Vortex and Bound Vortex
Consider



From Kelvin's Theorem

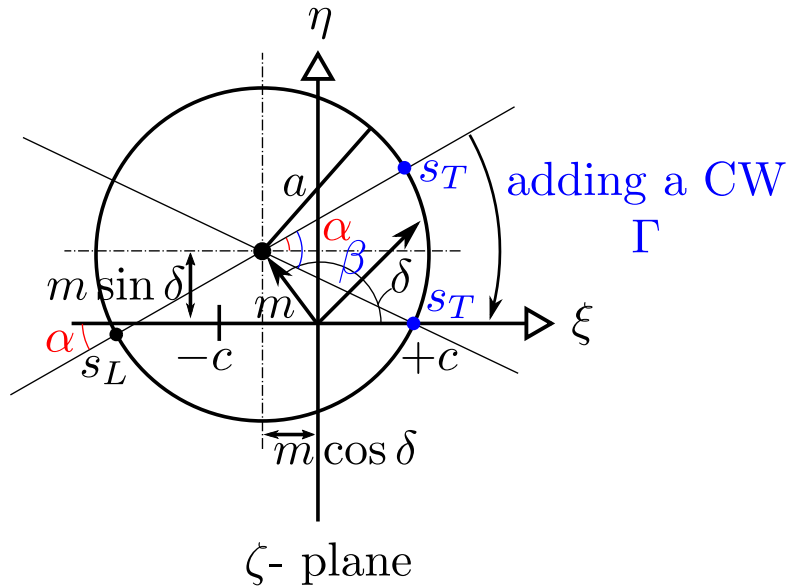
$$C \frac{D\Gamma_{ABCD}}{Dt} = 0 \implies \Gamma_{ABCD} = 0$$

$$\Gamma_b - \Gamma_s = \Gamma_{ABCD} = 0$$

$$\Gamma_b = -\Gamma_s$$

- Back to the Joukowski-airfoil problems

CHAPTER 5. TWO-DIMENSIONAL POTENTIAL FLOWS



Recall

$$F(\zeta) = V \left[(\zeta - \zeta_0) e^{-i\alpha} + \frac{a^2 e^{i\alpha}}{\zeta - \zeta_0} \right] + \frac{i\Gamma}{2\pi} \ln (\zeta - \zeta_0)$$

$$W(\zeta) = \frac{dF}{d\zeta} = V e^{-i\alpha} + \frac{i\Gamma}{2\pi} \frac{1}{\zeta - \zeta_0} - V e^{i\alpha} a^2 \frac{1}{(\zeta - \zeta_0)^2}$$

(5.75)

From geometry: $c - \zeta_0 = a e^{-i\beta}$

at TE : $\zeta = c$

$$W(c) = V e^{-i\alpha} + \frac{i\Gamma}{2\pi} e^{i\beta} - V e^{i(\alpha+2\beta)}$$

KC require $W(c) = 0$

$$\implies 2\pi a V \left[1 - e^{2i(\alpha+\beta)} \right] + i\Gamma e^{i(\alpha+\beta)} = 0$$

5.3. POTENTIAL FLOWS GENERATED BY TRANSFORMATIONS

$$\implies \Gamma = 4\pi aV \sin(\alpha + \beta)$$

The lift on the Joukowski airfoil is

$$L = \rho V \Gamma = 4\pi a \rho V^2 \sin(\alpha + \beta) \quad (5.76)$$

the lift coefficient C_L is defined as

$$C_L = \frac{L}{\frac{\rho}{2} V^2 l} = 8\pi \left(\frac{a}{l} \right) \sin(\alpha + \beta) \quad (5.77)$$

For small angle-of-attack, $\alpha \ll 1$, and small β (which corresponds to *thin* airfoil) $\beta \ll 1$,

$$\sin(\alpha + \beta) \approx \alpha + \beta$$

hence

$$L \approx 4\pi \rho V^2 a (\alpha + \beta) \quad (5.78)$$

$$C_L \approx 8\pi \left(\frac{a}{l} \right) (\alpha + \beta) \quad (5.79)$$

Furthermore, if $\beta \ll 1$, $l \approx 4c \approx 4a$, hence

$$L \approx \pi \rho l V^2 (\alpha + \beta) \quad (5.80)$$

$$\boxed{C_L \approx 2\pi (\alpha + \beta)} \quad (5.81)$$

where α is angle-of-attack and β is related to camber $\approx \frac{2h}{l}$

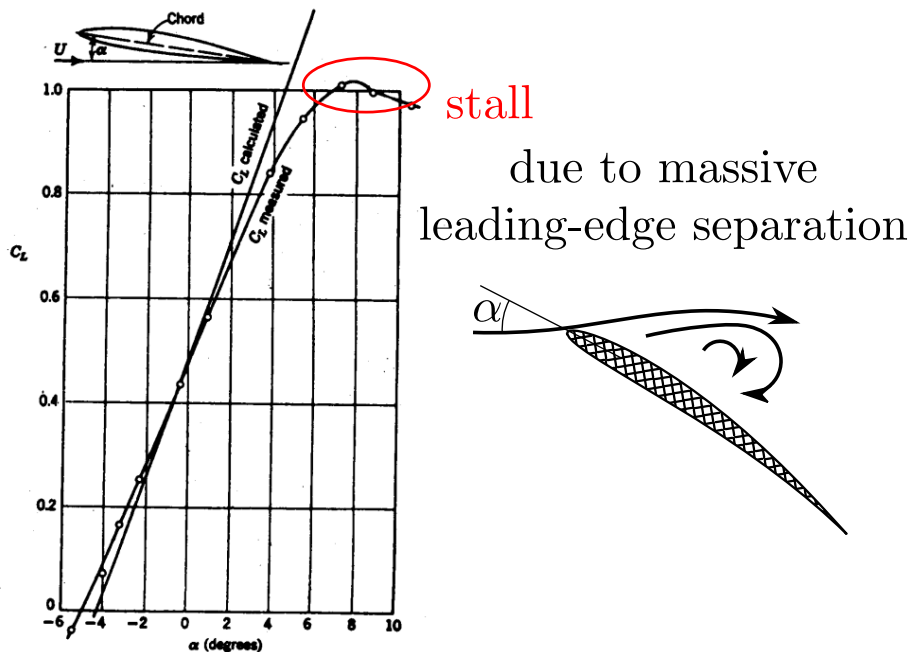
Note.

CHAPTER 5. TWO-DIMENSIONAL POTENTIAL FLOWS

- ① $L \neq 0$ at $\alpha = 0$ (due to camber)
- ② $L = 0$ when $\alpha = -\beta$, $-\beta \equiv \alpha_0$, where α_0 is zero-left angle

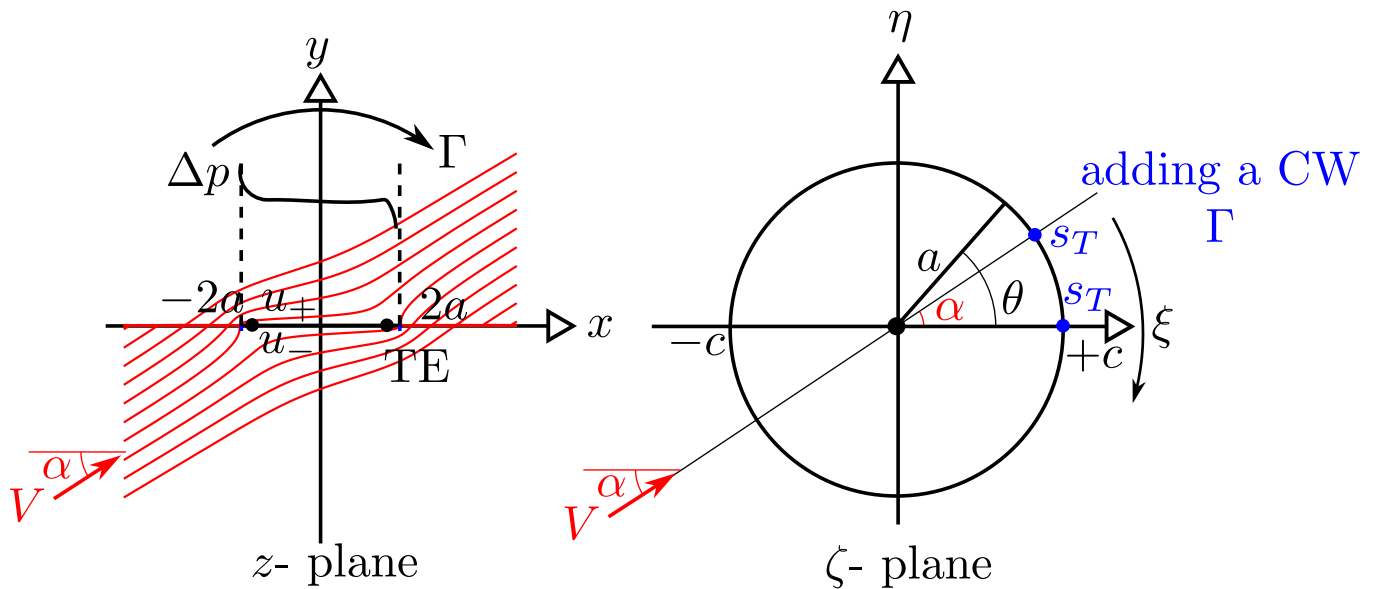
Define an effective angle-of-attack α_e

$$\alpha_e = \alpha - \alpha_0 = \alpha + \beta, \quad C_L \approx 2\pi\alpha_e$$



Example.

5.3. POTENTIAL FLOWS GENERATED BY TRANSFORMATIONS



$$F(\zeta) = V \left(\zeta e^{-i\alpha} + \frac{a^2}{\zeta} e^{i\alpha} \right) + \frac{i\Gamma}{2\pi} \ln \zeta$$

$$\frac{dF}{dz} = \frac{dF}{d\zeta} \frac{dz}{d\zeta} \equiv \text{finite at TE } (\zeta = a)$$

i.e.

$$\left. \frac{dF}{d\zeta} \right|_{\zeta=a} = 0, \quad \frac{dF}{d\zeta} = V e^{-i\alpha} - V \frac{a^2 e^{i\alpha}}{\zeta^2} + \frac{i\Gamma}{2\pi} \frac{1}{\zeta}$$

$$\left. \frac{dF}{d\zeta} \right|_{\zeta=a} = 0 \implies V e^{-i\alpha} - V e^{i\alpha} + \frac{i\Gamma}{2\pi a} = 0$$

$$\implies \boxed{\Gamma = 4\pi V a \sin \alpha}$$

CHAPTER 5. TWO-DIMENSIONAL POTENTIAL FLOWS

Velocity on the plate:

$$\begin{aligned}\left.\frac{dF}{d\zeta}\right|_{\text{plate}} &\equiv \left.\frac{dF}{d\zeta}\right|_{\zeta=ae^{i\theta}} = Ve^{-i\alpha} - Ve^{i(\alpha-2\theta)} + \frac{i\Gamma}{2\pi a}e^{-i\theta} = \dots \\ &= Ve^{-i\theta} \left[e^{-i(\alpha-\theta)} - e^{i(\alpha-\theta)} + i2\sin\alpha \right] \\ &= 2iVe^{-i\theta} [\sin\alpha - \sin(\alpha-\theta)]\end{aligned}$$

$$\left.\frac{dz}{d\zeta}\right|_{\text{plate}} \equiv \left.\frac{dz}{d\zeta}\right|_{\zeta=ae^{i\theta}} = 1 - e^{-2i\theta} = e^{-i\theta} [e^{i\theta} - e^{-i\theta}] = 2i\sin\theta e^{-i\theta}$$

$$\begin{aligned}\Rightarrow \left.\frac{dF}{dz}\right|_{\text{plate}} &= \frac{dF}{d\zeta} / \left.\frac{dz}{d\zeta}\right|_{\text{plate}} = \frac{2iVe^{-i\theta} [\sin\alpha - \sin(\alpha-\theta)]}{2i\sin\theta e^{-i\theta}} \\ &= \frac{V [\sin\alpha - \sin\alpha\cos\theta + \cos\alpha\sin\theta]}{\sin\theta} \\ &= \dots \\ &= V\cos\alpha \left(1 + \tan\alpha \frac{1 - \cos\theta}{\sin\theta} \right) \equiv u - iv|_{\text{plate}}\end{aligned}$$

on the plate

$$\begin{cases} u = V\cos\alpha \left(1 + \tan\alpha \frac{1 - \cos\theta}{\sin\theta} \right), 0 < \theta \leq \pi: \text{upper surface of plate} \\ v = 0, \pi < \theta \leq 2\pi: \text{lower surface of plate} \end{cases}$$

From Bernoulli's equation

$$p_+ + \frac{\rho}{2}u_+^2 = p_- + \frac{\rho}{2}u_-^2$$

5.3. POTENTIAL FLOWS GENERATED BY TRANSFORMATIONS

$$\begin{aligned}
 \Delta p &\equiv p_- - p_+ = \frac{\rho}{2} (u_+^2 - u_-^2) \\
 &= \frac{\rho}{2} V^2 \cos^2 \alpha \left[\left(1 + \tan \alpha \frac{1 - \cos \theta}{\sin \theta} \right)^2 - \left(1 - \tan \alpha \frac{1 - \cos \theta}{\sin \theta} \right)^2 \right] \\
 &= \rho V^2 \sin \alpha \frac{1 - \cos \theta}{\sin \theta}, \quad 0 \leq \theta \leq \pi
 \end{aligned}$$

Lift force Y

$$\begin{aligned}
 Y &= \int_{-2a}^{2a} \Delta p dx \stackrel{\P}{=} \int_{\theta=\pi}^{\theta=0} \Delta p d(2a \cos \theta) = 2a \int_0^\pi \Delta p \sin \theta d\theta \\
 &= 2\rho V^2 a \sin 2\alpha \int_0^\pi (1 - \cos \theta) d\theta = 2\pi \rho V^2 a \sin 2\alpha
 \end{aligned}$$

or from Kutta-Joukowski Theorem

$$X - iY = -i\rho V e^{-i\alpha} \Gamma, \quad \text{Here } \Gamma = 4\pi V a \sin \alpha$$

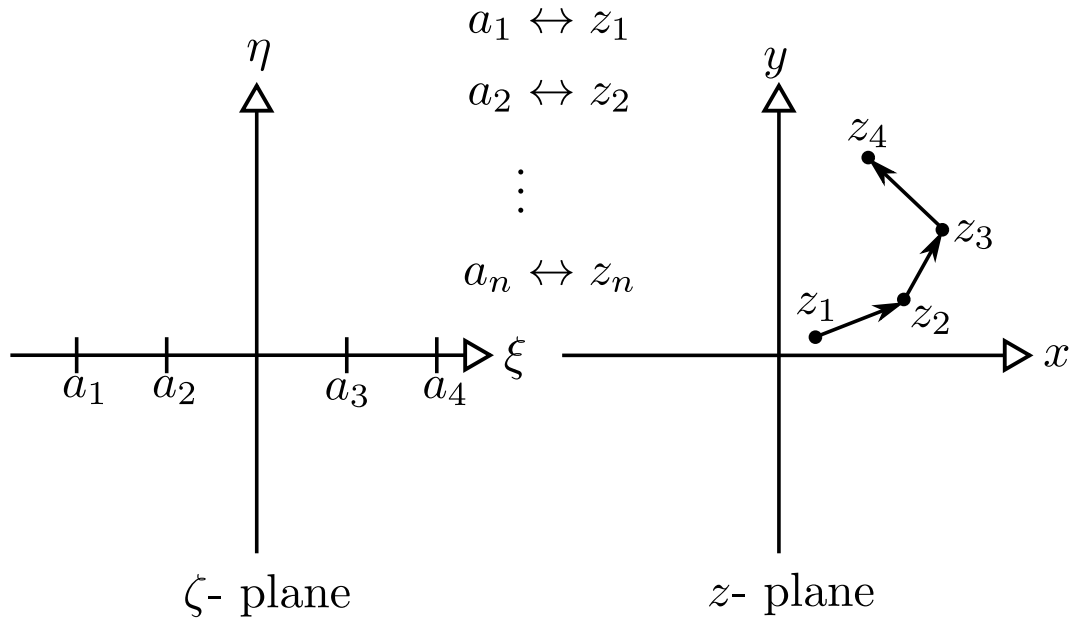
$$\therefore X - iY = -4\pi \rho V^2 a \sin^2 \alpha - i4\pi \rho V^2 a \sin \alpha \cos \alpha$$

$$\implies \begin{cases} X = -4\pi \rho V^2 a \sin^2 \alpha \neq 0 \\ Y = 4\pi \rho V^2 a \sin \alpha \cos \alpha \end{cases}$$

\P From J-T $z = \zeta + \frac{a^2}{\zeta}$, $\zeta = ae^{i\theta}$, $z = a^{i\theta} + ae^{-i\theta} = 2a \cos \theta = x + iy \implies \begin{cases} x = 2a \cos \theta \\ y = 0 \end{cases}$

CHAPTER 5. TWO-DIMENSIONAL POTENTIAL FLOWS

5.3.5 Schwarz-Christoffel Transformation



At critical point of transformation

A smooth curve $\iff$ A corner

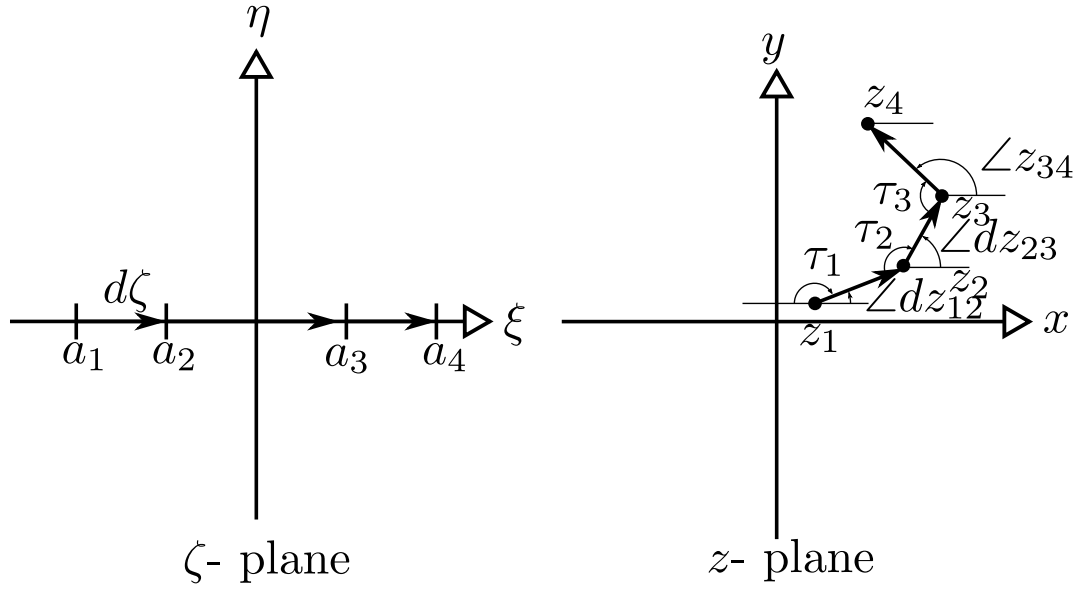
The transformation has the form

$$\frac{dz}{d\zeta} = K (\zeta - a_1)^{k_1-1} (\zeta - a_2)^{k_2-1} \cdots (\zeta - a_N)^{k_N-1} \quad (\text{S-C T}) \quad (5.82)$$

where $a_1, a_2, \cdots, a_N$ are real numbers, $k_1, k_2, \cdots, k_N$ are real numbers, and K is a complex number.

- General properties and rules of S-C T

5.3. POTENTIAL FLOWS GENERATED BY TRANSFORMATIONS



S-C T

$$\begin{aligned} \frac{dz}{d\zeta} &= K (\zeta - a_1)^{k_1-1} (\zeta - a_2)^{k_2-1} \dots \\ \implies \angle dz - \angle d\zeta &= \angle K + (k_1 - 1) \angle (\zeta - a_1) \\ &\quad + (k_2 - 1) \angle (\zeta - a_2) + \dots \end{aligned} \quad (*)$$

① ζ is between $\zeta = a_1$ and $\zeta = a_2$

$$\implies \angle d\zeta = 0 \quad \text{and} \quad \angle (\zeta - a_1) = 0, \angle (\zeta - a_2) = \angle (\zeta - a_3) = \dots = \pi$$

Between z_1 and z_2 , from (*)

$$\implies \angle dz_{12} - 0 = \angle K + \pi \sum_{i=2}^N (k_i - 1) = \text{constant}$$

So segment from z_1 and z_2 is a *straight line*.

CHAPTER 5. TWO-DIMENSIONAL POTENTIAL FLOWS

② ζ is between $\zeta = a_2$ and $\zeta = a_3$

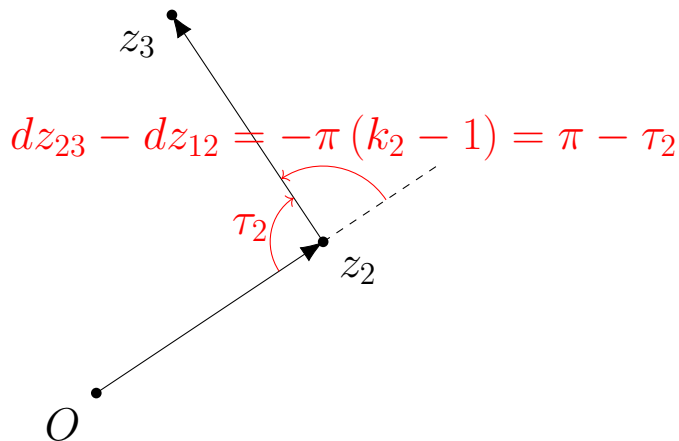
$$\angle(\zeta - a_1) = \angle(\zeta - a_2) = 0$$

$$\angle(\zeta - a_3) = \angle(\zeta - a_4) = \cdots = \pi$$

$$\begin{aligned} (*) \implies \angle dz_{23} - 0 &= \angle K + \pi \sum_{i=3}^N (k_i - 1) = \text{constant} \\ &= \angle K + \pi \sum_{i=2}^N (k_i - 1) - \pi(k_2 - 1) \\ &= \angle dz_{12} - \pi(k_2 - 1) \end{aligned}$$

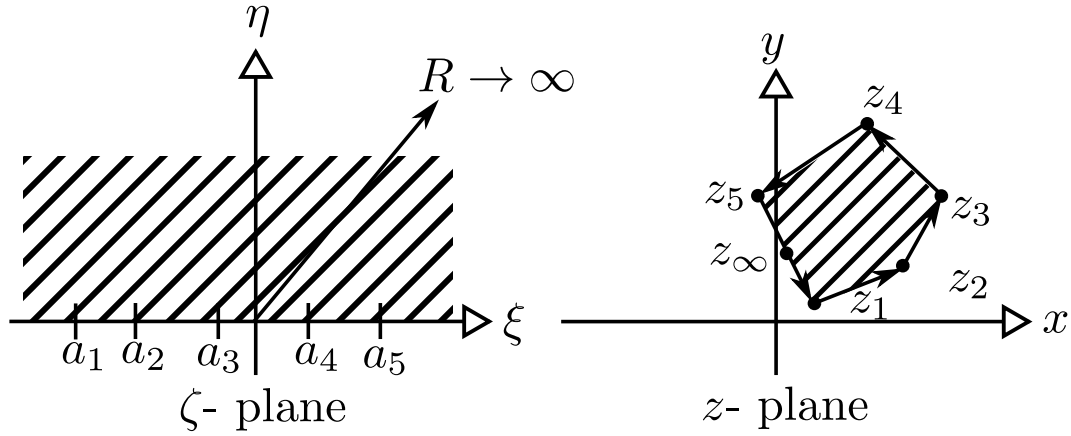
$$\implies \angle dz_{23} - \angle dz_{12} = -\pi(k_2 - 1) = \pi - \tau_2$$

$$\implies \boxed{k_2 = \frac{\tau_2}{\pi}}$$



Example. For a Pentagon, the S-C T is

5.3. POTENTIAL FLOWS GENERATED BY TRANSFORMATIONS



$$\frac{dz}{d\zeta} = K (\zeta - a_1)^{k_1-1} (\zeta - a_2)^{k_2-1} \dots (\zeta - a_5)^{k_5-1}$$

The points ∞ on ζ - plane, say $\zeta = Re^{i\varphi}$ ($R \rightarrow \infty$), will map to a *point* z_∞ on z - plane.

From (5.82)

$$\begin{aligned} \frac{dz}{d\zeta} &= K \zeta^{k_1-1} \zeta^{k_2-1} \dots \zeta^{k_N-1} \\ &\left(1 - \frac{a_1}{\zeta}\right)^{k_1-1} \left(1 - \frac{a_2}{\zeta}\right)^{k_2-1} \dots \left(1 - \frac{a_N}{\zeta}\right)^{k_N-1} \\ &= K \zeta^{\sum_{i=1}^N (k_i-1)} \left[1 - \frac{\sum_{i=1}^N (k_i-1)}{\zeta} + \mathcal{O}\left(\frac{1}{\zeta^2}\right)\right] \end{aligned}$$

Integral $\Rightarrow$

$$z = K \frac{\zeta^{1+\sum_{i=1}^N (k_i-1)}}{1 + \sum_{i=1}^N (k_i-1)} \left[1 + \mathcal{O}\left(\frac{1}{\zeta}\right)\right] + C$$

CHAPTER 5. TWO-DIMENSIONAL POTENTIAL FLOWS

but

$$\begin{aligned} 1 + \sum_{i=1}^N (k_i - 1) &= 1 + \sum_{i=1}^N \left(\frac{\tau_i}{\pi} - 1 \right) = 1 - N + \sum_{i=1}^N \frac{\tau_i}{\pi} \\ &= 1 - N + \left[\frac{(N-2)\pi}{\pi} \right] = -1 \end{aligned}$$

$$\therefore z = K \left(-\frac{1}{\zeta} \right) \left[1 + \mathcal{O}\left(\frac{1}{\zeta}\right) \right] + C \quad (5.83)$$

hence as $\zeta \rightarrow \infty$, $z \rightarrow C$

- ③ The action of K in $\frac{dz}{d\zeta}$ is to magnify $K = \sigma e^{i\psi}$ and rotate the polygon on z - plane

- Some considerations in constructing the S-C T

Q: Given $z_1, z_2, \dots, z_N$, (and hence $\tau_1, \tau_2, \dots, \tau_N$), how to choose $a_1, a_2, \dots, a_N$?

Note.

integrate (5.82)

$$\Rightarrow z = K \int_{a_1}^{\zeta} (\zeta - a_1)^{k_1-1} (\zeta - a_2)^{k_2-1} \dots (\zeta - a_N)^{k_N-1} d\zeta + z_1 \quad (5.84)$$

Observations:

- ① a_1 is arbitrary

5.3. POTENTIAL FLOWS GENERATED BY TRANSFORMATIONS

② Having chosen a_1 , (5.84) becomes

$$\begin{aligned}
 z_2 - z_1 &= K \int_{a_1}^{a_2} (\zeta - a_1)^{k_1-1} (\zeta - a_2)^{k_2-1} \cdots (\zeta - a_N)^{k_N-1} d\zeta \\
 &= K F_2(a_2, a_3, \cdots, a_N) \\
 z_3 - z_2 &= K \int_{a_2}^{a_3} (\zeta - a_1)^{k_1-1} (\zeta - a_2)^{k_2-1} \cdots (\zeta - a_N)^{k_N-1} d\zeta \\
 &= K F_3(a_2, a_3, \cdots, a_N) \\
 &\vdots \\
 z_N - z_{N-1} &= K \int_{a_{N-1}}^{a_N} (\zeta - a_1)^{k_1-1} (\zeta - a_2)^{k_2-1} \cdots (\zeta - a_N)^{k_N-1} d\zeta \\
 &= K F_N(a_2, a_3, \cdots, a_N)
 \end{aligned}$$

where are $N - 1$ constraints and $N - 1$ equations
(redundant: $z_1 - z_N$)

but have N unknowns, $a_2, a_3, \cdots, a_N, N$ hence another, say a_2 , can be chosen arbitrary. Only $N - 2$ constraints, not $N - 1$, hence still another, say a_3 , can be chosen arbitrary.

Conclusion. Among a_1 and a_N can arbitrary choose 3 of them (say a_1, a_2, a_3). Since arbitrary, for convenience, can take on of then, say $\boxed{a_1} \rightarrow -\infty$

$$\begin{aligned}
 \frac{dz}{d\zeta} &= K (-a)^{k_1-1} \left(1 - \frac{\zeta}{a_1}\right)^{k_1-1} (\zeta - a_2)^{k_2-1} \cdots \\
 &= K (-a)^{k_1-1} (\zeta - a_2)^{k_2-1} (\zeta - a_3)^{k_3-1} \cdots \\
 &= K' (\zeta - a_2)^{k_2-1} (\zeta - a_3)^{k_3-1} \cdots
 \end{aligned}$$

Chapter 6

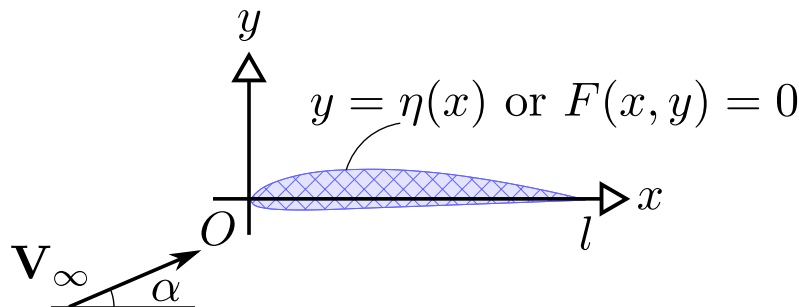
Elements of Thin-Airfoil Theory (Linearized 2D Potential Flow)

Assume airfoil is *thin* and *elongated*.

6.1. PERTURBATION FLOW- SMALL PERTURBATION THEORY (THIN-AIRFOIL APPROXIMATION)

6.1 Perturbation Flow- Small Perturbation Theory (Thin-Airfoil Approximation)

6.1.1 Thin airfoil assumption



$$\alpha \ll 1, |\eta(x)| \ll 1 \quad \text{or} \quad \eta/l \ll 1, \alpha \ll 1$$

Governing equation is

$$\nabla^2 \Phi = \frac{\partial^2 \Phi}{\partial x^2} + \frac{\partial^2 \Phi}{\partial y^2} = 0 \quad (6.1)$$

Boundary conditions are

$$\mathbf{v} \cdot \nabla F = 0 \quad \text{on } F(x, y) = 0 \quad (6.2a)$$

$$\nabla \Phi \rightarrow V_\infty \quad \text{at } \infty \quad (6.2b)$$

$$+ \text{KC(Kutta condition): } (\nabla \Phi)_{\text{TE}} \text{ is finite and continuous} \quad (6.3)$$

- Perturbations

CHAPTER 6. ELEMENTS OF THIN-AIRFOIL THEORY (LINEARIZED 2D POTENTIAL FLOW)

Since $\alpha \ll 1, |\eta| \ll 1$, can split the flow into

$$\mathbf{v} = \mathbf{V}_\infty + \mathbf{v}' \quad \text{except } |\mathbf{v}'| \ll 1$$

where $\mathbf{v}'$ is perturbation velocity

i.e.

$$u = V_\infty \cos \alpha + u', \quad v = V_\infty \sin \alpha + v'$$

Define perturbation potential to be $\mathbf{v}' = \nabla \phi$

hence $\Phi = \Phi_\infty + \phi$

$$\nabla^2 \Phi = 0 \implies \nabla^2 (\Phi_\infty + \phi) = 0 \implies \nabla^2 \phi = 0$$

$$\therefore \nabla^2 \phi \equiv \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 0 \quad (6.4)$$

Boundary conditions are

$$(\mathbf{V}_\infty + \nabla \phi) \cdot \nabla F = 0 \quad \text{on } F(x, y) = 0 \quad (6.5a)$$

$$\nabla^2 \phi \rightarrow 0 \quad \text{at } \infty \quad (6.5b)$$

$$+ \text{KC (Kutta condition): } (\nabla \phi)_{\text{TE}} \text{ is finite and continuous} \quad (6.6)$$

6.1.2 Linearized Boundary Condition

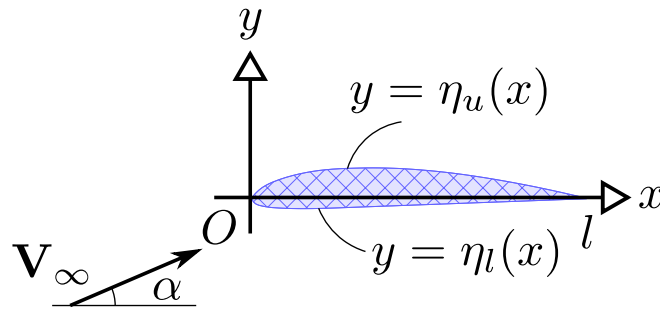
From B.C. (6.5a)

$$(\mathbf{V}_\infty + \nabla \phi) \cdot \nabla F = 0$$

6.1. PERTURBATION FLOW- SMALL PERTURBATION THEORY (THIN-AIRFOIL APPROXIMATION)

$$\implies (V_\infty \cos \alpha + u') \frac{\partial F}{\partial x} + (V_\infty \sin \alpha + v') \frac{\partial F}{\partial y} = 0 \text{ on } F(x, y) = 0 \quad (6.7)$$

From now on we drop prime ' and the subscript represent perturbation term.



$$(6.7) \implies \frac{V_\infty \sin \alpha + v}{V_\infty \cos \alpha + u} = \frac{-\frac{\partial F}{\partial x}}{\frac{\partial F}{\partial y}}^* = \left(\frac{dy}{dx} \right)_{\text{body}} \equiv \frac{d\eta}{dx} \quad (6.8)$$

(flow tangency condition on airfoil surface)

- Small Perturbation assumption

Since $|\eta(x)| \ll 1$, $\alpha \ll 1$ expect velocities u and v be *small*, i.e.

$$\frac{u}{V_\infty}, \frac{v}{V_\infty} \ll 1$$

therefore

$$V_\infty \cos \alpha \approx V_\infty + \mathcal{O}(\alpha^2)$$

$$V_\infty \sin \alpha \approx V_\infty \alpha + \mathcal{O}(\alpha^2)$$

$$^* dF = \frac{\partial F}{\partial x} dx + \frac{\partial F}{\partial y} dy = 0$$

CHAPTER 6. ELEMENTS OF THIN-AIRFOIL THEORY (LINEARIZED 2D POTENTIAL FLOW)

$$(6.8) \implies \frac{v + V_\infty \alpha}{V_\infty} \approx \frac{d\eta}{dx} \implies$$

$$v(x, \eta) \approx V_\infty \frac{d\eta}{dx} - V_\infty \alpha \text{ on } y = \eta(x), 0 \leq x \leq l \quad (6.9)$$

$$v(x, \eta) = v(x, 0) + \underbrace{\left(\frac{\partial v}{\partial y} \right)_{y=0}}_{\approx \mathcal{O}(\eta^2)} \eta + \mathcal{O}(\eta^2)$$

hence

$$v(x, 0) \approx V_\infty \frac{d\eta}{dx} - V_\infty \alpha, \quad 0 \leq x \leq l \quad (6.10)$$

(transfer of B.C.'s)

for non-symmetry geometry, (6.10) becomes

$$v(x, 0^+) = V_\infty \frac{d\eta_u}{dx} - V_\infty \alpha \quad (6.11a)$$

$$v(x, 0^-) = V_\infty \frac{d\eta_l}{dx} - V_\infty \alpha \quad (6.11b)$$

for $0 \leq x \leq l$

6.1.3 Linearized Pressure Coefficient

$$p(x, y) = p_\infty + \frac{\rho}{2} V_\infty^2 \left(1 - \frac{V^2}{V_\infty^2} \right)$$

where $V^2 = (V_\infty \cos \alpha + u)^2 + (V_\infty \sin \alpha + v)^2$

$$\implies p = p_\infty - \frac{\rho V_\infty^2}{2} \left(2 \frac{u}{V_\infty} \cos \alpha \overset{1}{\cancel{\rightarrow}} + \frac{2v}{V_\infty} \sin \alpha \overset{0}{\cancel{\rightarrow}} + \frac{u^2 + v^2 \overset{0}{\cancel{\rightarrow}}}{V_\infty^2} \right)$$

6.1. PERTURBATION FLOW- SMALL PERTURBATION THEORY (THIN-AIRFOIL APPROXIMATION)

$$\begin{aligned}
 &\Rightarrow p \approx p_\infty - \rho V_\infty u \\
 &\Rightarrow C_p = \frac{p - p_\infty}{\frac{\rho}{2} V_\infty^2} \\
 &\Rightarrow C_p \approx -\frac{2u}{V_\infty} = -\frac{2}{V_\infty} \frac{\partial \phi}{\partial x}
 \end{aligned} \tag{6.12}$$

6.1.4 Framework of the Thin-Airfoil Theory

G.E.

$$\nabla^2 \phi = \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 0 \tag{6.13}$$

B.C's

$$v(x, 0^+) = \left(\frac{\partial \phi}{\partial y} \right)_{x, 0^+} = V_\infty \frac{d\eta_u}{dx} - V_\infty \alpha \tag{6.14a}$$

$$v(x, 0^-) = \left(\frac{\partial \phi}{\partial y} \right)_{x, 0^-} = V_\infty \frac{d\eta_l}{dx} - V_\infty \alpha \tag{6.14b}$$

for $0 \leq x \leq l$

$$\frac{\partial \phi}{\partial x}, \frac{\partial \phi}{\partial y} \rightarrow 0 \quad \text{as} \quad x, y \rightarrow \infty \tag{6.15}$$

K.C.

$$\left(\frac{\partial \phi}{\partial x} \right)_{\text{TE}}, \left(\frac{\partial \phi}{\partial y} \right)_{\text{TE}} \text{ are } \textit{finite} \text{ and } \textit{continuous} \text{ at TE} \tag{6.16}$$

From the geometry linearlization

CHAPTER 6. ELEMENTS OF THIN-AIRFOIL THEORY (LINEARIZED 2D POTENTIAL FLOW)

introduce a camberline function η_c

$$\eta_c = \frac{1}{2}(\eta_u + \eta_l) \quad (6.17)$$

and thickness function η_t

$$\eta_t = \frac{1}{2}(\eta_u - \eta_l) \quad (6.18)$$

from (6.17) and (6.18)

$$\begin{aligned} \eta_u &= \eta_c + \eta_t \\ \eta_l &= \eta_c - \eta_t \end{aligned} \quad (6.19)$$

B.C. (6.14) becomes

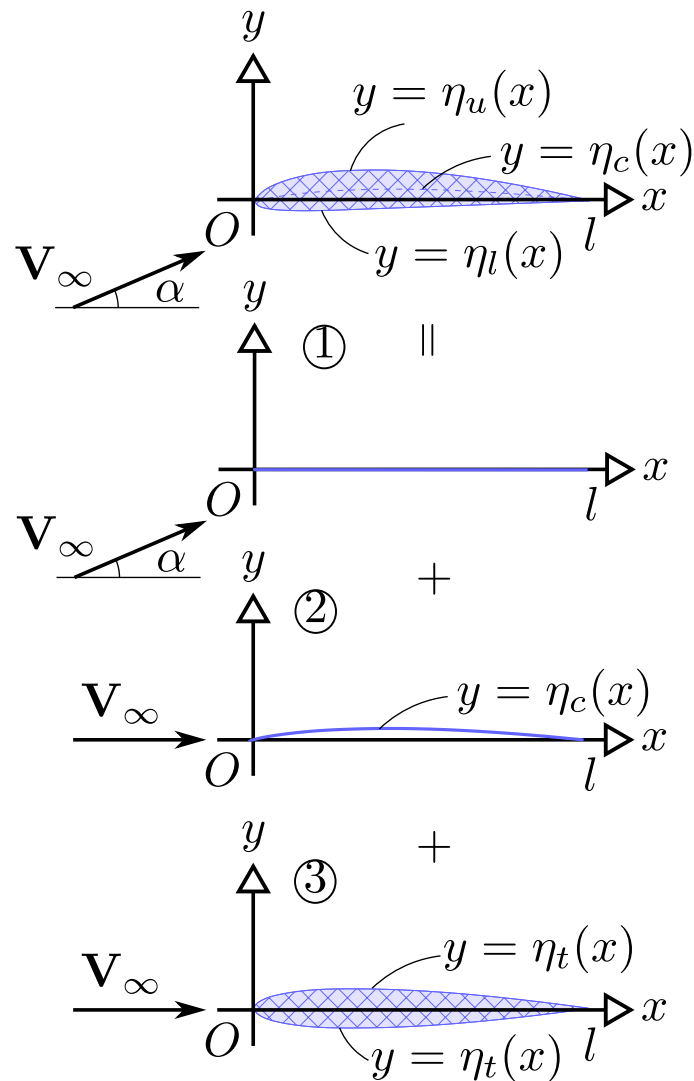
$$v(x, 0^+) = \left(\frac{\partial \phi}{\partial y} \right)_{x, 0^+} = -V_\infty \alpha + V_\infty \left(\frac{d\eta_c}{dx} + \frac{d\eta_t}{dx} \right) \quad (6.20a)$$

$$v(x, 0^-) = \left(\frac{\partial \phi}{\partial y} \right)_{x, 0^-} = -V_\infty \alpha + V_\infty \left(\frac{d\eta_c}{dx} - \frac{d\eta_t}{dx} \right) \quad (6.20b)$$

for $0 \leq x \leq l$

- ① flow past a zero-thickness flat plate at an angle-of-attack α
- ② flow past a zero-thickness camber at zero angle-of-attack
- ③ flow past a symmetric thickness airfoil at zero angle-of-attack

6.1. PERTURBATION FLOW- SMALL PERTURBATION THEORY (THIN-AIRFOIL APPROXIMATION)



CHAPTER 6. ELEMENTS OF THIN-AIRFOIL THEORY (LINEARIZED 2D POTENTIAL FLOW)

6.2 Solutions to Elementary Thin Airfoil Problems

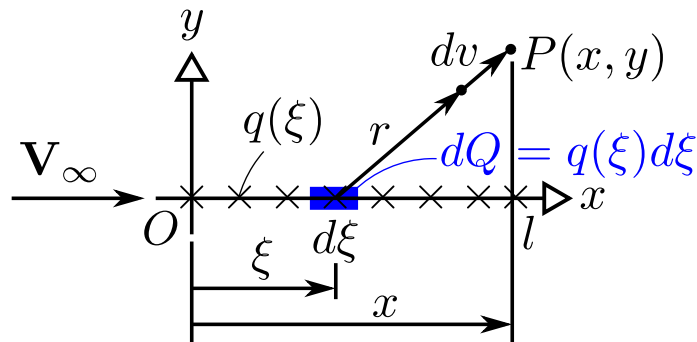
6.2.1 Symmetric Airfoil at Zero Angle of Attack- solution by source/sink Distribution

- Governing equation

$$\nabla^2 \phi = 0 \quad (6.21)$$

$$v(x, 0^\pm) = \left(\frac{\partial \phi}{\partial y} \right)_{x, 0^\pm} = \pm V_\infty \frac{d\eta_t}{dx}, 0 \leq x \leq l \quad (6.22a)$$

$$\nabla \phi \rightarrow 0 \text{ as } x, y \rightarrow 0 \quad (6.22b)$$



$$\begin{aligned} \phi(x, y) &= \int d\phi = \int \frac{dQ}{2\pi} \ln r \\ &= \frac{1}{2\pi} \int_{\xi=0}^l q(\xi) \ln \sqrt{(x - \xi)^2 + y^2} d\xi \end{aligned} \quad (6.23)$$

6.2. SOLUTIONS TO ELEMENTARY THIN AIRFOIL PROBLEMS

and perturbation velocities u, v are

$$u = \frac{\partial \phi}{\partial x} = \frac{1}{2\pi} \int_0^l q(\xi) \frac{x - \xi}{(x - \xi)^2 + y^2} d\xi \quad (6.24a)$$

$$v = \frac{\partial \phi}{\partial y} = \frac{1}{2\pi} \int_0^l q(\xi) \frac{y}{(x - \xi)^2 + y^2} d\xi \quad (6.24b)$$

The unknown strength $q(x)$ can be determined from the B.C. (6.22a)

$$\begin{aligned} v(x, 0^\pm) &= \lim_{y \rightarrow 0^\pm} \left[\frac{1}{2\pi} \int_0^l q(\xi) \frac{y}{(x - \xi)^2 + y^2} d\xi \right] \\ &= \pm V_\infty \frac{d\eta_t}{dx} \quad (0 \leq x \leq 1) \end{aligned} \quad (6.25)$$

which is *Integral equation*

For upper surface equation (6.25)

$$\begin{aligned} v(x, 0^+) &\stackrel{\dagger}{=} \frac{1}{2\pi} \left\{ \lim_{y \rightarrow 0^+} \left[\int_0^{x-\delta} + \int_{x-\delta}^l + \int_{x-\delta}^{x+\delta} \right] \frac{yq(\xi)}{(x - \xi)^2 + y^2} d\xi \right\} \\ &= \frac{1}{2\pi} \left[\lim_{y \rightarrow 0^+} \int_{x-\delta}^{x+\delta} \frac{yq(\xi)}{(x - \xi)^2 + y^2} d\xi \right] \end{aligned}$$

First, shift origin to point x , i.e. $\xi = x + \bar{\xi}$

$$v(x, 0^+) = \frac{1}{2\pi} \left[\lim_{y \rightarrow 0^+} \int_{\bar{\xi}=-\delta}^{\delta} \frac{yq(x + \bar{\xi})}{\bar{\xi}^2 + y^2} d\bar{\xi} \right], |\bar{\xi}| \ll 1 \quad (6.26)$$

[†]for small positive δ

CHAPTER 6. ELEMENTS OF THIN-AIRFOIL THEORY (LINEARIZED 2D POTENTIAL FLOW)

then re-scale the variable $\bar{\xi}$, i.e. let $\sigma = \bar{\xi}/y \sim \mathcal{O}(1)$

$$\begin{aligned}
 \Rightarrow v(x, 0^+) &= \frac{1}{2\pi} \left[\lim_{y \rightarrow 0^+} \int_{-\delta/y}^{\delta/y} \frac{q(x + y\sigma)}{\sigma^2 + 1} d\sigma \right] \\
 &\stackrel{\ddagger}{=} \frac{1}{2\pi} \lim_{y \rightarrow 0^+} \left\{ \int_{-\delta/y}^{\delta/y} \frac{q(x)}{\sigma^2 + 1} d\sigma + y \int_{-\delta/y}^{\delta/y} \frac{q'(x)\sigma}{\sigma^2 + 1} d\sigma \right\} + \dots \\
 &= \frac{q(x)}{2\pi} \left\{ \lim_{y \rightarrow 0^+} \int_{-\delta/y}^{\delta/y} \frac{d\sigma}{\sigma^2 + 1} \right\} \\
 &= \frac{q(x)}{2\pi} \left\{ \lim_{y \rightarrow 0^+} [\arctan \sigma]_{-\delta/y \rightarrow -\infty}^{\delta/y \rightarrow \infty} \right\} \\
 &= \frac{q(x)}{2}
 \end{aligned}$$

Similarly,

$$v(x, 0^-) = -\frac{q(x)}{2}$$

i.e.

$$q(x) = 2v(x, 0^+) = 2V_\infty \frac{d\eta_t}{dx} \quad (6.27)$$

Having known the $q(x)$

$$\Rightarrow u(x, y) = ?, \quad v(x, y) = ?$$

The pressure coefficient $C_p(x, y)$ is

$$C_p(x, y) = -\frac{2u(x, y)}{V_\infty} \quad (6.28)$$

[‡]hence $q(x + y\sigma) = q(x) + q'(x)y\sigma + \mathcal{O}(y^2)$

6.2. SOLUTIONS TO ELEMENTARY THIN AIRFOIL PROBLEMS

Specifically, the C_p on the surface of the airfoil is

$$C_p(x, 0^\pm) = -\frac{2u(x, 0^\pm)}{V_\infty} \quad (6.29a)$$

with

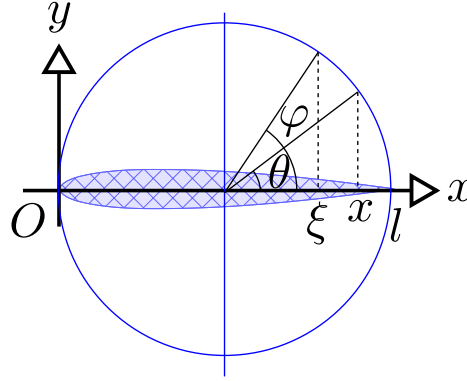
$$\begin{aligned} u(x, 0) &= \frac{1}{2\pi} \left\{ \int_{\xi=0}^l q(\xi) \frac{x - \xi}{(x - \xi)^2 + y^2} d\xi \right\}_{y=0} \\ &= \frac{1}{2\pi} \int_0^l \frac{q(\xi)}{x - \xi} d\xi = \frac{V_\infty}{\pi} \int_{\xi=0}^l \frac{d\eta_t(\xi)}{d\xi} \frac{1}{x - \xi} d\xi \end{aligned} \quad (6.29b)$$

(Hilbert integral)

Often, transform x, ξ into θ, φ by using

$$\begin{aligned} x &= \frac{l}{2} (1 + \cos \theta), \quad 0 \leq \theta \leq \pi (\text{upper}), \quad -\pi \leq \theta \leq 0 (\text{lower}) \\ \xi &= \frac{l}{2} (1 + \cos \varphi), \quad 0 \leq \varphi \leq \pi (\text{upper}), \quad -\pi \leq \varphi \leq 0 (\text{lower}) \end{aligned} \quad (6.30)$$

CHAPTER 6. ELEMENTS OF THIN-AIRFOIL THEORY (LINEARIZED 2D POTENTIAL FLOW)



then (6.29b) becomes Poisson's Integral Formula

$$\begin{aligned} u(\theta) &= -\frac{1}{2\pi} \int_{\varphi=\pi}^0 q(\varphi) \frac{\sin \varphi}{\cos \theta - \cos \varphi} d\varphi \\ &= -\frac{1}{2\pi} \int_0^\pi q(\varphi) \frac{\sin \varphi}{\cos \varphi - \cos \theta} d\varphi \end{aligned} \quad (6.31)$$

$$= -\frac{V_\infty}{\pi} \int_0^\pi \frac{d\eta_t(\varphi)}{d\xi} \frac{\sin \varphi}{\cos \varphi - \cos \theta} d\varphi \quad (6.32)$$

(Poisson's Integral Formula)

notations:

$$u(\theta) \equiv u(x(\theta), 0^\pm)$$

$$\frac{\partial \eta_+(\theta)}{\partial x} = \frac{\partial \eta_+(x)}{\partial x} \Big|_{x=x(\theta)}$$

- The pressure coefficient C_p on the surface of the airfoil

Since $\frac{d\eta_t}{dx}$ (odd function in x by expanding $\frac{d\eta_t}{dx}$ into a

6.2. SOLUTIONS TO ELEMENTARY THIN AIRFOIL PROBLEMS

Fourier Sine Series), using $x = \frac{l}{2}(1 + \cos \theta)$

$$\frac{d\eta_t(\theta)}{dx} = \sum_{n=1}^{\infty} A_n \sin n\theta \quad (6.33)$$

where

$$A_n = \frac{2}{\pi} \int_0^{\pi} \frac{d\eta_t(\theta)}{dx} \sin n\theta d\theta \quad (6.34)$$

then (6.32) becomes

$$\frac{u(\theta)}{V_{\infty}} = -\frac{1}{\pi} \int_0^{\pi} \underbrace{\left(\sum_{n=1}^{\infty} A_n \sin n\varphi \right)}_{\text{Given}} \underbrace{\frac{\sin \varphi}{\cos \varphi - \cos \theta}}_{\text{Kernal}} d\varphi \quad (6.35)$$

Now what is $\int_0^{\pi} \frac{\sin n\varphi \sin \varphi}{\cos \varphi - \cos \theta} d\varphi = ?$ first

$$\sin n\varphi \sin \varphi = -\frac{1}{2} [\cos(n+1)\varphi - \cos(n-1)\varphi]$$

Now define

$$I_n(\theta) = \int_0^{\pi} \frac{\cos n\varphi}{\cos \varphi - \cos \theta} d\varphi$$

$$\textcircled{1} \quad I_0(\theta) = \int_0^{\pi} \frac{1}{\cos \varphi - \cos \theta} d\varphi = 0$$

CHAPTER 6. ELEMENTS OF THIN-AIRFOIL THEORY (LINEARIZED 2D POTENTIAL FLOW)

$$\begin{aligned} \textcircled{2} \quad I_1(\theta) &= \int_0^\pi \frac{\cos \varphi}{\cos \varphi - \cos \theta} d\varphi = \\ &= \int_0^\pi \left[1 + \frac{\cos \theta}{\cos \varphi - \cos \theta} \right] d\varphi = \pi \end{aligned}$$

$$\textcircled{3} \quad I_2(\theta) = \cdots = 2 \cos \theta I_1 = 2\pi \cos \theta$$

for general n

$$I_n = \pi \frac{\sin n\theta}{\sin \theta} \quad (6.36)$$

Hence in (6.35)

$$\begin{aligned} \int_0^\pi \frac{\sin n\varphi \sin \varphi}{\cos \varphi - \cos \theta} d\varphi &= -\frac{1}{2} \left\{ \underbrace{\int_0^\pi \frac{\cos(n+1)\varphi}{\cos \varphi - \cos \theta} d\varphi}_{I_{n+1}} - \underbrace{\int_0^\pi \frac{\cos(n-1)\varphi}{\cos \varphi - \cos \theta} d\varphi}_{I_{n-1}} \right\} \\ &= -\frac{1}{2} [I_{n+1} - I_{n-1}] \\ &= -\frac{1}{2} \frac{\pi}{\sin \theta} [2 \cos n\theta \sin \theta] = -\pi \cos n\theta \end{aligned}$$

$$\therefore \frac{u(\theta)}{V_\infty} = \sum_{n=1}^{\infty} A_n \cos n\theta \quad (6.37)$$

where

$$A_n = \frac{2}{\pi} \int_0^\pi \frac{d\eta_t(\theta)}{dx} \sin n\theta d\theta$$

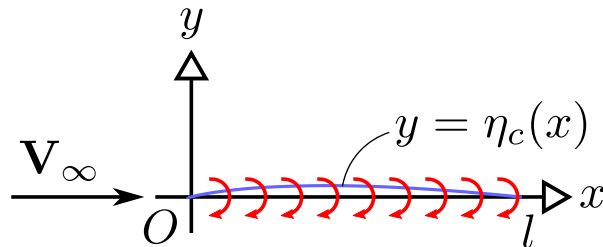
6.2. SOLUTIONS TO ELEMENTARY THIN AIRFOIL PROBLEMS

$$\begin{aligned}
 C_p(x, 0^\pm) &\equiv C_p(\theta) = -\frac{2u(\theta)}{V_\infty} \\
 &= -2 \sum_{n=1}^{\infty} A_n \cos n\theta
 \end{aligned} \tag{6.38}$$

where

$$\begin{aligned}
 0 \leq \theta \leq \pi &: && \text{upper surface} \\
 -\pi \leq \theta \leq 0 &: && \text{lower surface}
 \end{aligned}$$

6.2.2 Cambered Airfoil at zero Angle-of-Attack- solution by vortices Distribution (Vortex sheet)



- Governing equation

$$\nabla^2 \phi = 0 \tag{6.39}$$

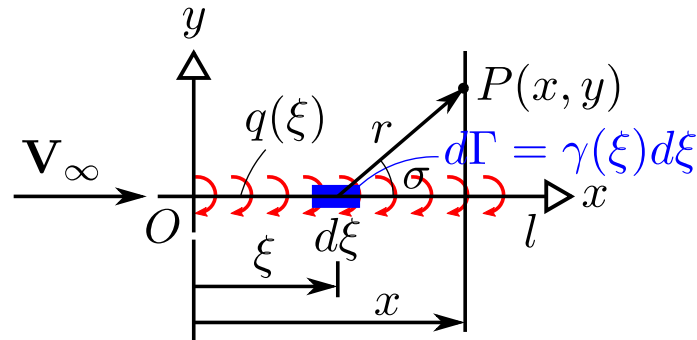
$$v(x, 0^\pm) = \left(\frac{\partial \phi}{\partial y} \right)_{x, 0^\pm} = V_\infty \frac{d\eta_c}{dx}, 0 \leq x \leq l \tag{6.40a}$$

$$\nabla \phi \rightarrow 0 \text{ as } x, y \rightarrow \infty \tag{6.40b}$$

CHAPTER 6. ELEMENTS OF THIN-AIRFOIL THEORY (LINEARIZED 2D POTENTIAL FLOW)

K.C.

$$u = \frac{\partial \phi}{\partial x}, v = \frac{\partial \phi}{\partial y} \text{ is finite and continuous at TE} \quad (6.41)$$



$$\begin{aligned} \phi(x, y) &= \int d\phi = \int -\frac{d\Gamma}{2\pi} \sigma = -\frac{1}{2\pi} \int_{\xi=0}^l \gamma(\xi) \sigma d\xi \\ &= -\frac{1}{2\pi} \int_0^l \gamma(\xi) \arctan \frac{y}{x - \xi} d\xi \end{aligned} \quad (6.42)$$

velocities u, v

$$u(x, y) = \frac{\partial \phi}{\partial x} = \frac{1}{2\pi} \int_0^l \gamma(\xi) \frac{y}{(x - \xi)^2 + y^2} d\xi \quad (6.43a)$$

$$v(x, y) = \frac{\partial \phi}{\partial y} = \frac{1}{2\pi} \int_0^l \gamma(\xi) \frac{x - \xi}{(x - \xi)^2 + y^2} d\xi \quad (6.43b)$$

Now what is $\gamma(x)$?

6.2. SOLUTIONS TO ELEMENTARY THIN AIRFOIL PROBLEMS

look at $u(x, 0^\pm)$ first

$$(6.43a) \implies u(x, 0^\pm) = \lim_{y \rightarrow 0^\pm} \left[\underbrace{\frac{1}{2\pi} \int_0^l \gamma(\xi) \frac{y}{(x - \xi)^2 + y^2} d\xi}_{\text{Improper integral}} \right] \quad (*)$$

Again, an improper integral

$$\langle \text{cf} \rangle (6.25), (6.26) \implies v(x, 0^+) = q(x)/2$$

Hence $(*)$ becomes

$$u(x, 0^+) = \frac{\gamma(x)}{2} \quad (6.44a)$$

$$u(x, 0^-) = -\frac{\gamma(x)}{2} \quad (6.44b)$$

$$(6.44a), (6.44b) \implies \gamma(x) = u(x, 0^+) - u(x, 0^-) \quad (6.45)$$

From (6.45), deduce

$$\text{KC} \implies \gamma(x = l) = 0 \quad (6.46)$$

The pressure coefficient C_p on the camber surface

$$C_p(x, 0^\pm) = -\frac{2u(x, 0^\pm)}{V_\infty} = \mp \frac{\gamma(x)}{V_\infty} \quad (6.47)$$

CHAPTER 6. ELEMENTS OF THIN-AIRFOIL THEORY (LINEARIZED 2D POTENTIAL FLOW)

where $+$ is *upper surface* and $-$ is *lower surface* The lift force on the camber airfoil

$$\begin{aligned} L &= \int_0^l [p(x, 0^-) - p(x, 0^+)] dx \\ &= \frac{\rho}{2} V_\infty^2 \int_0^l [C_p(x, 0^-) - C_p(x, 0^+)] dx \\ &= \rho V_\infty \int_0^l \gamma(x) dx = \rho V_\infty \Gamma \end{aligned}$$

The moment (about $x = 0$) on the camber airfoil is

$$\begin{aligned} M &= \int_0^l [p(x, 0^+) - p(x, 0^-)] x dx \\ &= -\rho V_\infty \int_0^l \gamma(x) x dx \end{aligned} \tag{6.48}$$

lift and moment coefficients

$$C_L = \frac{L}{\frac{\rho}{2} V_\infty^2 l} = \frac{2}{V_\infty l} \int_0^l \gamma(x) dx \tag{6.49}$$

$$C_M = \frac{M}{\frac{\rho}{2} V_\infty^2 l^2} = \frac{-2}{V_\infty l^2} \int_0^l \gamma(x) x dx \tag{6.50}$$

6.2. SOLUTIONS TO ELEMENTARY THIN AIRFOIL PROBLEMS

Now $\gamma(x)$ has to be determined from B.C. (6.40a)

$$\begin{aligned}
 v(x, 0^\pm) &= \lim_{y \rightarrow 0^\pm} \left[-\frac{1}{2\pi} \int_0^l \gamma(\xi) \frac{x - \xi}{(x - \xi)^2 + y^2} d\xi \right] = \underbrace{V_\infty \frac{d\eta_c}{dx}}_{\text{Given}} \quad (0 \leq x \leq l) \\
 &= -\frac{1}{2\pi} \int_0^l \gamma(\xi) \frac{d\xi}{x - \xi} = V_\infty \frac{d\eta_c}{dx} \quad (0 \leq x \leq l)
 \end{aligned}
 \tag{6.51}$$

(Singular integral/Hilbert integral)

- As before; can transform x, ξ into θ, φ by

$$\begin{aligned}
 x &= \frac{l}{2} (1 + \cos \theta) \quad 0 \leq \theta, \varphi \leq \pi & : & \text{upper surface} \\
 \xi &= \frac{l}{2} (1 + \cos \varphi) \quad -\pi \leq \theta, \varphi \leq 0 & : & \text{lower surface}
 \end{aligned}$$

$$\begin{aligned}
 (6.51) \implies \frac{1}{2\pi V_\infty} \int_{\varphi=0}^{\pi} \underbrace{\gamma(\varphi)}_{\text{unknown}} \frac{\sin \varphi}{\cos \varphi - \cos \theta} d\varphi &= \underbrace{\frac{d\eta_c(\theta)}{dx}}_{\text{Given}} \\
 & \tag{6.52}
 \end{aligned}$$

For arbitrary shape of camber

Since $\frac{d\eta_c}{dx}$ is an even function $\implies \frac{d\eta_c(\theta)}{dx}$ is an even

CHAPTER 6. ELEMENTS OF THIN-AIRFOIL THEORY (LINEARIZED 2D POTENTIAL FLOW)

function in θ

$$\Rightarrow \frac{d\eta_c(\theta)}{dx} = \frac{B_0}{2} + \sum_{n=1}^{\infty} B_n \cos n\theta \quad (6.53)$$

where

$$\underbrace{B_0 = \frac{2}{\pi} \int_0^\pi \frac{d\eta_c(\theta)}{dx} d\theta}_{\text{(average of slope)}}, \quad B_n = \frac{2}{\pi} \int_0^\pi \frac{d\eta_c(\theta)}{dx} \cos n\theta d\theta$$

Using (6.53), (6.52) $\Rightarrow$

$$\frac{1}{2\pi V_\infty} \int_0^\pi \underbrace{\gamma(\varphi)}_{\text{unknown}} \frac{\sin \varphi}{\cos \varphi - \cos \theta} d\varphi = \underbrace{\frac{B_0}{2} + \sum_{n=1}^{\infty} B_n \cos n\theta}_{\text{Given}} \quad (6.54)$$

wish to solve $\gamma(\theta) = ?$

Assume the solution to $\gamma(\theta)$ is

$$\gamma(\theta) = \gamma_1(\theta) + \gamma_2(\theta) + \gamma_h(\theta)$$

where

$$\frac{1}{2\pi V_\infty} \int_0^\pi \gamma_1(\varphi) \frac{\sin \varphi}{\cos \varphi - \cos \theta} d\varphi = \sum_{n=1}^{\infty} B_n \cos n\theta \quad (6.55)$$

$$\frac{1}{2\pi V_\infty} \int_0^\pi \gamma_2(\varphi) \frac{\sin \varphi}{\cos \varphi - \cos \theta} d\varphi = \frac{B_0}{2} \quad (6.56)$$

6.2. SOLUTIONS TO ELEMENTARY THIN AIRFOIL PROBLEMS

$$\frac{1}{2\pi V_\infty} \int_0^\pi \gamma_h(\varphi) \frac{\sin \varphi}{\cos \varphi - \cos \theta} d\varphi = 0 \quad (6.57)$$

Solution to $\gamma_1(\theta), \gamma_2(\theta), \gamma_h(\theta)$

$$\gamma_1(\varphi) = -2V_\infty \sum_{n=1}^{\infty} B_n \sin n\varphi \quad (\text{Recall (6.35)})$$

or

$$\gamma_1(\theta) = -2V_\infty \sum_{n=1}^{\infty} B_n \sin n\theta \quad (6.58)$$

Recall that

$$I_n = \int_0^\pi \frac{\cos n\varphi}{\cos \varphi - \cos \theta} d\varphi = \pi \frac{\sin n\theta}{\sin \theta}, \quad n = 0, 1, 2, \dots$$

which suggest

$$\gamma_2(\varphi) = V_\infty B_0 \frac{\cos \theta}{\sin \theta} \quad (6.59)$$

Also $I_0 = 0$, which suggest

$$\gamma_h(\theta) = \frac{K}{\sin \theta} \quad (6.60)$$

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Hence

$$\begin{aligned}\gamma(\theta) &= \gamma_1(\theta) + \gamma_2(\theta) + \gamma_h(\theta) \\ &= -2V_\infty \sum_{n=1}^{\infty} B_n \sin n\theta + \frac{V_\infty B_0}{\sin \theta} \left(\frac{\overbrace{K}^{\text{undetermined}}}{V_\infty B_0} + \cos \theta \right)\end{aligned}\tag{6.61}$$

Apply K.C.

$$\gamma(x=l) = \gamma(\theta=0) = 0 \implies \frac{1}{\sin \theta} \left(\frac{K}{V_\infty B_0} + \cos \theta \right) \bigg|_{\theta=0} = 0$$

$$\implies K = -V_\infty B_0$$

$$\therefore \gamma(\theta) = -2V_\infty \left(\frac{B_0}{2} \frac{1 - \cos \theta}{\sin \theta} + \sum_{n=1}^{\infty} B_n \sin n\theta \right) \tag{6.62}$$

6.2. SOLUTIONS TO ELEMENTARY THIN AIRFOIL PROBLEMS

- Pressure coefficient along the camber

$$\begin{aligned}
 C_p(\theta) &\equiv C_p(x(\theta), 0^\pm) = -\frac{2u(x(\theta), 0^\pm)}{V_\infty} \\
 &= -\frac{\gamma(\theta)}{V_\infty}, 0 \leq \theta \leq \pi (\text{upper}), -\pi \leq \theta \leq 0 (\text{lower}) \\
 &= B_0 \frac{1 - \cos \theta}{\sin \theta} + 2 \sum_{n=1}^{\infty} B_n \sin n\theta
 \end{aligned} \tag{6.63}$$

$$\begin{aligned}
 C_L &= \frac{2}{V_\infty l} \int_{x=0}^l \gamma(x) dx, x = \frac{l}{2} (1 + \cos \theta) \\
 &= \frac{1}{V_\infty} \int_{\theta=0}^{\pi} \gamma(\theta) \sin \theta d\theta
 \end{aligned} \tag{6.64}$$

$$\begin{aligned}
 C_M &= -\frac{2}{V_\infty l^2} \int_{x=0}^l \gamma(x) x dx \\
 &= -\frac{1}{2V_\infty} \int_{\theta=0}^{\pi} \gamma(\theta) (1 + \cos \theta) \sin \theta d\theta
 \end{aligned} \tag{6.65}$$

Use (6.62)

$$C_L = -(B_0 + B_1) \pi \tag{6.66}$$

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$$\begin{aligned}
 C_M &= B_0 \frac{\pi}{4} + B_1 \frac{\pi}{2} + B_2 \frac{\pi}{4} \\
 &= \frac{\pi}{4} (B_0 + B_1) + \frac{\pi}{4} (B_1 + B_2) \\
 &= -\frac{C_L}{4} + \frac{\pi}{4} (B_1 + B_2)
 \end{aligned} \tag{6.67}$$

Note.

$$\begin{aligned}
 \int_0^\pi \sin n\theta \sin m\theta d\theta &= 0 && \text{for } n \neq m \\
 &= \frac{\pi}{2} && \text{for } n = m
 \end{aligned}$$

Recall

$$\begin{aligned}
 B_0 &= \frac{2}{\pi} \int_0^\pi \frac{d\eta_c(\theta)}{dx} d\theta \\
 B_n &= \frac{2}{\pi} \int_0^\pi \frac{d\eta_c(\theta)}{dx} \cos n\theta d\theta, \quad \text{for } n = 1, 2, \dots
 \end{aligned} \tag{6.68}$$

6.2.3 Flat-Plate Airfoil at an Angle-of-Attack α - solution by vortices Distribution (Vortex sheet)

- Governing equation

$$\nabla^2 \phi = 0 \tag{6.69}$$

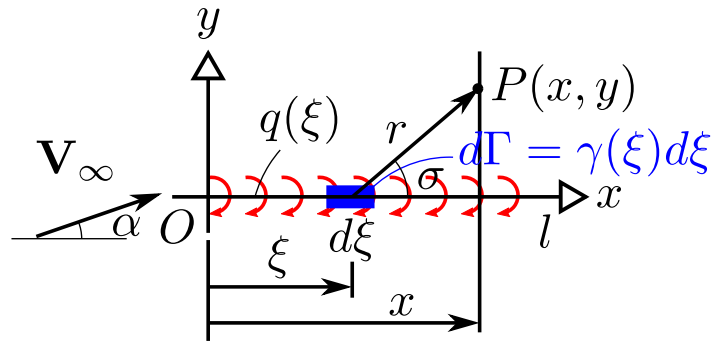
$$v(x, 0^\pm) = \left(\frac{\partial \phi}{\partial y} \right)_{x, 0^\pm} = -V_\infty \alpha, 0 \leq x \leq l \tag{6.70a}$$

$$\nabla \phi \rightarrow 0 \text{ as } x, y \rightarrow 0 \tag{6.70b}$$

6.2. SOLUTIONS TO ELEMENTARY THIN AIRFOIL PROBLEMS

K.C.

$$u = \frac{\partial \phi}{\partial x}, v = \frac{\partial \phi}{\partial y} \text{ is finite and continuous at TE} \quad (6.71)$$



From previous cambered airfoil case

$$v(x, 0^\pm) = -\frac{1}{2\pi} \int_0^l \gamma(\xi) \frac{d\xi}{x - \xi} \quad (0 \leq x \leq l) \quad (6.72)$$

As before, transform $\begin{cases} x \\ \xi \end{cases} \Rightarrow \begin{cases} \theta \\ \varphi \end{cases}$ by $\begin{cases} x = \frac{l}{2}(1 + \cos \theta) \\ \xi = \frac{l}{2}(1 + \cos \varphi) \end{cases}$

$$(6.72) \Rightarrow v(x, 0^\pm) = \frac{1}{2\pi} \int_0^l \gamma(\varphi) \frac{\sin \varphi}{\cos \varphi - \cos \theta} d\varphi = -V_\infty \alpha \quad (6.73)$$

As before, can get $\gamma(\theta) = \gamma_2(\theta) + \gamma_h(\theta)$ <cf> (6.54)

$$\Rightarrow \gamma(\theta) = \underbrace{\frac{K}{\sin \theta}}_{\gamma_h(\theta)} - \underbrace{2V_\infty \alpha \frac{\cos \theta}{\sin \theta}}_{\gamma_2(\theta)} \quad (6.74)$$

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To satisfy $\gamma(\theta = 0) = 0 \implies K = 2V_\infty\alpha$, i.e.

$$\gamma(\theta) = 2V_\infty\alpha \frac{1 - \cos \theta}{\sin \theta} \quad (6.75)$$

- The pressure coefficient along the flat plate

$$C_p(\theta) = -\frac{\gamma(\theta)}{V_\infty} = -2\alpha \frac{1 - \cos \theta}{\sin \theta} \quad (6.76)$$

$0 \leq \theta \leq \pi$: upper surface, $-\pi \leq \theta \leq 0$: lower surface

$$C_L = \frac{1}{V_\infty} \int_{\theta=0}^{\pi} \gamma(\theta) \sin \theta d\theta = 2\alpha \int_0^{\pi} \frac{1 - \cos \theta}{\sin \theta} \sin \theta d\theta = 2\pi\alpha \quad (6.77)$$

$$C_M = -\frac{1}{2V_\infty} \int_{\theta=0}^{\pi} \gamma(\theta) (1 + \cos \theta) \sin \theta d\theta = -\frac{\pi}{2}\alpha = -\frac{C_L}{4} \quad (6.78)$$

center of pressure

$$C_L = \frac{L}{\frac{\rho}{2}V_\infty^2 l} (= 2\pi\alpha), \quad C_M = \frac{M}{\frac{\rho}{2}V_\infty^2 l^2} (= -\frac{C_L}{4})$$

$$M = -Lx_{cp}$$

$$\implies \frac{\rho}{2}V_\infty^2 l^2 C_M = \frac{\rho}{2}V_\infty^2 l^2 \left(-\frac{C_L}{4}\right) = -L\frac{l}{4}$$

$$\therefore \boxed{x_{cp} = \frac{l}{4}} \quad (6.79)$$

Aerodynamic center (x_{ac}):

6.2. SOLUTIONS TO ELEMENTARY THIN AIRFOIL PROBLEMS

About which the moment is independent of angle-of-attack α . Take moment about $x = l/4$

$$\implies \overline{C_M} = 0 \quad \text{independent of } \alpha \quad (6.80)$$

so in the case: $x_{ac} = x_{cp} = l/4$

6.2.4 Aerodynamic Characteristic of a Complete Thin Airfoil (Superposition)

Superpose the above 3 elementary cases:
camber + thickness

$$\implies \frac{d\eta_{u,l}}{dx} = \frac{d\eta_c}{dx} \pm \frac{d\eta_t}{dx}$$

or

$$\frac{d\eta(\theta)}{dx} = \frac{d\eta_c(\theta)}{dx} + \frac{d\eta_t(\theta)}{dx}, \quad \left(x = \frac{l}{2} (1 + \cos \theta) \right) \quad (6.81)$$

from previous results

$$\frac{d\eta(\theta)}{dx} = \underbrace{\frac{B_0}{2} + \sum_{n=1}^{\infty} B_n \cos n\theta}_{\text{camber}} + \underbrace{\sum_{n=1}^{\infty} A_n \sin n\theta}_{\text{thickness}} \quad (6.82)$$

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where

$$\begin{cases} A_n = \frac{2}{\pi} \int_0^\pi \frac{d\eta_t(\theta)}{dx} \sin n\theta d\theta, n = 1, 2, 3, \dots \\ B_0 = \frac{2}{\pi} \int_0^\pi \frac{d\eta_c(\theta)}{dx} d\theta \\ B_n = \frac{2}{\pi} \int_0^\pi \frac{d\eta_c(\theta)}{dx} \cos n\theta d\theta, n = 1, 2, 3, \dots \end{cases} \quad (6.83)$$

Pressure distribution

$$\begin{aligned} C_p(\theta) &= -2 \frac{u(\theta)}{V_\infty} \\ &= \underbrace{-2\alpha \frac{1 - \cos \theta}{\sin \theta}}_{\text{due to flat plat at } \alpha \text{ angle-of-attack}} + \underbrace{\left[B_0 \frac{1 - \cos \theta}{\sin \theta} + 2 \sum_{n=1}^{\infty} B_n \sin n\theta \right]}_{\text{due to camber at zero angle-of-attack}} - \underbrace{2 \sum_{n=1}^{\infty} A_n \cos n\theta}_{\text{due to thickness at zero angle-of-attack}} \end{aligned} \quad (6.84)$$

$$C_L = \underbrace{2\pi\alpha}_{\text{due to flat plate at } \alpha} - \underbrace{(B_0 + B_1)\pi}_{\text{due to camber at zero}} \quad (6.85)$$

6.2. SOLUTIONS TO ELEMENTARY THIN AIRFOIL PROBLEMS

$$\begin{aligned}
 C_M &= \underbrace{-\frac{\pi}{2}\alpha}_{\text{due to flat plate at } \alpha} + \underbrace{\frac{\pi}{4}(B_0 + B_1) + \frac{\pi}{4}(B_1 + B_2)}_{\text{due to camber at zero}} \\
 &= -\frac{C_L}{4} + \frac{\pi}{4}(B_1 + B_2)
 \end{aligned} \tag{6.86}$$

Take moment about $x_{ac} = l/4$

$$\Rightarrow \overline{C_M} = \frac{\pi}{4}(B_1 + B_2) \quad \text{independent of } \alpha \tag{6.87}$$

Example. a parabolic camber airfoil

$$y = 4\epsilon \frac{x}{l} (l - x), \quad (\epsilon \ll 1)$$

$$\begin{aligned}
 \frac{d\eta_c}{dx} &\equiv \frac{dy}{dx} = 4\epsilon - \frac{8\epsilon}{l}x = 4\epsilon - \frac{8\epsilon l}{l} \frac{1}{2} (1 + \cos \theta) \\
 &= -4\epsilon \cos \theta = \frac{B_0}{2} + \sum_{n=1}^{\infty} B_n \cos n\theta
 \end{aligned}$$

$$\Rightarrow B_0 = 0, \quad B_1 = -4\epsilon, \quad B_n = 0 \quad (n \geq 2)$$

$$C_L = 2\pi\alpha - (B_0 + B_1)\pi = 2\pi\alpha + 4\epsilon\pi$$

$$C_M = -\frac{C_L}{4} + \frac{\pi}{4}(B_1 + B_2) = -\frac{\pi\alpha}{2} - 2\epsilon\pi$$

$$\overline{C_M} = \frac{\pi}{4}(B_1 + B_2) = -\epsilon\pi \quad \text{independent of } \alpha$$

also

$$\begin{aligned}
 \gamma(\theta) &= 2V_{\infty}\alpha \frac{1 - \cos \theta}{\sin \theta} - 2V_{\infty} \left(\frac{B_0}{2} \frac{1 - \cos \theta}{\sin \theta} + \sum_{n=1}^{\infty} B_n \sin n\theta \right) \\
 &= 2V_{\infty}\alpha \frac{1 - \cos \theta}{\sin \theta} + 8\epsilon V_{\infty} \sin \theta
 \end{aligned}$$

CHAPTER 6. ELEMENTS OF THIN-AIRFOIL THEORY (LINEARIZED 2D POTENTIAL FLOW)

and

$$\Gamma = \int_{x=0}^l \gamma(x) dx = \frac{l}{2} \int_{\theta=0}^{\pi} \gamma(\theta) \sin \theta d\theta = \pi V_{\infty} l (\alpha + 2\epsilon)$$

$$C_p(\theta) = -2\alpha \frac{1 - \cos \theta}{\sin \theta} - 8\epsilon \sin \theta$$

6.2. SOLUTIONS TO ELEMENTARY THIN AIRFOIL PROBLEMS

Chapter 7

Three-Dimensional Potential Flows

$$\left. \begin{array}{l} \text{Incompressible: } \nabla \cdot \mathbf{v} = 0 \\ \text{Irrotational: } \nabla \times \mathbf{v} = 0 \end{array} \right\} \implies \nabla^2 \Phi = 0, \mathbf{v} = \nabla \Phi \quad (7.1)$$

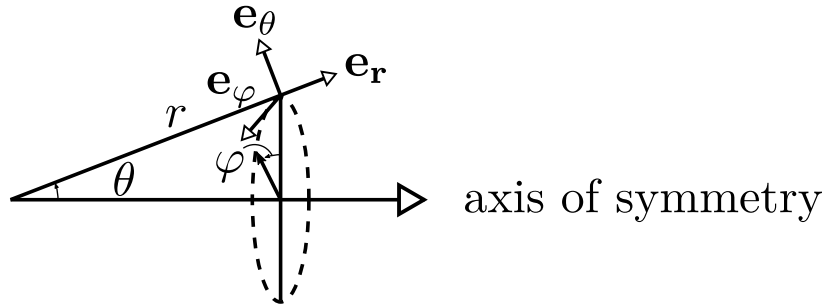
7.1 General Solution to 3D Axisymmetric Potential Flows

7.1.1 Governing Equations

$$\nabla^2 \Phi = 0 \quad (7.2)$$

Since axisymmetric, adopt spherical coordinate (r, θ, φ)

7.1. GENERAL SOLUTION TO 3D AXISYMMETRIC POTENTIAL FLOWS



$$\nabla^2 \Phi = 0, \quad \left(\frac{\partial}{\partial \varphi} = 0 \right)$$

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \Phi}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial \Phi}{\partial \theta} \right) = 0 \quad (7.3)$$

(Laplace equation)

- Stream function- Stokes' stream function
Recall vector potential in spherical coordinates

$$\mathbf{\Psi} = \mathbf{e}_\varphi \frac{1}{r \sin \theta} \underbrace{\psi}_{\text{Stokes' stream function}}$$

where

$$v_r = \frac{1}{r^2 \sin \theta} \frac{\partial \psi}{\partial \theta} \quad (7.4a)$$

$$v_\theta = -\frac{1}{r \sin \theta} \frac{\partial \psi}{\partial r} \quad (7.4b)$$

and governing equation for $\mathbf{\Psi}$

$$\nabla^2 \mathbf{\Psi} = 0 \quad \text{i.e.} \quad \nabla^2 \left(\mathbf{e}_\varphi \frac{1}{r \sin \theta} \psi \right) = 0$$

CHAPTER 7. THREE-DIMENSIONAL POTENTIAL FLOWS

spelling out

$$\Rightarrow \underbrace{\frac{\partial^2 \psi}{\partial r^2} + \frac{\sin \theta}{r^2} \frac{\partial}{\partial \theta} \left(\frac{1}{\sin \theta} \frac{\partial \psi}{\partial \theta} \right)}_{\text{Not Laplace equation}} = 0 \quad (7.5)$$

7.1.2 General Solution to the Potential Equation for Axisymmetric Flows

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \Phi}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial \Phi}{\partial \theta} \right) = 0$$

Use separation of variables

$$\text{Let } \Phi(r, \theta) = f(r)g(\theta) \quad (7.6)$$

Substitute into (7.3)

$$\frac{1}{f} \frac{d}{dr} \left(r^2 \frac{df}{dr} \right) = - \frac{1}{g \sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{dg}{d\theta} \right) = \text{constant} \equiv \lambda$$

For $f(r)$:

$$\frac{d}{dr} \left(r^2 \frac{df}{dr} \right) - \lambda f = 0 \quad (7.7)$$

Let $f(r) = Ar^\alpha$ subs into (7.7)

$$\alpha(\alpha + 1) Ar^\alpha - \lambda Ar^\alpha = 0 \Rightarrow \alpha(\alpha + 1) - \lambda = 0$$

$$\alpha_1 = \frac{-1 + \sqrt{1 + 4\lambda}}{2} \quad (7.8a)$$

$$\alpha_2 = \frac{-1 - \sqrt{1 + 4\lambda}}{2} \quad (7.8b)$$

7.1. GENERAL SOLUTION TO 3D AXISYMMETRIC POTENTIAL FLOWS

$$\implies f(r) = A_1 r^{\alpha_1} + A_2 r^{\alpha_2} \quad (7.9)$$

For $g(\theta)$:

$$\frac{1}{\sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{dg}{d\theta} \right) + \lambda g = 0 \quad (7.10)$$

Let $x = \cos \theta$

$$\implies \frac{d}{dx} \left[(1 - x^2) \frac{dg}{dx} \right] + \lambda g = 0, \quad -1 \leq x \leq 1 \quad (7.11)$$

This is a Legendre differential equation

Solution is Legendre polynomials with $\lambda = n(n+1)$,
 $n = 0, 1, 2, \dots$ (eigenvalues)

$$n = 0 \implies P_0(x) = 1$$

$$n = 1 \implies P_1(x) = x = \cos \theta$$

$$n = 2$$

$$\implies P_2(x) = \frac{1}{2} (3x^2 - 1) = \frac{1}{2} (3 \cos^2 \theta - 1) = \frac{1}{4} (3 \cos 2\theta + 1)$$

$$n = 2$$

$$\begin{aligned} \implies P_2(x) &= \frac{1}{2} (3x^2 - 1) = \frac{1}{2} (3 \cos^2 \theta - 1) \\ &= \frac{1}{4} (3 \cos 2\theta + 1) \end{aligned}$$

$$n = 3$$

$$\begin{aligned} \implies P_3(x) &= \frac{1}{2} (5x^3 - 3x) = \frac{1}{2} (5 \cos^3 \theta - 3 \cos \theta) \\ &= \frac{1}{8} (5 \cos 3\theta + 3 \cos \theta) \end{aligned}$$

CHAPTER 7. THREE-DIMENSIONAL POTENTIAL FLOWS

hence

$$g(x) = \sum_{n=0}^{\infty} a_n P_n(x) \quad \text{or} \quad g(\theta) = \sum_{n=0}^{\infty} a_n P_n(\cos \theta)$$

Combine $f(r), g(\theta)$, since $\lambda = n(n+1)$

$$(7.8) \implies \begin{cases} \alpha_1 = n \\ \alpha_2 = -(n+1) \end{cases}$$

i.e.

$$f(r) = A_1 r^n + \frac{A_2}{r^{n+1}}$$

so

$$\Phi = f(r)g(\theta) = \boxed{\sum_{n=0}^{\infty} \left(A_n r^n + \frac{B_n}{r^{n+1}} \right) P_n(\cos \theta)} \quad (7.12)$$

7.2 Some Elementary 3D Potential Flows

$$\Phi(r, \theta) = \sum_{n=0}^{\infty} \left(A_n r^n + \frac{B_n}{r^{n+1}} \right) P_n(\cos \theta) \quad (7.13)$$

- Uniform flow

In (7.13), take $B_n = 0$ for all n 's and $P_1(\cos \theta) = \cos \theta$

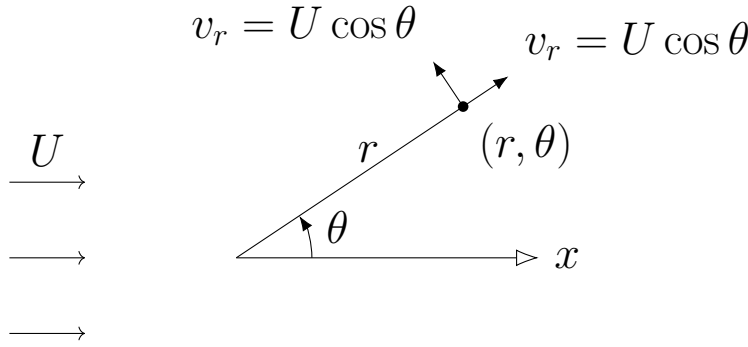
$$A_n = U \text{ for } n = 1 \quad A_n = 0 \text{ for } n \neq 1$$

7.2. SOME ELEMENTARY 3D POTENTIAL FLOWS

$\therefore$ (7.13) gives $\implies$

$$\Phi(r, \theta) = Ur \cos \theta \quad (7.14)$$

$$\begin{cases} v_r = \frac{\partial \Phi}{\partial r} = U \cos \theta & (7.15a) \\ v_\theta = \frac{1}{r} \frac{\partial \Phi}{\partial \theta} = -U \sin \theta & (7.15b) \end{cases}$$



Exercise. Show that the Stokes' stream function for uniform flow is

$$\psi(r, \theta) = \frac{1}{2} U r^2 \sin^2 \theta \quad (7.16)$$

using

$$v_r = \frac{1}{r^2 \sin \theta} \frac{\partial \psi}{\partial \theta}, \quad v_\theta = -\frac{1}{r \sin \theta} \frac{\partial \psi}{\partial r}$$

- Point source/sink

In (7.13), take $A_n = 0$ for all n' 's and $P_0(\cos \theta) = 1$

$$B_n = B_0 \text{ for } n = 0 \quad B_n = 0 \text{ for } n \neq 0$$

CHAPTER 7. THREE-DIMENSIONAL POTENTIAL FLOWS

$\therefore$ (7.13) gives $\implies$

$$\Phi(r, \theta) = \frac{B_0}{r} \quad (7.17)$$

Velocity components are

$$\begin{cases} v_r = \frac{\partial \Phi}{\partial r} = -\frac{B}{r^2} & (7.18a) \\ v_\theta = \frac{1}{r} \frac{\partial \Phi}{\partial \theta} = 0 & (7.18b) \end{cases}$$

volume flow rate is

$$\begin{aligned} Q &= \oiint_S \mathbf{v} \cdot \mathbf{n} dS = \int_{\varphi=0}^{\pi} d\varphi \int_{\theta=0}^{\pi} \left(-\frac{B_0}{r^2} \right) r^2 \sin \theta d\theta \\ &= -4\pi B_0 \end{aligned}$$

$$\therefore \Phi = \frac{B_0}{r} = -\frac{Q}{4\pi r} \quad (7.19)$$

where

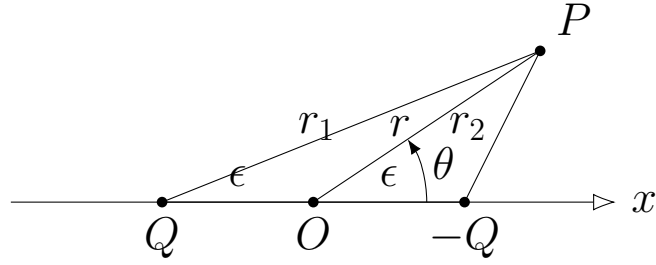
$$\begin{aligned} B_0: \quad " - " &\implies Q: \quad " + " \implies \text{source} \\ B_0: \quad " + " &\implies Q: \quad " - " \implies \text{sink} \end{aligned}$$

Exercise. Show that the Stokes' stream function for source/sink is

$$\begin{aligned} \psi(r, \theta) &= \frac{1}{2} U r^2 \sin^2 \theta \\ &= -\frac{Q}{4\pi} (1 + \cos \theta) \end{aligned} \quad (7.20)$$

7.2. SOME ELEMENTARY 3D POTENTIAL FLOWS

- Doublet



$$\text{at } P: \quad \Phi(r, \theta) = -\frac{Q}{4\pi r_1} + \frac{Q}{4\pi r_2} \quad (7.21)$$

but

$$\begin{aligned} r_1^2 &= r^2 + \epsilon^2 - 2\epsilon r \cos(\pi - \theta) \\ &= r^2 + \epsilon^2 + 2\epsilon r \cos \theta \\ &= r^2 \left[1 + \left(\frac{\epsilon}{r} \right)^2 + \left(\frac{2\epsilon}{r} \right) \cos \theta \right] \end{aligned}$$

$$\therefore \frac{1}{r_1^2} \approx \frac{1}{r^2} \left[1 - \frac{2\epsilon}{r} \cos \theta + \mathcal{O}\left(\frac{\epsilon}{r}\right)^2 \right]$$

Similarly

$$\therefore \frac{1}{r_2^2} \approx \frac{1}{r^2} \left[1 + \frac{2\epsilon}{r} \cos \theta + \mathcal{O}\left(\frac{\epsilon}{r}\right)^2 \right]$$

CHAPTER 7. THREE-DIMENSIONAL POTENTIAL FLOWS

$$\begin{aligned}
 \therefore \Phi(r, \theta) &= -\frac{Q}{4\pi} \left(\frac{1}{r_1} - \frac{1}{r_2} \right) \\
 &\approx -\frac{Q}{4\pi} \frac{1}{r} \left[\sqrt{1 - \frac{2\epsilon}{r} \cos \theta} - \sqrt{1 + \frac{2\epsilon}{r} \cos \theta} \right] \\
 &\approx -\frac{Q}{4\pi r} \left[\left(1 - \frac{\epsilon}{r} \cos \theta \right) - \left(1 + \frac{\epsilon}{r} \cos \theta \right) \right] \\
 &= \frac{2\epsilon Q}{4\pi r^2} \cos \theta
 \end{aligned}$$

let $\epsilon \rightarrow 0$ and $Q \rightarrow \infty$, such that $2\epsilon Q \rightarrow \mu$, then

$$\Phi(r, \theta) = \frac{\mu \cos \theta}{4\pi r^2} \quad (7.22)$$

where μ is strength of the doublet
velocity components are

$$\begin{cases} v_r = \frac{\partial \Phi}{\partial r} = -\frac{\mu}{2\pi r^3} \cos \theta & (7.23a) \\ v_\theta = \frac{1}{r} \frac{\partial \Phi}{\partial \theta} = -\frac{\mu}{4\pi r^3} \sin \theta & (7.23b) \end{cases}$$

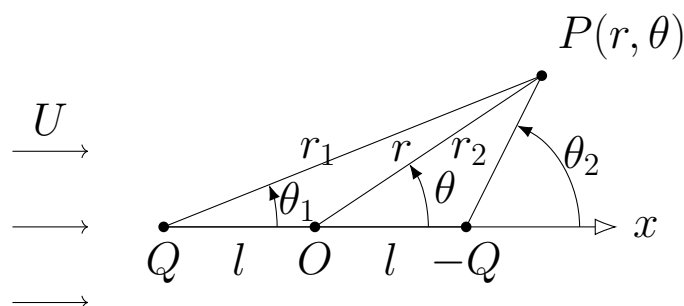
Exercise. Show that the Stokes' stream function for doublet is

$$\psi(r, \theta) = -\frac{\mu}{4\pi r} \sin^2 \theta \quad (7.24)$$

7.3. FLOWS GENERATED BY SUPERPOSITION

7.3 Flows Generated by Superposition

7.3.1 Rankine Bodies



$$\Phi = Ur \cos \theta - \frac{Q}{4\pi r_1} + \frac{Q}{4\pi r_2}, \quad Q > 0 \quad (7.25)$$

Stream function is

$$\psi = \frac{1}{2}Ur^2 \sin^2 \theta - \frac{Q}{4\pi} \cos \theta_1 + \frac{Q}{4\pi} \cos \theta_2 \quad (7.26)$$

Exercise. Show that

$$\begin{cases} \left(L^2 - l^2 \right)^2 - \frac{Ql}{\pi U} L = 0 \\ h^2 \sqrt{h^2 + l^2} - \frac{Ql}{\pi U} = 0 \end{cases}$$

7.3.2 Uniform Flow Past a sphere

Stokes' stream function is

$$\psi = \frac{U}{2}r^2 \sin^2 \theta - \frac{\mu}{4\pi r} \sin^2 \theta$$

CHAPTER 7. THREE-DIMENSIONAL POTENTIAL FLOWS

require $\psi = 0$ at $r = a \implies \mu = 4\pi U a^3$, i.e.

$$\psi = \frac{U}{2} \left(r^2 - \frac{a^3}{r} \right) \sin^2 \theta \quad (7.27)$$

Exercise. Show that

$$\Phi = U \left(r + \frac{a^3}{2r^2} \right) \cos \theta \quad (7.28)$$

velocity components

$$\begin{cases} v_r = \frac{\partial \Phi}{\partial r} = U \left(1 - \frac{a^3}{r^3} \right) \cos \theta & (7.29a) \\ v_\theta = \frac{1}{r} \frac{\partial \Phi}{\partial \theta} = -U \left(1 + \frac{a^3}{2r^3} \right) \sin \theta & (7.29b) \end{cases}$$

velocities on the sphere

$$\begin{cases} v_r = 0 \\ v_\theta = -\frac{3}{2}U \sin \theta \end{cases}$$

<cf> 2D cylinder

$$\begin{cases} v_r = 0 \\ v_\theta = -2U \sin \theta \end{cases}$$

7.3. FLOWS GENERATED BY SUPERPOSITION

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